

CONCEPTS OF PROBABILITY

6.1 Introduction to Probability

In daily life, many situations involve uncertainty. People often say things like:

- "Our cricket team is likely to win the World Cup."
- "I have a good chance of getting selected in the college team."
- "Probably I may meet the Prime Minister today."

Words such as likely, chance, and probably show that there is some doubt or uncertainty about whether an event will occur. At first, it may seem that such uncertainties cannot be measured. But in Mathematics, there are methods to calculate and express uncertainty in numerical form. This study belongs to a branch of Mathematics called **Probability**.



Meaning : Probability

Probability is the study of uncertainty, chance, or likelihood of events. It is a branch of Mathematics that helps in measuring how likely an event is to occur. The value of probability always lies between 0 and 1.

- **Probability = 1 (Certain Event)** : Probability 1 means the event is certain.
Example: The Sun rises in the east. This is always true. So the probability is 1.
- **Probability = 0 (Impossible Event)** : Probability 0 means the event is impossible.
Example: Getting a 7 when a standard dice is thrown. Since a dice has only numbers 1 to 6, getting a 7 is impossible. So the probability is 0.
- **Probability between 0 and 1 (Uncertain Event)** : Every probability value between 0 and 1 shows the degree of chance of an event happening.
Example: Tossing a coin and getting a head. There are two possible outcomes: head or tail. The probability of getting a head is $1/2 = 0.5$, which lies between 0 and 1.
 - A probability of 0.8 means the event has an 80% chance of happening.
 - A probability of 0.2 means the event has only a 20% chance of happening.

The importance of probability is seen in many areas. It gives a numerical measure of uncertainty and helps in making decisions when the outcome is not sure. It is widely used in fields like finance, insurance, medicine, and even sports.

The history of probability goes back to the 18th century. It developed mainly through the study of games of chance such as rolling a dice, drawing a card, or tossing a coin. Questions like "What are the chances of getting a six on a dice?" or "What is the chance of getting a heart in a card game?" led to the development of the theory.

A simple example can be taken from a coin toss. When a coin is tossed, there are two possible outcomes: head or tail. The number of total outcomes is two, and the favorable outcome for head is one. So, the probability of getting a head is $1/2$. Similarly, the probability of getting a tail is also $1/2$. This shows that probability helps to calculate the likelihood of uncertain events in a clear and logical way.

Probability is therefore a fundamental concept that makes uncertain situations measurable. It allows us to understand chance, take calculated risks, and apply the idea to real-life situations ranging from simple games to complex decision-making.

Examples Probability

1. **Coin Toss** : When tossing a fair coin, the probability of getting a head is $1/2$. This is because there are two equally likely outcomes - heads or tails.
2. **Rolling a Die** : When rolling a fair six-sided die, the probability of getting a 3 is $1/6$. This is because there are six equally likely outcomes - numbers 1 through 6.
3. **Drawing a Card** : If we have a standard deck of 52 playing cards, the probability of drawing a heart from the deck is $1/4$, as there are 13 hearts out of 52 cards.
4. **Weather Forecast** : The probability of rain tomorrow might be 0.3, indicating a 30% chance of rain based on historical data and meteorological forecasts.
5. **Product Defects** : In a manufacturing process, the probability of a product being defective might be 0.05, indicating a 5% chance of a defect occurring.

6.2 Experiment and Sample Space



Meaning : Experiment

In probability, an experiment is an action or process that is carried out to observe possible results. The outcome of an experiment may not always be known in advance, but the set of all possible outcomes can be listed.

Examples :

- Tossing a coin is an experiment. The possible outcomes are Head (H) or Tail (T).
- Throwing a die is another experiment. The possible outcomes are numbers from 1 to 6.

Thus, an experiment in probability means performing an action and noting the result.



Meaning : Random Experiment

A random experiment is one in which:

- The process can be repeated under the same conditions.
- The result is not certain and cannot be predicted before it happens.
- The set of all possible outcomes is known.

Examples of Random Experiments:

- **Tossing a Coin:** It may give Head or Tail, but which one will occur is not known before tossing.
- **Throwing a Die:** The numbers 1 to 6 are possible, but the exact number is uncertain.
- **Drawing a Card from a Pack:** The total set of 52 cards is known, but which card will appear is unknown.



Meaning : Trial

A trial is a single performance of an experiment. Each time the experiment is performed once, it is called a trial.

Examples:

- Tossing a coin once is a trial.
- Throwing a die once is a trial.
- Picking one card from a pack is a trial.

If the same process is repeated, we get multiple trials. For example, tossing a coin 10 times means 10 trials of the same experiment.



Meaning : Outcome

An outcome is the result of a trial. Each trial gives one outcome.

Examples:

- Tossing a coin may give Head as an outcome.
- Throwing a die may give 4 as an outcome.
- Drawing a card may give a King of Hearts as an outcome.



Meaning : Sample Space

The sample space is the collection of all possible outcomes of an experiment. It is denoted by S . Each element in the sample space is called a **sample point**. A sample space with finite number of outcomes is a **finite sample space**. A sample space with infinite number of outcomes is an **infinite sample space**.

Examples:

- Tossing a coin: $S = \{H, T\}$. This is a finite sample space.
- Throwing a die: $S = \{1, 2, 3, 4, 5, 6\}$. This is a finite sample space.
- Tossing two coins: $S = \{HH, HT, TH, TT\}$. This is a finite sample space.
- Throwing two dice: $S = \{(1,1), (1,2), \dots, (6,6)\}$. This is a finite sample space.
- Tossing a coin until a head appears : The possible outcomes are based on the number of tosses needed to get the first head. $S = \{1, 2, 3, 4, 5, \dots\}$. Here, 1 means head appears on the first toss, 2 means head appears on the second toss, and so on. Since there is no limit, this is an infinite sample space.
- Counting the number of goals scored in a football match : The possible outcomes could be 0, 1, 2, 3, 4, and so on. $S = \{0, 1, 2, 3, 4, \dots\}$. There is no upper limit on how many goals can be scored, so this forms an infinite sample space.

Problems on Experiment and Sample Space

Illustration 1

A coin is tossed three times. Write its sample space.

Solution:

We denote the head by H and tail by T.

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Illustration 2

A die is thrown twice. Find out the sample space.

Solution:

$$\begin{aligned} \text{Sample space} = & \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ & (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ & (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ & (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ & (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ & (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\} \end{aligned}$$

Illustration 3

From a leap year selected at random, what would be the sample space for the two odd days.

Solution:

Since the leap year has 366 days which means 52 weeks and two extra days, which are known as odd days.

$$\begin{aligned} \text{Sample space} = & \{(\text{Sunday, Monday}), (\text{Monday, Tuesday}), (\text{Tuesday, Wednesday}), \\ & (\text{Wednesday, Thursday}), (\text{Thursday, Friday}), (\text{Friday, Saturday}), \\ & (\text{Saturday, Sunday})\} \end{aligned}$$

6.2 Events

An event is a subset of the sample space S , denoted by A, B, C , etc. Since events are subsets of S , concepts of algebra of sets such as union, intersection, and complements are applicable to events in a random experiment.



Definition : Event

An event is a subset of the sample space that consists of one or more outcomes of a random experiment.

Example : If a die is thrown, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$

- Event $A =$ "getting an even number" $= \{2, 4, 6\}$
- Event $B =$ "getting a number less than 4" $= \{1, 2, 3\}$
- Event $C =$ "getting a 5" $= \{5\}$

Here A, B , and C are different events formed from the sample space. So, an event means a collection of outcomes from the sample space, depending on the condition described.

Examples Event

1. **Throwing a Die** : Let's say we define the event A as the set of outcomes where the die roll results in an even number. Then, $A = \{2, 4, 6\}$. This is a compound event since it consists of multiple outcomes. If we roll the die and get a 2, 4, or 6, then event A occurs.
2. **Tossing Two Coins** : Let's say we define the event B as the set of outcomes where at least one of the coins lands on heads. Then, $B = \{HH, HT, TH\}$. This is a compound event since it consists of multiple outcomes. If we toss the two coins and get either HH, HT, or TH, then event B occurs.
3. **Drawing Cards** : Let's say we define the event C as the set of outcomes where we draw a red card from a standard deck of 52 playing cards. Then, $C = \{\text{hearts, diamonds}\}$. This is a compound event since it consists of multiple outcomes. If we draw a heart or a diamond from the deck, then event C occurs.

6.3.1 Types of Events

Events are subsets of the sample space. Based on their nature, they can be classified into the following types:

1. Null Event (Impossible Event)



Definition : Null Event

A null event is an event that contains no outcomes at all. Such an event is impossible to occur. It is denoted by the symbol ϕ or sometimes by $\{\}$.

Example: Null Event

When a die is thrown, the possible outcomes are $\{1, 2, 3, 4, 5, 6\}$. The event "getting a 7" is a null event, because 7 is not in the sample space. Hence, the probability of this event is always 0.

2. Sure Event (Certain Event)



Definition : Sure Event

A sure event is an event that contains all the outcomes of the sample space. This means the event is bound to happen and cannot be avoided.

Example: Sure Event

When a die is thrown, the event "getting a number between 1 and 6" is a sure event. Since all possible outcomes are included in the event, its probability is always 1.

3. Simple or Elementary Event



Definition : Simple or Elementary Event

A simple event (or elementary event) is one that consists of only a single outcome from the sample space. These are the smallest possible events in probability.

Example: Simple or Elementary Event

When a die is thrown, the event "getting a 4" is a simple event because it includes only one outcome $\{4\}$. Similarly, when a coin is tossed, the event "getting a head" is a simple event.

4. Compound Event



Definition : Compound Event

A compound event is one that consists of more than one outcome of the sample space. It is the opposite of a simple event.

Example: Compound Event

- When a die is thrown, the event A "getting an even number" = $\{2, 4, 6\}$ is a compound event since it contains three outcomes.
- In tossing two coins, the event A "getting at least one head" = $\{HH, HT, TH\}$ is a compound event since it contains three outcomes.

5. Sub-event



Definition : Sub-event

An event B is said to be a sub-event of another event A if every outcome of B is also an outcome of A. In other words, B is contained inside A.

Example: Sub-event

Consider the sample space $S = \{1, 2, 3, 4, 5, 6\}$ when a die is thrown.

- Let Event A $\rightarrow$ "getting an even number." = $\{2, 4, 6\}$
- Let Event B $\rightarrow$ "getting an even number less than 5." = $\{2, 4\}$

Here, B is a sub-event of A because all outcomes of B are also included in A.

6. Mutually Exclusive Events (Disjoint Events)



Definition : Mutually Exclusive Events

Two events are called mutually exclusive if they cannot occur at the same time. If one event happens, the other is automatically ruled out.

Example: Mutually Exclusive Events

1. Drawing a card from a standard deck:

- Event A $\rightarrow$ {drawing a heart} = 13 cards
- Event B $\rightarrow$ {drawing a spade} = 13 cards

These two events are mutually exclusive because a single card cannot be both a heart and a spade at the same time.

2. Throwing two dice:

- Event A = {sum of numbers is 7}
- Event B = {sum of numbers is 11}

These two events are mutually exclusive because the sum of the two dice cannot be 7 and 11 at the same time.

3. Tossing a coin twice:

- Event A $\rightarrow$ "both heads" = {HH}
- Event B $\rightarrow$ "both tails" = {TT}
- Event C $\rightarrow$ "one head and one tail" = {HT, TH} $\rightarrow$ "one head and one tail"

These events are mutually exclusive because if one occurs, the others cannot occur in the same trial.

7. Exhaustive Events



Definition : Exhaustive Events

Events are exhaustive if they together cover the entire sample space. In other words, all possible outcomes are included in the collection of these events.

Example: Exhaustive Events

1. Throwing a fair die : Sample space: $S = \{1, 2, 3, 4, 5, 6\}$

- Event A $\rightarrow$ "getting an even number" = {2, 4, 6}
- Event B $\rightarrow$ "getting an odd number" = {1, 3, 5}

Both A and B are exhaustive events because together they cover the entire sample space {1, 2, 3, 4, 5, 6}.

2. Throwing two coins: Sample Space $S = \{HH, HT, TH, TT\}$

- Event A $\rightarrow$ "getting a Head on the first coin" = {HH, HT}
- Event B $\rightarrow$ "getting a Tail on the first coin" = {TH, TT}

Together, $A \cup B = \{HH, HT, TH, TT\}$, which is the whole sample space. Hence, A and B are exhaustive.

3. Drawing a card from a well-shuffled deck of 52 cards:

- Event A = {drawing a red card} (26 red cards)
- Event B = {drawing a black card} (26 black cards)

Together, $A \cup B$ covers all 52 cards in the deck. Hence, A and B are exhaustive.

8. Equally Likely Events



Definition : Equally Likely Events

Events are equally likely if each of them has the same chance of occurring. In other words equally likely events are such that none of them has greater chance of occurrence than the others. This condition usually applies to fair games of chance such as coins, dice, and cards.

Example: Equally Likely Events

1. Throwing a fair die : Sample space: $S = \{1, 2, 3, 4, 5, 6\}$

- Event A = {getting 3}
- Event B = {getting 5}

Here, $P(A) = 1/6$ and $P(B) = 1/6$. Both have the same chance, so A and B are equally likely.

2. Tossing a fair coin : Sample space: $S = \{H, T\}$

- Event $A = \{\text{getting Head}\}$
- Event $B = \{\text{getting Tail}\}$

Here, $P(A) = 1/2$ and $P(B) = 1/2$. Both are equally likely.

3. Drawing a card from a well-shuffled pack : A standard deck has 52 cards (26 red, 26 black)

- Event $A = \{\text{drawing a red card}\}$
- Event $B = \{\text{drawing a black card}\}$

Here, $P(A) = 26/52 = 1/2$ and $P(B) = 26/52 = 1/2$. Both have the same chance, so these events are equally likely.

9. Independent Events



Definition : Independent Events

Two events are said to be independent if the occurrence of one does not influence or change the probability of the other. Each event happens on its own.

Example: Independent Events

1. Tossing two coins:

- Event $A = \{\text{first coin shows Head}\}$
- Event $B = \{\text{second coin shows Tail}\}$

These two events are independent because the result of the first coin does not affect the result of the second coin.

2. Throwing a die and tossing a coin:

- Event $A = \{\text{die shows an even number}\}$
- Event $B = \{\text{coin shows Head}\}$

These two events are independent because the outcome of the die has no effect on whether the coin shows Head or Tail.

3. Drawing marbles with replacement: A bag has 3 red and 2 blue marbles.

- Event $A = \{\text{drawing a red marble in the first draw}\}$
- Event $B = \{\text{drawing a red marble in the second draw}\}$

If one marble is taken out and then put back into the bag before the next draw, the second draw does not depend on the first. For example, if a red marble is taken out first and replaced, the bag still has 3 red and 2 blue marbles. So, the probability of getting a red marble on the second draw is again $3/5$. Since the first draw has no impact on the second draw, events A and B are independent events.

10. Dependent Events



Definition : Dependent Events

Two events are said to be dependent if the occurrence of one event affects or changes the probability of the other. In such cases, the events are not independent because the result of the first trial influences the result of the next.

Example: Dependent Events
1 Drawing two Cards without Replacement:

- Event A = {drawing an Ace on the first draw}
- Event B = {drawing an Ace on the second draw}

If a card is drawn from a pack of 52 cards and not replaced, the total number of cards left in the pack becomes 51. Out of the 52 cards, there are 4 Aces, so the probability of drawing an Ace in the first draw is $4/52$ or $1/13$. If an Ace is drawn first and not replaced, then only 3 Aces remain in the 51 cards. Now the probability of drawing an Ace in the second draw becomes $3/51$ or $1/17$. Since the probability of the second draw depends on the result of the first draw, the two events are called dependent events.

2. Picking Marbles without Replacement:

A bag has 3 red and 2 blue marbles (total 5).

- Event A = {drawing a red marble in the first draw}
- Event B = {drawing a red marble in the second draw}

If a bag contains 3 red marbles and 2 blue marbles, the probability of drawing a red marble in the first draw is $3/5$. If a red marble is drawn first and not replaced, the bag then has 2 red and 2 blue marbles, making a total of 4. Now, the probability of drawing a red marble in the second draw becomes $2/4$ or $1/2$. Since the probability of the second draw is not the same as the first draw and depends on what happened earlier, events A and B are dependent events.

6.3.2 Operations with Events

Since events are subsets of the sample space, the operations of set theory such as union, intersection, complement, and difference can be applied to them. These operations help us combine or separate events to study their relationships.

1. Complement of an Event:

Definition : Complement of an Event

The complement of event A is the event that A does not occur. It includes all outcomes in the sample space except those in A. The complement of A is denoted by A' or $\bar{A}$.

Example: Complement of an Event

- Sample space $S = \{1, 2, 3, 4, 5, 6\}$
- Event $A \rightarrow$ "getting an even number" = $\{2, 4, 6\}$
- $A' = \{1, 3, 5\}$
- The complement includes all outcomes from S that are not in A.

2. Union of Events:

Definition : Union of Events

Union of two or more events refers to the event that occurs if either of any events occurs. The union of two events A and B is the event that either A occurs or B occurs or both occur. Union of Events A and B is denoted by $A \cup B$.



Example: Union of Event

- (i) Let's consider an example for an Union of Events, let two events be $E_1 = \{2, 3, 4, 5, 6\}$ and $E_2 = \{3, 4, 7, 8, 10\}$. Also, assume that the Union of both events is represented by E i.e., $E = E_1 \cup E_2$. Thus, $E = E_1 \cup E_2 = \{2, 3, 4, 5, 6, 7, 8, 10\}$
- (ii) While tossing two coins simultaneously, let $A = \{HH\}$ and $B = \{TT\}$ be the two events. Then clearly their union is $A \cup B = \{HH, TT\}$. Here, A is the event of occurrence of two heads and B is the event of occurrence of two tails. Their union $A \cup B$ is the event of occurrence of two heads or two tails.
- (iii) While throwing a die, let $P = \{1, 3, 5\}$, $Q = \{2, 6\}$ and $R = \{3, 5, 6\}$ be three events. Then, their union is $P \cup Q \cup R = \{1, 2, 3, 5, 6\}$

3. Intersection of Events:**Definition : Intersection of Events**

The intersection of two or more events refers to the event that occurs if all of the events occur simultaneously i.e., let's say we have two events A and B then the intersection of A and B represents the event that occurs if both A and B occur. The intersection of Events A and B is denoted by $A \cap B$.

**Example: Intersection of Events**

- (i) Let's consider an example for an Intersection of Events, let two events be $E_1 = \{2, 3, 4, 5, 6\}$ and $E_2 = \{3, 4, 7, 8, 10\}$. Also, assume that the Intersection of both events is represented by E i.e., $E = E_1 \cap E_2$. Thus, $E = E_1 \cap E_2 = \{3, 4\}$
- (ii) While tossing two coins, let $A = \{HH, TT\}$ and $B = \{HH, HT, TH\}$ be two events. Then, their intersection is $A \cap B = \{HH\}$.
- (iii) While throwing a die, let $A = \{1, 3, 5\}$, $B = \{3, 5\}$ and $C = \{2, 5, 6\}$ be three events. Then, their intersection is $A \cap B \cap C = \{5\}$.

4. Difference of Events**Definition : Difference of Events**

The difference of two events A and B refers to the event that occurs if A happens but B does not happen. In other words, it includes all outcomes that are in A but not in B . The difference of Events A and B is denoted by $A - B$.

Example: Difference of Events

- (i) Let two events be $E_1 = \{2, 3, 4, 5, 6\}$ and $E_2 = \{3, 4, 7, 8, 10\}$. The difference of both events is represented by E , i.e., $E = E_1 - E_2$. Thus, $E = \{2, 5, 6\}$.

- (ii) While tossing two coins, let $A = \{HH, HT\}$ and $B = \{HT, TT\}$ be two events. Then, their difference is $A - B = \{HH\}$.
- (iii) While throwing a die, let $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6\}$, and $C = \{1, 2, 5\}$. Then, their differences are:
 $A - B = \{1, 3\}$
 $B - C = \{4, 6\}$

6.4 Probability of an Event

The probability of an event is a numerical measure of how likely it is for that event to occur. Since an event is a subset of the sample space, its probability is calculated by considering how many outcomes in the sample space belong to that event. The value of probability always lies between 0 and 1, where 0 means the event is impossible and 1 means the event is certain. The probability of any event A is given by:

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

This value always lies between 0 and 1. If $P(A) = 0$, the event is impossible. If $P(A) = 1$, the event is certain. If $0 < P(A) < 1$, the event is uncertain but possible.

Example: Probability of an Event

When a fair die is thrown, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$. If Event $A = \{2, 4, 6\} \rightarrow$ "getting an even number," then

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{3}{6} = \frac{1}{2} = 0.5$$

So, the probability of getting an even number when a fair die is thrown is 0.5 or 50%, which means there is an equal chance of getting an even number or an odd number.

6.5 Approaches to Probability

Probability can be studied in different ways. The three main approaches are the **Classical approach**, the **Empirical approach**, and the **Axiomatic approach**. Each approach gives a clear rule to assign numbers to events and to reason about uncertainty.

I. The Classical Approach (Mathematical or a Priori)



Definition: The Classical Approach (Mathematical or a Priori)

The classical approach works when all outcomes are equally likely, mutually exclusive, and exhaustive. The probability of an event A in a classical (mathematical or a priori) approach is given by the ratio of the number of favorable outcomes to the total number of equally likely, mutually exclusive, and exhaustive outcomes. The probability of an event A is,

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

where, 'm' is the number of favourable outcomes and 'n' is the total number of equally likely outcomes.

Note: This definition is applicable only when the outcomes are equally likely, mutually exclusive and exhaustive.

Example: Classical Approach

- Consider the probability of rolling a 6 on a fair six-sided die. In this case, there is only one favourable outcome (rolling a 6) out of six possible outcomes (rolling a 1, 2, 3, 4, 5, or 6). Using the classical approach, the probability of rolling a 6 can be calculated as the number of favorable outcomes (1) divided by the total number of possible outcomes (6), resulting in a probability of $1/6$.
 - Similarly, when tossing a fair coin, there are two equally likely outcomes: heads or tails. Using the classical approach, the probability of getting heads can be calculated as the number of favorable outcomes (1) divided by the total number of possible outcomes (2), resulting in a probability of $1/2$.
- A bag has 5 white and 5 red marbles. Probability of drawing a white marble is $5/10$ is 0.5.



Key Points

1. $P(A)$ is a value between 0 and 1. That is, $0 \leq P(A) \leq 1$

- $P(A) = 0$: If the probability of an event is 0, it means the event will never happen. Such an event is called an impossible event.

Example : When a die is rolled, the probability of getting a number greater than 6 is 0 (impossible).

- $P(A) = 1$: If the probability of an event is 1, it means the event will definitely happen. Such an event is called a sure event.

Example : The probability of getting a number less than or equal to 6 is 1 (certain).

- $P(A)$ = between 0 and 1: Any value between 0 and 1 shows the degree of chance of the event occurring.

Example : The probability of getting an even number is $3/6 = 0.5$, which lies between 0 and 1.

2. $P(A') = 1 - P(A)$, That is $P(A) = 1 - P(A')$.

The complement of an event A (written as A') means that event A does not occur. The probability of the complement event is equal to 1 minus the probability of the event.

Example:

- Suppose a coin is tossed. Let Event A = {getting a Head}. Then, $P(A) = 1/2$.
- The complement event $A' = \{\text{getting a Tail}\}$. So, $P(A') = 1 - 1/2 = 1/2$.
- This shows that the chances of not getting a Head are equal to the chances of getting a Tail.

3. Total Exhaustive Cases in Probability

In probability, every outcome of a random experiment is either favourable to a given event or not favourable (failure).

- Let m = number of favourable cases for an event A.
- Let n = number of cases where event A does not occur.

Then the total exhaustive cases (all possible outcomes of the experiment) = $m + n$. This means the complete sample space is made up of both the favourable and unfavourable outcomes together.

$$P(A) = \frac{m}{m+n} \text{ and } P(A') = \frac{n}{m+n}$$

So, $P(A) + P(A') = 1$

Example: A bag has 3 red marbles and 2 blue marbles.

- Favourable cases for drawing red (m) = 3.
- Unfavourable cases (not red) (n) = 2.
- Total exhaustive cases = $m + n = 3 + 2 = 5$.
- $P(\text{red}) = 3/5$, $p(\text{not red}) = 2/5$
- So, $P(\text{red}) + P(\text{not red}) = 3/5 + 2/5 = 1$

The classical approach works very well for simple experiments such as tossing a fair coin, rolling a fair die, or drawing a card from a well-shuffled pack. These situations have a finite number of outcomes, and each outcome is equally likely, mutually exclusive, and exhaustive. However, in many real-life situations, the outcomes are not equally likely or may be too complex to handle with this method. In such cases, alternative approaches such as the empirical approach or the axiomatic approach may be more appropriate for calculating probabilities.

Problems on Classical Approach

Illustration 1

Find the probability of drawing one white marble from a bag containing 5 white and 5 red marbles.

Solution

Here the total number of marbles is $5W + 5R = 10$, and there are five white marbles. Hence

Probability of drawing a white marble

$$= \frac{\text{Favourable number of cases}}{\text{Total number of cases}} = \frac{5}{5+5} = \frac{5}{10} = \frac{1}{2}$$

Therefore, the probability of drawing one white marble from the bag is 0.5 or 50%.

Here, there is no need for combinations because we are not choosing a group. We are selecting just 1 marble, so favourable cases = number of white marbles.

Illustration 2

Find the probability of drawing 4 black marbles from a bag containing 6 white and 6 black marble.

Solution

When only one marble is drawn, probability can be found directly as favourable outcomes divided by total outcomes. But when four marbles are drawn at the same time, the outcome is not a single marble but a group of four marbles. Hence, we must count how many such groups are possible in total $C(12,4)$ and how many of those groups contain all black marbles $C(6,4)$. The probability is then $C(6,4) / C(12,4)$.

- Favourable case for drawing 4 black marbles from 6 black marbles is 6C_4 i.e., 6C_2 ($\because {}^nC_r = {}^nC_{n-r}$)
- Total number of ways in which 4 marbles can be drawn from a total of 12 marbles is

$${}^{12}C_4 = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1}$$

- The probability of getting four black marbles = $\frac{{}^6C_4}{{}^{12}C_4}$ or $\frac{{}^6C_2}{{}^{12}C_4}$
- $$\left(\frac{6 \cdot 5}{1 \cdot 2}\right) / \left(\frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1}\right) = \frac{15}{495}$$

Therefore, the probability of drawing 4 black marbles from the bag is approximately 0.0303 or 3.03%.

Illustration 3

Find the probability of drawing a red ball from a bag containing 4 white and 6 black balls.

Solution:

$$n(S) = 4 + 6$$

$$n(S) = 10$$

Let A denote the event of drawing a red ball.

$$n(A) = 0$$

Because the event of drawing a red ball is an impossible event.

$$P(A) = \frac{n(A)}{n(S)} = \frac{0}{10} = 0$$

Illustration 4

What is the probability that a number selected from the first 25 natural numbers is a prime number.

Solution:

$$S = \{1, 2, 3, 4, 5, \dots, 25\} \quad n(S) = 25$$

Let A denote the event of getting a prime number

$$A = \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$$

$$n(A) = 9$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{9}{25} = 0.36$$

Illustration 5

A coin is tossed twice. Find the probability of getting exactly one head.

Solution:

Sample Space $S = \{HH, TT, TH, HT\}$

Let A denote the event of getting exactly one head.

$$A = \{TH, HT\}$$

$$n(A) = 2$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = \frac{1}{2}$$

II. The Empirical Approach

The classical approach is totally abstract in which the probabilities are obtained without actually conducting experiments. In the empirical approach, the probabilities are found by actually conducting the experiment. The empirical approach to probability is based on observed data or experimentation. It involves calculating the probability of an event based on the frequency of its occurrence in a series of trials or experiments. The empirical approach is useful for situations where data is available and can be used to estimate probabilities.

Example: Empirical Approach

Suppose a bus company wants to know the probability that a bus on a certain route arrives late. Using the classical approach is not possible because the outcomes (late or on-time) are not equally likely and depend on many factors like traffic, weather, and road conditions. So, the company actually collects data. They observe the buses for 100 days. Out of these, the bus arrives late on 28 days and on time on 72 days.

- Total trials (n) = 100
- Number of times bus was late (frequency f) = 28

$$\text{Therefore, } P(\text{Bus late}) = f/n = 28/100 = 0.28$$

This means, based on real data, the probability that the bus will be late is about 28%.

Example: Empirical Approach

In the empirical approach, the probability of getting a Head when tossing a coin is estimated by actually performing the experiment many times. For example, if a coin is tossed 100 times and Heads appear 47 times, then the probability of getting a Head is taken as $47/100 = 0.47$. If the number of tosses is increased to 1000 or more, the result will get closer to the theoretical value of 0.5, but it is always based on the observed data from repeated trials.

Definition Empirical (Statistical or Posterior) Approach

Let a random experiment be repeated 'n' times specifically under identical condition. Let f of these repetitions result in the occurrence of an event A . In simple words, the probability of an event is the relative frequency of its occurrence in a large number of trials, assuming this relative frequency approaches a stable, finite value.

That is,
$$P(A) = \lim_{n \rightarrow \infty} \frac{f}{n}$$
 Here, it is assumed that a unique and finite limit exists.

Note:

In practice, it is not possible to repeat an experiment infinite number of times. Therefore, the probability is obtained through a sufficiently high number of repetitions.

Definition : Relative Frequency of an Event

Let S be a sample space associated with an experiment Let A be an event. Suppose an experiment is repeated N times and the event A occurs f times. Then $\frac{f}{N}$ is called the relative frequency of the event A . Clearly $0 \leq \frac{f}{N} \leq 1$

Example:

Consider the sample space S of throwing two dice, simultaneously:

$$S = \{(x, y) : x, y \in N, 1 \leq x \leq 6, 1 \leq y \leq 6\}$$

Consider an event $A = \{(1, 6), (2, 5), (3, 4), (4, 3), (6, 1)\}$ which consists of the sample points (a, b) for which $a + b = 7$.

Let the die be thrown 400 times. $\Rightarrow N = 400$.

Suppose that the number of times the sum seven occurs is 80 then $f = 80$ and the relative frequency with which the outcome was in A is $\frac{f}{N} = \frac{80}{400} = 0.2$.



Definition : Probability of an Event $P(A)$ based on the Relative Frequency of the Event A

The probability $P(A)$ of an event A is the value around which the relative frequency ' f ' of that event becomes stable after a large number of trials.

Problems on Empirical Approach

Illustration 1

- (i) Out of 600 babies born in a hospital in a year, 260 were male. find the probability that a new-born baby is male.
- (ii) Team A has played 100 cricket matches with team B. Among them, 20 matches were won by team A. Find the probability that 101 the match that team A plays with team B will be won by team A.

Solution

- (i) Here, $n = 600$ births were observed and among them $f = 260$ have resulted in birth to male children. Therefore probability that a new-born baby is male is -

$$P[\text{male baby}] = \frac{f}{n} = \frac{260}{600} = 0.433$$

- (ii) Here, out of $n = 100$ matches played, $f = 20$ were won by team A. Therefore probability that team A wins the 101 match is -

$$P[A \text{ wins}] = \frac{f}{n} = \frac{20}{100} = 0.20$$

Illustration 2

The following is an extract from a life table.

Age (Years)	0	10	20	40	60	80
No. of Survivors	2,00,000	1,80,000	1,60,000	1,20,000	80,000	40,000

Find the probability that

- (i) a person aged 20 lives till he is 80
- (ii) a new-born baby survives up to the age of 20 years
- (iii) a person aged 60 lives for 20 more years.

Solution

- (i) In the given table, out of the 1,60,000 persons aged 20 years, 40,000 survived till the age 80. Therefore, Probability of a person aged 20 living till he is 80 is

$$P[\text{person aged 20 lives till he is 80}] = \frac{40,000}{1,60,000} = 0.25$$

- (ii) Out of 2,00,000 new-born babies, 1,60,000 survived till the age 20. Therefore, probability of a new-born baby surviving till the 20 is

$$P[\text{new-born baby survives till 20}] = \frac{1,60,000}{2,00,000} = 0.8$$

- (iii) Out of 80,000 persons aged 60 years, 40,000 lived till the age 80. Therefore, probability of a person aged 60 living for 20 more years is

$$P[\text{Person aged 60 lives for 20 more years}] = \frac{40,000}{80,000} = 0.5$$

III. The Axiomatic Approach

The axiomatic approach to probability is based on a set of mathematical axioms or rules that define the properties of probability. This approach is often used in more complex scenarios where the outcomes are not equally likely or where empirical data is not available to estimate probabilities.

The axiomatic approach starts with a set of axioms or rules that define the properties of probability. These axioms include the probability of an event being between 0 and 1, the probability of the entire sample space being 1, and the probability of the union of two mutually exclusive events being the sum of their individual probabilities. Using these axioms, probabilities can be calculated for more complex scenarios where the outcomes are not equally likely.



Definition : Axiomatic Approach

The axiomatic approach defines probability as a function P on events that follows simple rules. These rules hold for any experiment and for any sample space.

- (i) **Axiom 1 ($P(A) \geq 0$)**: The probability of an event is between 0 and 1.
- (ii) **Axiom 2 ($P(S) = 1$)**: The probability of the entire sample space is 1.
- (iii) **Axiom 3 : Probability of two or more mutually exclusive events being the sum of their individual probabilities.**

If $A, B, C, \dots$ is a sequence of mutually exclusive events in S , the $P(A \cup B \cup C, \dots) = P(A) + P(B) + P(C), \dots$

The axiomatic approach to probability is a powerful tool for calculating probabilities in more complex scenarios where the outcomes are not equally likely or where empirical data is not available to estimate probabilities. It provides a rigorous and systematic framework for understanding and analyzing random phenomena and is widely used in various fields such as science, engineering, finance, and social sciences.

Example: Axiomatic Approach - Probability of Drawing a Red Card

In a standard deck of 52 playing cards:

- Red cards = 26 (13 hearts + 13 diamonds)
- Black cards = 26 (13 spades + 13 clubs)
- Sample space $S = 52$ cards

Let Event $A =$ "drawing a red card."

Using the Axioms:

1. Axiom 1 ($P(A) \geq 0$):

$$P(A) = \frac{26}{52} = \frac{1}{2} = 0.5, \text{ which lies between 0 and 1.}$$

2. Axiom 2 ($P(S) = 1$):

$$\text{Probability of drawing any card from the full deck} = P(S) = \frac{52}{52} = 1$$

3. Axiom 3 (Additivity for disjoint events):

Event $A =$ "drawing a red card"

Event $B =$ "drawing a black card"

These two events are mutually exclusive, so:

$$P(A \cup B) = P(A) + P(B) = 0.5 + 0.5 = 1$$

$\therefore$ The axiomatic approach, the probability of drawing a red card from a standard deck is 0.5 or 50%.

6.6 Basic Probability Rules**1. Rule of Non-Negativity**

The probability of any event can never be negative. Probability values start from 0 (impossible) and go upward but cannot be less than 0.

Example

When a die is rolled, Event $A =$ {getting a 3}.

$$P(A) = \frac{1}{6} = 0.166\dots$$

This is a positive number. We cannot say the probability is -0.2 or any negative value, because that has no meaning in real life.

2. Rule of Maximum Value

The probability of an event can never exceed 1. The maximum value is 1, which represents complete certainty.

Example

When tossing a coin, Event $A =$ {Head}.

$$P(A) = \frac{1}{2} = 0.5$$

This is less than 1. If we said $P(A) = 1.2$, it would mean the event is more than certain, which is not possible.

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3. Rule of Certainty (Sure Event)

The probability of the whole sample space (the sure event) is always 1, because when we conduct an experiment, something from the sample space must definitely happen.

Example

When rolling a die, sample space $S = \{1, 2, 3, 4, 5, 6\}$.

Event $A =$ "getting a number from 1 to 6."

$$P(A) = \frac{6}{6} = 1$$

This shows that it is certain one of these outcomes will occur.

4. Rule of Impossible Event

The probability of an impossible event (something that can never happen) is 0.

Example

When rolling a die, Event $A = \{\text{getting a 7}\}$.

Since 7 is not in the sample space,

$$P(A) = \frac{0}{6} = 0$$

This shows the event is impossible.

5. Complement Rule

The probability that an event does not occur is equal to 1 minus the probability that it does occur.

$$P(A') = 1 - P(A)$$

Example

When tossing a coin, Event $A = \{\text{Head}\}$.

$$P(A) = \frac{1}{2}, P(A') = P(\text{not Head, i.e., Tail}) = 1 - \frac{1}{2} = \frac{1}{2}$$

This means the chance of getting a Tail is exactly what remains after considering the chance of a Head.

6. Addition Rule (Mutually Exclusive Events)

If two events cannot happen at the same time, then the probability of either happening is the sum of their probabilities.

$$P(A \cup B) = P(A) + P(B)$$

Example

When rolling a die:

- Event $A = \{2\}$
- Event $B = \{5\}$

These are mutually exclusive because you cannot roll both 2 and 5 in one trial.

$$P(A \cup B) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

So, the chance of getting either a 2 or a 5 is $\frac{1}{3}$.

7. Addition Rule (General Case)

If two events are not mutually exclusive, we must subtract the overlap (intersection).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example

When rolling a die:

- Event A = {even numbers} = {2, 4, 6}, $P(A) = 3/6$
- Event B = {numbers greater than 3} = {4, 5, 6}, $P(B) = 3/6$.
- Overlap $A \cap B = \{4, 6\}$, $P(A \cap B) = 2/6$.

Now,

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3}$$

So, the probability of getting either an even number or a number greater than 3 is $2/3$.

8. Multiplication Rule (Independent Events)

If two events are independent, the probability that both happen is the product of the probabilities.

$$P(A \cap B) = P(A) \times P(B)$$

Example

When tossing two coins:

- Event A = {first coin is Head}, $P(A) = 1/2$.
- Event B = {second coin is Head}, $P(B) = 1/2$.

Since the tosses are independent,

$$P(A \cap B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

So, the probability of both coins showing Head is $1/4$.

9. Multiplication Rule (Dependent Events)

When events are dependent, the probability of both happening is the probability of the first multiplied by the conditional probability of the second (given the first has occurred).

$$P(AB) = P(A) \times P(B|A)$$

Example 1

A bag has 3 red and 2 blue marbles (total 5). Two marbles are drawn without replacement.

- Event A = {first marble is red}.

$$P(A) = \frac{3}{5}$$

- After drawing one red, 2 red and 2 blue marbles remain (total 4).
Event B = {second marble is red given first is red}.

$$P(B|A) = \frac{2}{4} = \frac{1}{2}$$

So,

$$P(A \cap B) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10} = 0.3$$

$\therefore$ The probability of drawing 2 red marbles in succession without replacement is 0.3.

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Example 2

A deck of 52 cards. Two cards are drawn without replacement.

- Event $A = \{\text{first card is an Ace}\}$,

$$P(A) = \frac{4}{52} = \frac{1}{13}$$

- After removing one Ace, 3 Aces remain and total cards = 51.

Event $B = \{\text{second card is an Ace given first is an Ace}\}$,

$$P(B|A) = \frac{3}{51} = \frac{1}{17}$$

So,

$$P(A \cap B) = \frac{1}{13} \times \frac{1}{17} = \frac{1}{221}$$

∴ The probability of drawing two Aces in succession without replacement is $1/221$.

6.7 Conditional Probability

In daily life, the chance of an event often depends on some condition or extra information. This is where conditional probability comes into play. It tells us the probability of an event happening given that another event has already occurred.

Examples:

- In cricket, the probability that a team wins may change if we know the team is batting first.
- In weather forecasting, the probability of traffic jams is higher given that it is raining.
- In health studies, the probability that a person has a disease changes if we know the test result is positive.
- In dice rolling, the probability of getting an even number changes if we are told that the number rolled is greater than 3.

In all these examples, the condition (batting first, raining, test positive, number > 3) affects the probability of the event.

Definition : Conditional Probability

If A and B are two events of a random experiment and $P(B) > 0$, then the conditional probability of event A given that B has occurred is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A|B)$ = Probability of A given B has happened
- $P(A \cap B)$ = Probability of both A and B happening
- $P(B)$ = Probability of B happening

Example 1 Die Example

A fair die is rolled. Find the probability of getting an even number given that the outcome is greater than 3.

Solution:

- Sample space $S = \{1, 2, 3, 4, 5, 6\}$
- Event $A = \{\text{even number}\} = \{2, 4, 6\}$
- Event $B = \{\text{number greater than 3}\} = \{4, 5, 6\}$

Now,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- $A \cap B = \{4, 6\}$, so $P(A \cap B) = 2/6$.
- $P(B) = 3/6$.

$$P(A|B) = \frac{2/6}{3/6} = \frac{2}{3}$$

$\therefore$ The probability is $2/3$.

Example 2 Marble Example

A bag contains 3 red and 2 blue marbles. A marble is drawn at random. Find the probability that the marble is blue given that it is not red.

Solution:

- Sample space = 5 marbles (3 red + 2 blue)
- Event $A = \{\text{blue marble}\} = 2$ outcomes
- Event $B = \{\text{not red marble}\} = 2$ outcomes (only blue marbles are not red)

Now,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- $P(B) = 2/5$
- $P(A \cap B) = 2/5$

$$P(A|B) = \frac{2/5}{2/5} = 1$$

$\therefore$ The probability is 1, because if a marble is not red, it must definitely be blue.

Problems on Conditional Probability**Illustration 1**

Three horses A, B and C are in a race. A is twice as likely to win as B and B is twice as likely to win as C. What are their respective probabilities of winning?

Solution:

Let the probability of winning of horse C = P . Then the probability of winning of B and will be $2P$ and $4P$ respectively. Now, only one horse can win the race (so the events are mutually exclusive and exhaustive). Since the events of winning of horse A, B and C are mutually exclusive and exhaustive, we have

$$P + 2P + 4P = 1 \Rightarrow P = \frac{1}{7}$$

$\therefore$ Respective probabilities of winning of A, B, C are $\frac{4}{7}$, $\frac{2}{7}$ and $\frac{1}{7}$.

Illustration

A, B and C are in a race. A is twice as likely to win as B and B is twice as likely to win as C.

Solution:

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Since there are three events, A, B, C

$$\therefore P(A) + P(B) + P(C) = 1$$

$\Rightarrow$

Illustration

From 20 tickets, 10 are marked with the number 1 to 10 and the rest are marked with the number 11 to 20.

Solution:

In first 20 numbers, 10 are marked with the number 1 to 10 and the rest are marked with the number 11 to 20.

The multiples of 5 are 5, 10, 15, 20.

Required probability is

Illustration

In a town, the probability that a person is a doctor is $\frac{1}{100}$, the probability that a person is a teacher is $\frac{1}{200}$, and the probability that a person is a lawyer is $\frac{1}{300}$.

Solution:

Total number of people in the town is 1000.

So, total ways = 1000.

Now, the number of people who are doctors, teachers, and lawyers are

- First person is a doctor
- Second person is a teacher
- Third person is a lawyer

Illustration 2

A, B and C are bidding for a contract. It is believed that A has exactly half the chance that B has, B in turn is $\frac{4}{5}$ as likely as C to win the contract. What is the probability for each to win?

Solution:

$P(A)$ = probability that A wins the contract

$P(B)$ = probability that B wins the contract

$P(C)$ = probability that C wins the contract

Let $P(C) = p$, then $P(B) = \frac{4}{5}p$, $P(A) = \frac{2}{5}p$

Since there are only three persons bidding for a contract, and one and only one of them must win, the events, A, B, C can be taken as mutually exclusive and exhaustive.

$$P(A) + P(B) + P(C) = 1 \Rightarrow p + \frac{4}{5}p + \frac{2}{5}p = 1$$

$$\Rightarrow \frac{11p}{5} = 1 \Rightarrow p = \frac{5}{11}$$

$$P(A) = \frac{2}{11}, P(B) = \frac{4}{11}, P(C) = \frac{5}{11}$$

Illustration 3

From 20 tickets numbered 1 to 20, one ticket is drawn at random. What is the probability that the number on it is a multiple of 3 or 7?

Solution:

In first 20 numbers, the multiples of 3 are 3, 6, 9, 12, 15, 18. Total numbers = 6

The multiples of 7 are 7, 14. Total numbers = 2

$$\text{Required Probability} = \frac{6+2}{20} = \frac{8}{20} = \frac{2}{5}$$

Illustration 4

In a town, there are four hotels. If three men check into hotels on the same day, what is the probability that each man chooses a different hotel?

Solution:

Total number of ways in which three persons can check into 4 hotels: Each person has 4 choices. So, total ways = $4 \times 4 \times 4 = 64$ ways

Now, the number of ways in which the three persons check into different hotels:

- First person can choose any of the 4 hotels.
- Second person can then choose from the remaining 3 hotels.

• Third person can choose from the remaining 2 hotels.
So, favourable ways = $4 \times 3 \times 2 = 24$.

$$\text{Required Probability} = \frac{24}{64} = \frac{3}{8}$$

The probability that each person checks into a different hotel is $3/8$.

Illustration 5

Three coins are tossed together. Find the probability that either only 2 heads or only 2 tails turns up.

Solution:

Sample space (S) = {HHH, TTT, THH, HTT, HTH, THT, HHT, TTH}

Let A denote the event of getting only 2 heads. $A = \{THH, HTH, HHT\}$ and $n(A) = 3$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{8}$$

Let B denote the event of getting only 2 tails. $B = \{HTT, THT, TTH\}$ and $n(B) = 3$

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{8}$$

Both the events are mutually exclusive.

$$P(A \cup B) = p(A) + p(B)$$

$$= \frac{3}{8} + \frac{3}{8} = \frac{6}{8} = \frac{3}{4}$$

Probability of getting only two heads or only two tails is $3/4$.

Illustration 6

Two dice are thrown together. What is the probability of getting a total 8 or a product 12.

Solution:

When two dice thrown together, then

Sample space S = { (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6),
(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),
(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6),
(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),
(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6),
(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6) }

We have, $n(S) = 36$

Let A denote the event of getting a total of 8.

$A = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$ and $n(A) = 5$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{36}$$

Let B denote the event of getting a product of 12.

$$B = \{(2, 6), (3, 4), (4, 3), (6, 2)\} \text{ and } n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{36}$$

Now, events A and B are not mutually exclusive, because some outcomes belong to both sets (for example (2,6) and (6,2)). Since they are not mutually exclusive, we need to find their intersection in order to avoid counting the common outcomes twice.

$$\text{Now } A \cap B = \{(2, 6), (6, 2)\} \quad n(A \cap B) = 2$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{2}{36}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{5}{36} + \frac{4}{36} - \frac{2}{36} = \frac{7}{36}$$

The probability of getting a total of 8 or a product of 12 is $\frac{7}{36}$.

Illustration 7

A number is selected at random from 40 to 80. Find the probability that it is divisible by 6 or 9.

Solution:

Sample space (S) = {40, 41, 80} and $n(S) = 41$

Let A denote the event of getting a number divisible by 6.

$$A = \{42, 48, 54, 60, 66, 72, 78\} \text{ and } n(A) = 7$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{7}{41}$$

Let B denote the event of getting a number divisible by 9.

$$B = \{45, 54, 63, 72\} \text{ and } n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{41}$$

Now, events A and B are not mutually exclusive, because some outcomes belong to both sets (for example 54 and 72). Since they are not mutually exclusive, we need to find their intersection.

$$\text{We have, } A \cap B = \{54, 72\} \text{ and } n(A \cap B) = 2$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{2}{41}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{7}{41} + \frac{4}{41} - \frac{2}{41}$$

Required probability is $\frac{9}{41}$.

Illustration 8

In a city survey, 50% of the residents read the morning newspaper, 60% read the evening newspaper, and 20% read both. Find the probability that a randomly selected resident reads:

1. Either the morning or the evening newspaper, and
2. Both the morning and evening newspapers.

Solution:

Let A and B represent that the resident read morning and evening paper respectively.

Then $P(A) = 0.50$; $P(B) = 0.60$ and $P(A \cap B) = 0.20$

The probability that the resident reads either the morning or evening or both the papers is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.50 + 0.60 - 0.20 = 0.90$$

Illustration 9

A can solve 80% of the problems in a Statistics book, while B can solve 60% of the problems. If a problem is chosen at random, what is the probability that at least one of them will be able to solve it?

Solution:

This type of question can be solved in three ways.

Method 1:

Probability that A cannot solve the problem = $1 - 0.80 = 0.20$

Probability that B cannot solve the problem = $1 - 0.60 = 0.40$

Probability that both A and B cannot solve the problem = $0.20 \times 0.40 = 0.08$

Probability that at least one of them will solve the problem = $1 - 0.08 = 0.92$

Method 2:

Probability that both will solve the problem = $0.8 \times 0.6 = 0.48$

Probability that at least one of them will solve the problem

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.6 - 0.48 = 0.92$$

Method 3:

1. A solves, but B does not solve:

Probability A solves = 0.8

Probability B does not solve = $1 - 0.6 = 0.4$

Combined probability = $0.8 \times 0.4 = 0.32$

2. B solves, but A does not solve

Probability B solves = 0.6

Probability A does not solve = $1 - 0.8 = 0.2$

Combined probability = $0.6 \times 0.2 = 0.12$

3. Both
- Prob
- Prob
- Com
4. Rec
- B wi

Illustration

The probability that an employee was a trainee is 0.9. What is the probability that the trainee

Solution

Let A be the event that an employee earns more than Rs. 10,000 per month.

Thus, the probability that an employee stayed with the company is 0.8.

Note : This is a problem of conditional probability. The probability of an employee staying with the company is already stated as 0.8. The company wants to know how many employees will stay under the new probability.

Illustration

Suppose the outcomes of a random experiment are A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z. Find the probability that the outcome is A or B or C or D or E or F or G or H or I or J or K or L or M or N or O or P or Q or R or S or T or U or V or W or X or Y or Z.

3. Both A and B solve

Probability A solves = 0.8

Probability B solves = 0.6

Combined probability = $0.8 \times 0.6 = 0.48$

4. **Required probability** = Probability that A will solve the problem but not B + Probability that B will solve the problem but not A + probability that A and B both will solve the problem
- $$= 0.32 + 0.12 + 0.48 = 0.92.$$

Illustration 10

The probability that a trainee will remain with a company is 0.8. The probability that an employee earns more than Rs. 50,000 per year is 0.4. The probability that an employee, who was a trainee and remained with the company or who earns more than Rs. 50,000 per year is 0.9. What is the probability that an employee earns more than Rs. 50,000 per year, given that the trainee stayed with the company?

Solution :

Let A be the event that a trainee will remain with the company and B be the event that an employee earns more than Rs 50,000 per year. Then

$$P(A) = 0.8, P(B) = 0.4 \text{ and } P(A \cup B) = 0.9$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.8 + 0.4 - 0.9 = 0.3$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{0.3}{0.8} = 0.375$$

Thus, the probability that an employee earns more than Rs. 50,000 per year given that the trainee stayed with the company is 0.375 or 37.5%.

Note : This is a case of conditional probability because we are not asked about the general chance of an employee earning more than Rs. 50,000, but specifically about those employees who have already stayed with the company as trainees. The condition "given that the trainee stayed with the company" narrows down the sample space to only those who remained, and then we look at how many of them also earn more than Rs. 50,000. Such problems arise whenever we have extra information or a restriction on the situation, and the probability of one event needs to be calculated under the assumption that another event has already happened. This is why we apply the conditional probability formula.

Illustration 11

Suppose we flip two coins, one after the other. Let S be the sample space of all possible outcomes. Assign a probability of 0.25 to each element of the sample space. Let A be the event that the first coin lands heads, and B be the event that the second coin lands heads. Calculate the probability of the union of events A and B.

Solution

The sample space S consists of four possible outcomes: $S = \{(H, H), (H, T), (T, H), (T, T)\}$. Since each element of the sample space has been assigned a probability of 0.25, we have:

$$P((H, H)) = 0.25$$

$$P((H, T)) = 0.25$$

$$P((T, H)) = 0.25$$

$$P((T, T)) = 0.25$$

Event A is the event that the first coin lands heads: $A = \{(H, H), (H, T)\}$

Event B is the event that the second coin lands heads: $B = \{(H, H), (T, H)\}$

The union of events A and B , denoted as $A \cup B$, is the set of all elements that are in A , in B , or in both A and B .

$$A \cup B = \{(H, H), (H, T), (T, H)\}$$

To calculate the probability of the union of events A and B , we use the formula:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The intersection of events A and B is the outcome (H, H) , which is in both A and B : $A \cap B = \{(H, H)\}$

Therefore, the probability of the intersection is: $P(A \cap B) = 0.25$

Substituting the probabilities of A , B , and $A \cap B$ into the formula, we get:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.25 + 0.25 - 0.25 = 0.25$$

Therefore, the probability of the union of events A and B is 0.25, which means that the probability of getting at least one head in two coin flips is 0.25.

Illustration 12

A bag contains 15 white and 5 black marbles. Find the probability of drawing two marbles of the same colour.

Solution

Given, White marbles = 15, Black marbles = 5. Total marbles = $15 + 5 = 20$

We want the probability of drawing two marbles of the same colour (either both white or both black).

$$P(E_1) = P(\text{drawing two white}) = \frac{{}^{15}C_2}{{}^{20}C_2}$$

$$P(E_2) = P(\text{drawing two Black}) = \frac{{}^5C_2}{{}^{20}C_2}$$

$$P(E_1 \text{ or } E_2) = P(\text{The marbles which are white or black})$$

$$= P(E_1) + P(E_2) = \frac{{}^{15}C_2}{{}^{20}C_2} + \frac{{}^5C_2}{{}^{20}C_2} = \frac{15 \cdot 14}{20 \cdot 19} + \frac{5 \cdot 4}{20 \cdot 19} = \frac{23}{38}$$

The probability of drawing two marbles of the same colour is $23/38$.

Illustration 13

Three horses E_1 , E_2 and E_3 are in a race. The horse E_1 is twice as likely to win as E_2 and E_2 is twice as likely to win as E_3 . Find their respective probabilities $P(E_1)$, $P(E_2)$ and $P(E_3)$ of winning. What is the probability that E_2 , or E_3 wins?

Solution

Let $P(E_3) = x$
Then $P(E_2) = 2x$
and $P(E_1) = 2 P(E_2) = 2(2x) = 4x$
Since $P(E_1) + P(E_2) + P(E_3) = 1$

$$4x + 2x + x = 1$$

$$(or) \quad 7x = 1 \text{ or } x = \frac{1}{7}$$

$$\therefore P(E_1) = 4x = \frac{4}{7}, P(E_2) = 2x = \frac{2}{7}, P(E_3) = x = \frac{1}{7}$$

Since E_2 and E_3 are mutually exclusive events, we have

$\therefore$ Probability of E_2 or E_3 wins,

$$P(E_2 \cup E_3) = P(E_2) + P(E_3) = \frac{2}{7} + \frac{1}{7} = \frac{3}{7}$$

Illustration 14

A card is randomly drawn from a pack of playing cards. Find the probability that the drawn card is (i) a spade or a King (ii) a King or a Queen.

Solution

From the 52 cards of a pack, one card can be drawn in 52 equally likely, mutually exclusive and exhaustive ways.

Let A: the drawn card is a Spade out of 13 spades

B: the drawn card is a King out of 4 kings

C: the drawn card is a Queen out of 4 queens

$$\text{Then, } P(A) = \frac{13}{52}, P(B) = \frac{4}{52} \text{ and } P(C) = \frac{4}{52}$$

Here events A and B are not mutually exclusive because the drawn card can be Spade as well as King simultaneously. But events B and C are mutually exclusive because the drawn card cannot be King as well as Queen at a time, thus.

$$P(A \cap B) = P(\text{Spade-King}) = \frac{1}{52}$$

This is because the pack has one card which is Spade as well as King.

$$P(B \cap C) = 0 \text{ because B and C are mutually exclusive}$$

$$(i) P(\text{Spade or King}) = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

(ii) Since B and C are mutually exclusive,

$$P(\text{King or Queen}) = P(B \cup C)$$

$$= P(B) + P(C) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$$

Illustration 15

A and B are two candidates seeking admission in a college. The probability that A is selected is 0.7 and the probability that exactly one of them is selected is 0.6. Find the probability that B is selected.

Solution

Given, $P(A) = 0.7$ and $P(\text{Exactly one of A, B is selected}) = 0.6$

Let x be the probability that B gets selected. $P(B) = x$

$$\Rightarrow P(\text{Exactly one of A, B is selected}) = 0.6$$

$$\Rightarrow P(A \text{ is selected, B is not selected}) + P(B \text{ is selected, A is not selected}) = 0.6$$

$$\Rightarrow P(A \cap B') + P(A' \cap B) = 0.6$$

$$\Rightarrow P(A)P(B') + P(A')P(B) = 0.6$$

$$\Rightarrow (0.7)(1-x) + (0.3)x = 0.6$$

$$\Rightarrow x = 0.25$$

Thus the probability that B gets selected is 0.25.

Illustration 16

The probability of simultaneous occurrence of at least one of two events A and B is p . The probability that exactly one of A, B occurs is q , then prove that $P(A') + P(B') = 2 - 2p + q$.

Solution

Since $P(\text{exactly one of A, B occurs}) = q$ (given), we get

$$P(A \cup B) - P(A \cap B) = q$$

$$\Rightarrow p - P(A \cap B) = q$$

$$\Rightarrow P(A \cap B) = p - q$$

$$\Rightarrow 1 - P(A' \cup B') = p - q$$

$$\Rightarrow P(A' \cup B') = 1 - p + q$$

$$\Rightarrow P(A') + P(B') - P(A' \cap B') = 1 - p + q$$

$$\Rightarrow P(A') + P(B') = (1 - p + q) + P(A' \cap B')$$

$$= (1 - p + q) + (1 - P(A \cup B))$$

$$= (1 - p + q) + (1 - p)$$

$$= 2 - 2p + q.$$

Illustration

The probability that a student will not get a grade of 4/5. What is the probability that the student will get a grade of 4/5?

Solution :

Let A and B denote the events that the student will get a grade of 4/5.

$P(A)$ = The probability that the student will get a grade of 4/5.

$P(B)$ = The probability that the student will get a grade of 4/5.

The probability that the student will get a grade of 4/5 is

$$P(A)$$

$$P(A)$$

Thus the probability that the student will get a grade of 4/5 is

Illustration

The probability that a student will get a grade of 4/5 in the examination is p and the probability that a student will get a grade of 4/5 in the examination is q .

Solution :

Given $P(E) = p$

Where E stands for the event that the student will get a grade of 4/5.

$$P(E)$$

$$P(E)$$

Probability that the student will get a grade of 4/5 is

Illustration 17

The probability that a contractor will get a plumbing contract is $2/3$, and the probability that he will not get an electric contract is $5/9$. If the probability of getting at least one contract is $4/5$. What is the probability that he will get both the contracts?

Solution :

Let A and B denote the events of getting the plumbing and electric contracts respectively, then

$P(A)$ = The probability that the contractor will get plumbing contract = $\frac{2}{3}$

$P(B)$ = The probability that the contractor will get electric contract = $1 - \frac{5}{9} = \frac{4}{9}$

The probability that the contractor will get at least one contractor = $P(A \cup B) = \frac{4}{5}$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B) = \frac{4}{5}$$

$$P(A \cap B) \text{ or } P(AB) = \frac{2}{3} + \frac{4}{9} - \frac{4}{5} = \frac{30 + 20 - 36}{45} = \frac{14}{45}$$

Thus the probability of the contractor getting both the contracts is $14/45$.

Illustration 18

The probability that a student passes English test is $2/3$, the probability that he passes both English and Technical test is $11/15$. The probability that he passes atleast one test is $1/3$. What is the probability that he passes the Technical test?

Solution :

Given $P(E) = \frac{2}{3}$, $P(E \cap T) = \frac{11}{15}$ and $P(E \cup T) = \frac{1}{3}$

Where E stands for English T stands for Technical.

$$P(E \cup T) = \frac{1}{3}$$

$$P(E \cup T) = P(E) + P(T) - P(E \cap T) \Rightarrow \frac{1}{3} = \frac{2}{3} + P(T) - \frac{11}{15}$$

$$P(T) = \frac{2}{5}$$

Probability that the student passes the technical test is $2/5$.

Illustration 19

A family has 2 children. Given that one of the children is a boy, what is the probability the other child is also a boy?

Solution

Let event B = "at least one child is a boy," and event A = "both children are boys."

The possible outcomes for two children are: $\{bb, bg, gb, gg\}$ and each with probability $1/4$.

Probability of both being boys (event A) = $P(bb) = 1/4$

Probability of at least one boy (event B) = $\{bb, bg, gb\}$, so $P(B) = 3/4$.

Probability of both boys given at least one boy:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{3/4} = \frac{1}{3}$$

The probability that the other child is also a boy, given that one child is a boy, is $1/3$.

Illustration 20

Suppose 36% of families own a dog, 30% of families own a cat, and 22% of the families that have a dog also have a cat. A family is chosen at random and found to have a cat. What is the probability they also own a dog?

Solution

Let D be the families that own a dog, and C the families that own a cat.

We are given $P(D) = 0.36$, $P(C) = 0.30$, $P(C|D) = 0.22$. We want to know $P(D|C)$.

We know $P(D|C) = P(D \cap C)/P(C)$.

To find the numerator, we use $P(D \cap C) = P(C|D) \cdot P(D) = (0.22) \cdot (0.36) = 0.0792$.

So, $P(D|C) = 0.0792/0.3 = 0.264 = 26.4\%$.

Therefore, the probability that a family also owns a dog given that it has a cat is 0.264 or 26.4%.

Illustration 21

Suppose 30% of the women in a class received an 'A' on the test and 25% of the men received an 'A'. The class is 60% women. Given that a person chosen at random received an 'A', what is the probability this person is a woman?

Solution

Let A be the event of receiving an A, W be the event of being a woman, and

M the event of being a man.

We are given $P(A|W) = 0.30$, $P(A|M) = 0.25$, $P(W) = 0.60$ and we want $P(W|A)$.

From the definition $P(W|A) = P(W \cap A) / P(A)$.

As in the previous example, $P(W \cap A) = P(A|W) \cdot P(W) = (0.30) \cdot (0.60) = 0.18$.

To find $P(A)$, we write $P(A) = P(W \cap A) + P(M \cap A)$.

Since the class is 40% men, $P(M \cap A) = P(A | M) \cdot P(M) = (0.25) \cdot (0.40) = 0.10$.

$$P(A) = P(W \cap A) + P(M \cap A) = 0.18 + 0.10 = 0.28.$$

$$\text{Hence, } P(W | A) = P(W \cap A) / P(A) = 0.18 / 0.28.$$

Illustration 22

10% of the bulbs produced in a factory are of red colour and 2% are red and defective. If one bulb is picked up at random, determine the probability of its being defective if it is red.

Solution

Let A and B be the events that the bulb is red and defective, respectively.

$$P(A) = \frac{10}{100} = \frac{1}{10}, \quad P(A \cap B) = \frac{2}{100} = \frac{1}{50}, \quad P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{50} \times \frac{10}{1} = \frac{1}{5}$$

Thus the probability of the picked up bulb of its being defective, if it is red, is $\frac{1}{5}$.

Illustration 23

Two dice are thrown together. Let A be the event 'getting 6 on the first die' and B be the event 'getting 2 on the second die'. Are the events A and B independent?

Solution

When two dice are thrown together, then sample space contains 36 combinations. So, $n(S) = 36$.

$$A = \{(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\} \text{ and } n(A) = 6 \text{ and } P(A) = 6/36 = 1/6$$

$$B = \{(1, 2), (2, 2), (3, 2), (4, 2), (5, 2), (6, 2)\} \text{ and } n(B) = 6 \text{ and } P(B) = 6/36 = 1/6$$

$$A \cap B = \{(6, 2)\}, n(A \cap B) = 1, P(A \cap B) = 1/36$$

Events A and B will be independent if $P(A \cap B) = P(A) P(B)$

$$\text{i.e., } LHS = P(A \cap B) = \frac{1}{36}, \quad RHS = P(A)P(B) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Hence, A and B are independent.

Illustration 24

A committee of 4 students is selected at random from a group consisting of 8 boys and 4 girls. Given that there is at least one girl on the committee, calculate the probability that there are exactly 2 girls on the committee.

Solution

Let A denote the event that at least one girl will be chosen, and B the event that exactly 2 girls will be chosen. We require $P(B | A)$.

Since A denotes the event that at least one girl will be chosen, A^c denotes that no girl is chosen, i.e., 4 boys are chosen. Then

$$P(A') = \frac{{}^8C_4}{{}^{12}C_4} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = \frac{70}{495} = \frac{14}{99}$$

$$\Rightarrow P(A) = 1 - \frac{14}{99} = \frac{85}{99}$$

$$\text{Now } P(A \cap B) = P(2 \text{ boys and } 2 \text{ girls}) = \frac{{}^8C_2 \cdot {}^4C_2}{{}^{12}C_4} = \frac{2 \times 1 \cdot 2 \times 1}{12 \times 11 \times 10 \times 9} = \frac{216}{495}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{216}{495} \times \frac{99}{85} = \frac{24}{55} \times \frac{99}{85} = \frac{2376}{4675} = 0.50$$

Illustration 25

What is the chance of throwing a total of 6 or 5 or 11 with two dice?

Solution

The three events are mutually exclusive and exhaustive case are 36 events.

- (i) Favourable case for obtaining 6 is (1, 5), (2, 4), (3, 3), (5, 1), (4, 2) that is 5 events.
- (ii) Favourable case for obtaining 5 are (1, 4), (4, 1), (2, 3), (3, 2) i.e., 4 events
- (iii) Favourable case for obtaining 11 is (5, 6), (6, 5) i.e., only two events.

$$P(E_1) = P(\text{throwing } 6) = \frac{5}{36}$$

$$P(E_2) = P(\text{throwing } 5) = \frac{4}{36}$$

$$P(E_3) = P(\text{throwing } 11) = \frac{2}{36}$$

$$P(E_1 \text{ or } E_2 \text{ or } E_3) = P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) = \frac{5}{36} + \frac{4}{36} + \frac{2}{36} = \frac{11}{36}$$

Illustration 26

When a die is thrown, E_1 is event of getting odd number, and E_2 is the event getting number two. E_1 and E_2 are they independent event.

Solution

Favourable number of cases for E_1 is (1, 3, 5) which is 3.

Favourable number of cases for E_2 (2, 4, 6) which is 3.

$$P(E_1) = \frac{3}{6} = \frac{1}{2}, \quad P(E_2) = \frac{3}{6} = \frac{1}{2}$$

$E_1 \cap E_2 = \phi$, because a number cannot be both odd and even at the same time.

So, $P(E_1 \cap E_2) = 0$

Events E_1 and E_2 will be independent if

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$$

Since $P(E_1 \cap E_2) = 0$, we get $0 = P(E_1) \cdot P(E_2) = \frac{1}{4}$. [Here, LHS = 0 and RHS = 1/4, the condition fails]

Events E_1 (odd number) and E_2 (multiple of 2) are not independent. They are actually mutually exclusive, because they cannot occur together.

Illustration 27

A bag has 10 bolts out of which 4 are defective. Find the probability of drawing two non-defective bolts simultaneously.

Solution

Since 6 bolts are not defective, $P(\text{drawing two non defective bolts}) = \frac{{}^6C_2}{{}^{10}C_2} = \frac{6 \cdot 5}{10 \cdot 9} = \frac{30}{90} = \frac{1}{3}$

Note: It is same as drawing one bolt after the other without replacement.

$$P(E_1) = \text{Probability of Drawing One non defective bolt} = \frac{6}{10}$$

$$P\left(\frac{E_2}{E_1}\right) = \text{Probability of drawing 2}^{\text{nd}} \text{ Non Defective bolt with out replacement} = \frac{5}{9}$$

$$P(E_1 \text{ and } E_2) = P(E_1) P(E_2 / E_1) = \frac{6}{10} \cdot \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$$

Illustration 28

A problem is given to three students A, B and C whose probabilities of solving are $\frac{2}{3}$, $\frac{3}{4}$ and $\frac{1}{4}$

what is the probability that the problem is solved.

Solution

The probability of A (solving) is $P(A) = \frac{2}{3}$, Similarly $P(B) = \frac{3}{4}$ and $P(C) = \frac{1}{4}$

The problem will be solved if at least one student solves it. It is easier to use the complement rule:

$$P(\text{Solved}) = 1 - P(\text{Not solved by anyone})$$

$$\text{Probability that A does not solve} = 1 - P(A)$$

$$\text{Probability that B does not solve} = 1 - P(B)$$

$$\text{Probability that C does not solve} = 1 - P(C)$$

$$\text{Probability that none solve } P(\text{None solve}) = (1 - P(A)) \times (1 - P(B)) \times (1 - P(C))$$

$$P(\text{Solved}) = 1 - (1 - P(A))(1 - P(B))(1 - P(C))$$

$$= 1 - (1 - 2/3)(1 - 3/4)(1 - 1/4)$$

$$\begin{aligned}
 &= 1 - (1 - 2/3)(1 - 3/4)(1 - 1/4) \\
 &= 1 - (1/3)(1/4)(3/4) \\
 &= 1 - (1/3)(1/4)(3/4) \\
 &= 1 - (3/48) \\
 &= 1 - (1/16) \\
 &= 15/16
 \end{aligned}$$

The probability that the problem is solved is $15/16$ or 93.75% .

Illustration 29

Three machines A B and C produce 40%, 50% and 10% of the total productions out of which 4% and 1% are defective. Find the probability that an item selected at random is

(i) Defective

(ii) Is defective and manufactured by machine A.

Solution

Consider $P(A)$ = Probability that the item is made by machine A $= \frac{40}{100}$

Similarly $P(B) = \frac{50}{100}$, and $P(C) = \frac{10}{100}$

$P(D/A)$ = Probability of defective item made by machine A $= \frac{2}{100}$

Similarly $P(D/B) = \frac{4}{100}$ and $P(D/C) = \frac{1}{100}$

Further $P(A \cap D) = P(A) \cdot P(D/A)$ is defective item produced by A

and $P(D)$ = Probability of drawing defective item

$$= P(A \cap D) + P(B \cap D) + P(C \cap D)$$

$$P(D) = P(A) P(D/A) + P(B) P(D/B) + P(C) P(D/C)$$

$$= \frac{40}{100} \times \frac{2}{100} + \frac{50}{100} \times \frac{4}{100} + \frac{10}{100} \times \frac{1}{100} \Rightarrow P(D) = \frac{290}{10000}$$

But $P(A \cap D) = P(D) P(A/D)$

Probability of item is manufactured by A and defective $= P(A/D)$

$$= \frac{\text{Defective item by A}}{\text{Total defective item}} = \frac{P(A) \cdot P(D/A)}{P(D)} = \frac{80}{290} = \frac{8}{29}$$

Illustration 30

If $P(A \cup B) = 0.65$ and $P(A \cap B) = 0.15$, find $P(A) + P(B)$

Solution

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.65 = P(A) + P(B) - 0.15 \Rightarrow P(A) + P(B) = 0.80$$

Illustration 31

A number is chosen from the first 100 natural numbers. Find the probability that it is a number multiple of 4 or 6

Solution

Given that one number is selected from first 100 natural numbers, $n(S) = 100$,
Let A be the event of selecting a number which is multiple of 4

$$\text{Here } A = \{4, 8, 12, \dots, 100\}$$

$$\Rightarrow n(A) = 25;$$

Let B be the event of selecting a number which is multiple of 6;

$$B = \{6, 12, 18, \dots, 96\}$$

$$\Rightarrow n(B) = 16;$$

Here A, B are not exclusive

$(A \cap B)$ is the event of selecting a number which is a multiple of 4 and 6 i.e., it is a multiple of 12.

$$A \cap B = \{12, 24, \dots, 96\}$$

$$\Rightarrow n(A \cap B) = 8$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{25}{100} + \frac{16}{100} - \frac{8}{100} = \frac{33}{100}$$

Illustration 32

A candidate is selected for interview of management trainees for 3 companies. For the first company there are 12 candidates, for the second there are 15 candidates and for the third there are 10 candidates. What are the chances of his getting job at least at one of the company?

Solution :

The probability that the candidate does not get the job in any company.

$$\text{Probability that the candidate does not get the job in the first company} = 1 - \frac{1}{12} = \frac{11}{12}$$

$$\text{Probability that the candidate does not get the job in the second company} = 1 - \frac{1}{15} = \frac{14}{15}$$

$$\text{Probability that the candidate does not get the job in the third company} = 1 - \frac{1}{10} = \frac{9}{10}$$

$$\text{Probability that the candidate does not get the job in the third company} = 1 - \frac{1}{10} = \frac{9}{10}$$

Since the events are independent, therefore, the probability that the candidate does not get any of the three companies $= \frac{11}{12} \times \frac{14}{15} \times \frac{9}{10} = \frac{231}{300} = 0.77$

Hence the required probability $= 1 - 0.77 = 0.23$.

Illustration 33

A speaks the truth 2 out of 3 times and B, 4 out of 5 times. They agree that from a bag with balls of different colours, a black ball has been drawn. Find the probability that the statement is true.

Solution :

They will agree when both of them tell a lie or speak the truth.

$$\text{Probability that both of them tell a lie} = \frac{1}{3} \times \frac{1}{5} \times \frac{1}{6}$$

$$\text{Probability that both of them speak truth} = \frac{2}{3} \times \frac{4}{5} \times \frac{1}{6}$$

$$\text{Required Probability} = \frac{\frac{2}{3} \times \frac{4}{5} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{5} \times \frac{1}{6}}{\frac{2}{3} \times \frac{4}{5} \times \frac{1}{6} + \frac{1}{3} \times \frac{1}{5} \times \frac{1}{6}} = \frac{8}{9}$$

Illustration 34

A surveys show that 70% of all convenience store shoppers buy milk and 55% buy bread. 45% buy both bread and milk, What percentage buys neither?

Solution:

The **approach** is to use both, first the Complement Probability Rule and then the Addition Probability Rule, i.e.

$$P[\text{buy neither}] = 1 - P[\text{bread OR milk}] = 1 - [0.70 + 0.55 - 0.45] = 20\%$$

Illustration 35

- The traffic light at a certain road crossing is set in the following fashion. It starts green at 06:00 and continues to be green till 06:02 and again turns green at 06:06 and continues to be green till 06:08. This cycle is repeated throughout the day. What percentage of the time during any given hour, traffic light will be green?
- If your arrival time at this crossing is random and uniform over the interval 16:55 to 17:05, What is the probability that you will have to stop at this light?

Solution:

- In every 6 minutes the light remains green for 2 minutes.

$$\text{Required percentage} = \frac{2}{6} \times 100 = 33.3\%$$

- During 16:55 to 17:05 the light will remain green for three minutes and the total interval is 10 minutes. Required Probability $= \frac{3}{10} = 0.3$

Illustration 36

The probability that a car will be checked by the probability of a tyre check and have neither hit a plane.

Solution:

Let A be the event that a car is checked. P(A) =

Required Probability =

Illustration 37

Determine the probability of the following events.

- Both engines fail and that the aircraft is damaged.
- An aircraft is damaged by a 15 per cent probability of a problem.
- A citizen file return.

Solution:

- Probability of both engines failing and that the aircraft is damaged is the required probability.
- Required probability =
- Required probability =

Illustration 38

An Anti-aircraft gun is aimed at a plane. The probability of a hit is 0.8 respectively. What is the probability that the plane is hit?

Solution:

Plane gets hit if the gun is aimed at it. Required Probability =

Illustration 36

the probability that a driver stopping at a petrol station will have his car tyres checked is .012, the probability that he will have oil checked is 0.29 and probability that he will have both oil and tyres checked is 0.07. What is the probability that a driver stopping at this station will have neither his car tyres nor oil checked?

Solution:

Let A be the event that the driver will have tyres checked and B the event that he will have Petrol checked.

$$P(A \cup B) = \text{Probability that he will either get tyres checked or oil checked.}$$

$$= P(A) + P(B) - P(A \cap B) = 0.012 + 0.29 - 0.07 = 0.232$$

$$\text{Required Probability} = 1 - P(A \cup B)$$

$$= 1 - 0.232 = 0.768.$$

Illustration 37

Determine the probability that:

- Both engines on a small airplane fail, given that each engine fails with probability 0.05 and that an engine is twice as likely to fail when it is the only engine working.
- An automobile is recalled for brake failure and has steering problems, given that 15 percent of that model were recalled for brake failure and 2 percent has steering problems.
- A citizen files his or her tax return and cheats on it, given that 70 percent of all citizens file returns and 25 percent of those who file cheat.

Solution:

$$(i) \text{ Probability for the failure of any engine} = 0.05$$

$$\text{Probability that the second engine fails when one engine has already failed} = 0.05 \times 2 = 0.1$$

$$\text{Required Probability} = 0.05 \times 0.1 = 0.005$$

$$(ii) \text{ Required Probability} = 0.15 \times 0.02 = 0.03$$

$$(iii) \text{ Required Probability} = 0.7 \times 0.25 = 0.175$$

Illustration 38

An Anti-aircraft gun can take maximum of four shots of an enemy plane moving away from it. The probabilities the plane at the first, second, third and fourth shot are 0.4, 0.3, 0.2 and 0.2 respectively. What is the probability that the plane gets hit?

Solution:

Plane gets hit if at least one of the shots hits

$$\text{Required Probability} = 1 - \text{None of the shots hit}$$

$$= 1 - (1 - 0.4)(1 - 0.3)(1 - 0.2)(1 - 0.2)$$

$$= 1 - 0.6 \times 0.7 \times 0.8 \times 0.8 = 1 - 0.2688 = 0.7312.$$

Illustration 39

A University has to select an examiner from the list of 50 persons 20 of them women, 10 of them knowing Hindi and 40 not, 15 of the being teachers and the remaining 35 not. What is the probability of the university selecting a Hindi knowing woman teacher?

Solution:

$$\text{Probability that the selected candidate is a women} = \frac{20}{50} = \frac{2}{5}$$

$$\text{Probability that the selected candidate is a Hindi knowing} = \frac{10}{50} = \frac{1}{5}$$

$$\text{Probability that the selected candidate is a teacher} = \frac{15}{50} = \frac{3}{10}$$

$$\text{Probability of selecting a Hindi knowing woman teacher} = \frac{2}{5} \times \frac{1}{5} \times \frac{3}{10} = \frac{6}{250} = \frac{3}{125}$$

($\therefore$ Events are independent)

Illustration 40

A ball is drawn at random from a box containing 5 green, 6 red, and 4 yellow balls. Determine the probability that the ball drawn is (i) Green (ii) Red (iii) Yellow (iv) Green or red (v) not yellow.

Solution:

Total number of balls in the box = $5+6+4 = 15$ balls

(i) Probability of drawing a green ball = $5/15 = 1/3$

(ii) Probability of drawing a red ball = $6/15 = 2/5$

(iii) Probability of drawing a yellow ball = $4/15$

(iv) Probability of drawing a green or a Red ball = $5/15 + 6/15 = 11/15$

(v) Probability of getting not yellow = $1 - p(\text{yellow}) = 1 - 4/15 = 11/15$

Illustration 41

A test paper containing 10 problems is given to three students A, B, C. It is considered that student A can solve 60% problems, student B can solve 40% problems and student C can solve 30% problems. Find the probability that the problem chosen from the test paper will be solved by all three students.

Solution:

Probability of solving the problem by A = 60%

Probability of solving the problem by B = 40%

Probability of solving the problem by C = 30%

Solving the problem
Hence, $P(A \cap B \cap C)$

Illustration 42

A box containing
colour balls one

Solution:

(i) Selection of

Selection of

(ii) Selection of

number of

case are in

Probability

The event

$\therefore$ Probab

Properties of Inc

Property 1. If A and B are

Proof: A & B are

$\therefore P(A \cap B)$

$P(A \cap \bar{B})$

Hence A and B are

Property 2. If A and B are

Proof: Since A and B are

Now, $P(\bar{A} \cap \bar{B})$

Hence $\bar{A}$ and $\bar{B}$ are

solving the problem by a student is independent of solving the problem by the other students.

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C) \\ = \frac{60}{100} \times \frac{40}{100} \times \frac{30}{100} = 0.6 \times 0.4 \times 0.3 = 0.072$$

Illustration 42

A box containing 5 green balls and 3 red colour balls. Find the probability of selecting 3 green colour balls one by one (i) without replacement (ii) with replacement

Solution:

(i) **Selection without Replacement:** Selecting 3 balls out of 8 balls = 8C_3 ways
Selection of 3 green balls in 5C_3 ways

(ii) **Selection with Replacement:** When a ball is drawn and replaced before the next draw, the number of balls in the box remains the same. Also the 3 events of drawing a green ball in each case are independent.

Probability of drawing a green ball in each case is = $\frac{5}{8}$

The event of selecting a green ball in the first, second and third event are same,

$\therefore$ Probability of drawing 3 green balls

$$= \frac{5}{8} \times \frac{5}{8} \times \frac{5}{8} = \frac{125}{512}$$

Properties of Independent Events

Property 1. If A and B are independent events then A and $\bar{B}$ are also independent events.

Proof: A & B are disjoint events are $A = (A \cap \bar{B}) \cup (A \cap B)$

$$P(A) = P(A \cap \bar{B}) + P(A \cap B) \text{ Hence}$$

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

$$= P(A) P(A) P(B) \text{ (Here A and B are independent)}$$

$$= P(A) [1 - P(B)]$$

$$= P(A) P(\bar{B})$$

Hence A and $\bar{B}$ are independent events.

Property 2. If A and B are independent events then $\bar{A}$ and $\bar{B}$ are also independent events.

Proof: Since A and B are independent $P(A \cap B) = P(A) P(B)$

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) \text{ [By Demorgan's Theorem]}$$

$$= 1 - P(A \cup B)$$

$$= 1 - [P(A) + P(B) - P(A \cap B)]$$

$$= 1 - P(A) - P(B) + P(A) P(B) \text{ (using (1))}$$

$$= [1 - P(A)] [1 - P(B)]$$

$$= P(\bar{A}) P(\bar{B})$$

Hence $\bar{A}$ and $\bar{B}$ are independent events.

Property 3. If A, B, C are mutually independent events, then $A \cap B$ and C are independent.

$$\begin{aligned} \text{Proof: } P[(A \cap B) \cap C] &= P[(A \cap C) \cap (B \cap C)] \\ &= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C) \\ &= P(A)P(C) + P(B)P(C) - P(A)P(B)P(C) \\ &\quad (\text{Since A, B, C are mutually independent}). \\ &= [(P(A) + P(B) - P(A)P(B))]P(C) \\ &= P(A \cap B)P(C). \end{aligned}$$

Hence $A \cap B$ and C are independent events.



Note:

Let A, B be two events with $P(A) > 0$ and $P(B) > 0$. Then

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

and

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Substituting (2) in (1) we get,

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

This is known as Baye's formula.

Problems on Properties of Independent Events

Illustration 1

In a college, there are five lady lecturers. Among them, three are doctorates. If a committee consisting three of them is formed, what is the probability that at least two are doctorates?

Solution

Among the five lecturers, three lecturers can be selected in 5C_3 ways. Thus, there are 5C_3 equally mutually exclusive and exhaustive outcomes. Let events A and B be -

A : Two of the selected lecturers are doctorates.

B : All the three selected lecturers are doctorates.

Then, event A has ${}^3C_2 \times {}^2C_1$ favourable outcomes. And, event B has 3C_3 favourable outcomes. Then

$$\begin{aligned} P[\text{at least two doctorates}] &= P[\text{two or three doctorates}] \\ &= P(A \cup B) = P(A) + P(B) \end{aligned}$$

[Because A and B are mutually exclusive]

$$\frac{{}^3C_2 \times {}^2C_1}{{}^5C_3} + \frac{{}^3C_3}{{}^5C_3} = \frac{3 \times 2}{10} + \frac{1}{10} = \frac{7}{10} = 0.7$$

Illustration

A fair coin is tossed. Find the probability of the following events:

(i) head

Solution

There are $2^2 = 4$ possible outcomes. The outcomes are HH, HT, TH, TT.

(i) P[head]

(ii) P[at least one head]

Note: When a coin is tossed twice, the possible outcomes are HH, HT, TH, TT.

Illustration

Three unbiased dice are thrown. Find the probability that the sum of the numbers is 10.

Solution

When 3 dice are thrown, there are $6^3 = 216$ possible outcomes. Of these, the outcomes which give a sum of 10 are (4, 3, 3), (3, 4, 3), (3, 3, 4), (2, 4, 4), (4, 2, 4), (4, 4, 2), (1, 4, 5), (5, 1, 4), (4, 5, 1), (1, 5, 4), (4, 1, 5), (5, 4, 1).

P[at least one head]

Illustration

There are 20 persons. Find the probability that at least two are of the same age.

Solution

From 20 persons, 19 are of different ages. The remaining one is of the same age as one of the 19.

A : at least two are of the same age

B : no two are of the same age

Then, event A is the complement of event B.

Therefore, P[at least two are of the same age] = 1 - P[at least one is of the same age]

P[at least one is of the same age]

Illustration 2

A coin is tossed five times. Find the probability of obtained

(i) head in all the tosses,

(ii) head in at least one of the tosses

Solution

There are $2^5 = 32$ equally likely, mutually exclusive and exhaustive outcomes. Out of them, one outcome is HHHHH and another outcome is TTTTT. Therefore,

$$P[\text{head in all tosses}] = \frac{1}{32}$$

$$P[\text{at least one head}] = 1 - P[\text{tail in all tosses}] = 1 - \frac{1}{32} = \frac{31}{32}$$

Note Whenever probability of the event "at least one" has to be found, it is easier to find it by using the probability of the complementary event as follows.

$$P[\text{at least one}] = 1 - P[\text{none}]$$

Illustration 3

Three unbiased dice are thrown once. Find the probability that all the three dice show the number 6.

Solution

When 3 dice are thrown, there are $6 \times 6 \times 6 = 216$ equally likely, mutually exclusive and exhaustive outcomes. Of these 216 outcomes, and outcome, namely, (6, 6, 6) is favourable. Therefore, probability of all the three dice showing the number 6 is

$$P[\text{all the three result in the number 6}] = \frac{1}{216}$$

Illustration 4

There are 20 persons. 5 of them are graduates. 3 persons are randomly selected from these 20 persons. Find the probability that at least one of the selected persons is graduate.

Solution

From 20 persons, 3 persons can be selected in ${}^{20}C_3$ ways. Thus, there are ${}^{20}C_3$ equally likely, mutually exclusive and exhaustive outcomes. Let events A and B be -

A : at least one of the selected persons is graduate

B : none of the selected persons is graduate

Then, event B is the complement of event A. ${}^{15}C_3$ outcome are favourable to event B.

Therefore,

$$P[\text{at least one is graduate}] = 1 - P[\text{none is graduate}]$$

$$= 1 - \frac{{}^{15}C_3}{{}^{20}C_3} = 1 - \frac{15 \times 14 \times 13}{20 \times 19 \times 18} = 1 - \frac{455}{1140} = 0.6$$

6.8 Bayes' Theorem

Bayes' Theorem is a fundamental concept in probability theory that describes the probability of an event based on prior knowledge of related events. It is named after the Reverend Thomas Bayes, who is a powerful tool for updating probabilities based on new information.

Bayes' Theorem is particularly useful in situations where we have prior knowledge about the probabilities of events and want to revise those probabilities in light of new evidence. It has many applications in fields such as medical diagnosis, machine learning, statistics, and decision-making processes. By providing a systematic and principled way to update probabilities based on new information, Bayes' Theorem helps us make more informed decisions in situations where uncertainty and inference play a crucial role.

Bayes' Theorem is a mathematical formula used for calculating conditional probabilities. It allows us to determine the probabilities of an event occurring given that another event has already occurred. This theorem uses prior knowledge or evidence to predict the likelihood of related events.

The formula is:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A|B)$ is the probability of event A occurring given that B is true.
- $P(B|A)$ is the probability of event B occurring given that A is true.
- $P(A)$ is the probability of event A.
- $P(B)$ is the probability of event B.

Steps to Apply Bayes' Theorem

1. Define the Probabilities:

- Determine $P(A)$, the probability of event A.
- Determine $P(B)$, the probability of event B.
- Determine $P(B|A)$, the probability of event B occurring given event A is true.

2. Substitute the Values in Bayes' Formula:

- Insert the probabilities into the Bayes' formula.

3. Calculate $P(A|B)$:

- Perform the calculation to find $P(A|B)$, the probability of event A given event B.

Example:

Suppose we have a medical test to determine if a patient has a certain disease. The test has the following characteristics:

- The probability of having the disease (A) is 0.01 (1% of the general population has this disease).
- The probability of a positive test (B) given the disease (A) is 0.99 (99% sensitivity or true positive rate).
- The probability of a positive test (B) given no disease (A') is 0.05 (5% false positive rate).

We want to find out the probability that a person actually has the disease given they tested positive.

Step by Step Solution

1. Define the probabilities:

- $P(A)$
- $P(B)$
- $P(B|A)$

2. Calculate the probabilities:

- $P(B)$
- $P(A')$
- $P(B)$

3. Substitute the values in Bayes' formula:

- $P(A)$

- $P(A)$

4. Calculate the final probability:

- $P(A)$

Conclusion:

Given that a person tested positive, the probability that they actually have the disease is approximately 0.16. This result might be surprising, but it highlights the importance of considering the prevalence of the disease and the accuracy of the test as a tool for understanding probability.

Example:

Imagine we have a deck of 52 cards. What is the probability of drawing an ace, given that the card is a heart?

Definitions:

- Event A: Drawing an ace.
- Event B: Drawing a heart.

Probabilities:

- $P(A)$: The probability of drawing an ace is $\frac{4}{52}$.
- $P(B)$: The probability of drawing a heart is $\frac{13}{52}$.
- $P(B|A)$: The probability of drawing a heart, given that the card is an ace, is $\frac{1}{4}$.

Step Solution:**1. Define the Probabilities:**

- $P(A) = 0.01$ (Probability of having the disease)
- $P(B|A) = 0.99$ (Probability of testing positive given no disease)
- $P(B|A') = 0.05$ (Probability of testing positive given no disease)

2. Calculate $P(B)$:

- $P(B) = P(B|A) \cdot P(A) + P(B|A') \cdot P(A')$
- $P(A')$ is the probability of not having the disease, which is $1 - P(A) = 0.99$
- $P(B) = 0.99 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0099 + 0.0495 = 0.0594$

3. Substitute the Values into Bayes' Formula:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$P(A|B) = \frac{0.99 \cdot 0.01}{0.0594}$$

4. Calculate $P(A|B)$:

$$P(A|B) = \frac{0.0099}{0.0594} \approx 0.1667 \text{ or } 16.67\%$$

Decision:

Given that a person tested positive, there is approximately a 16.67% chance that they actually have the disease. This result might seem surprisingly low considering the test's high sensitivity, but it is due to the low overall prevalence of the disease and the relatively high false positive rate. Bayes' Theorem thus provides a powerful tool for understanding the true implications of test results within a population.

Example:

Suppose we have a standard deck of 52 playing cards, and we want to find the probability that a card drawn is an ace, given that it's already known to be a heart.

Definitions:

- Event A: The card is an ace.
- Event B: The card is a heart.

Probabilities:

- $P(A)$: The probability of drawing an ace from a standard deck, which is $\frac{4}{52}$ because there are 4 aces in 52 cards.
- $P(B)$: The probability of drawing a heart from a standard deck, which is $\frac{13}{52}$ because there are 13 hearts in 52 cards.
- $P(B|A)$: The probability that the card is a heart given that it's an ace. Since one out of the four aces is a heart, $P(B|A) = \frac{1}{4}$.

Bayes' Theorem Application:

We want to find $P(A|B)$, the probability that the card is an ace given it's a heart.

1. Using Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

2. Substituting the Known Values:

$$P(A|B) = \frac{\frac{1}{4} \cdot \frac{4}{52}}{\frac{13}{52}}$$

3. Simplifying the Expression:

The $\frac{52}{52}$ terms cancel out, simplifying to:

$$P(A|B) = \frac{1}{4} \cdot \frac{4}{13} \Rightarrow P(A|B) = \frac{1}{13}$$

Conclusion:

So, if we know the card drawn is a heart, the probability that it is also an ace is $\frac{1}{13}$, or approximately 7.7%.

This example demonstrates how Bayes' Theorem allows us to update the probability of an event based on new information. In this case, knowing the card is a heart changes the probability of it being an ace from the prior probability of $\frac{4}{52}$ to the posterior probability of $\frac{1}{13}$.

**Applications of Bayes' Theorem**

Bayes' Theorem has many applications in various fields, including:

- 1. Medical Diagnosis:** Bayes' Theorem is commonly used in medical diagnosis to calculate the probability of a patient having a disease given the results of a medical test.
- 2. Spam Filtering:** Bayes' Theorem is used in email spam filtering to classify emails as spam or not-spam based on their content.
- 3. Machine Learning:** Bayes' Theorem is used in machine learning algorithms, such as Naïve Bayes classifiers, to classify data based on probabilities.
- 4. Risk Assessment:** Bayes' Theorem is used in risk assessment to calculate the probability of an event occurring based on prior knowledge and new information.
- 5. Fraud Detection:** Bayes' Theorem is used in fraud detection to identify fraudulent transactions based on patterns and probabilities.
- 6. Natural Language Processing:** Bayes' Theorem is used in natural language processing to classify text based on probabilities.
- 7. Image Recognition:** Bayes' Theorem is used in image recognition to classify images based on probabilities.
- 8. Decision-making:** Bayes' Theorem is used in decision-making processes to update probabilities based on new information and make more informed decisions.

Theorem: If $A_1, A_2, \dots$

$$(1) P(A) = \sum_{i=1}^n P(A_i)$$

$$(2) P(A|A_i) P(A_i) = \frac{P(A \cap A_i)}{P(A)}$$

$$(3) P(A_i | A) = \frac{P(A \cap A_i)}{\sum_{i=1}^n P(A \cap A_i)}$$

Proof: Consider the Venn diagram and exhaustive

From the figure

$$A = (A \cap A_1) \cup (A \cap A_2) \cup \dots$$

and

$$P(A) = P(A \cap A_1) + P(A \cap A_2) + \dots$$

or

$$P(A) = P(A \cap A_1) + P(A \cap A_2) + \dots$$

$$\text{But } P(A \cap A_i) = P(A_i | A) P(A)$$

$$\Rightarrow P(A \cap A_i) = P(A_i | A) P(A)$$

Then using (2) and (3)

$$P(A) = P(A \cap A_1) + P(A \cap A_2) + \dots$$

Also

$$P(A) = P(A \cap A_1) + P(A \cap A_2) + \dots$$

$$\text{Again } P(A \cap A_i) = P(A_i | A) P(A)$$

From (3) and (1)

Finally, from (3) and

Theorem: If $A_1, A_2, \dots, A_n$ are n mutually exclusive exhaustive events and E any other event then

$$(1) P(A) = \sum_{i=1}^n P(A \cap A_i)$$

$$(2) P(A|A_i) P(A_i) = P(A \cap A_i) \quad P(A) = P(A \cap A_i) \text{ and}$$

$$(3) P(A_i|A) = \frac{P(A|A_i)P(A_i)}{\sum_{i=1}^n P(A|A_i) \cdot P(A_i)}$$

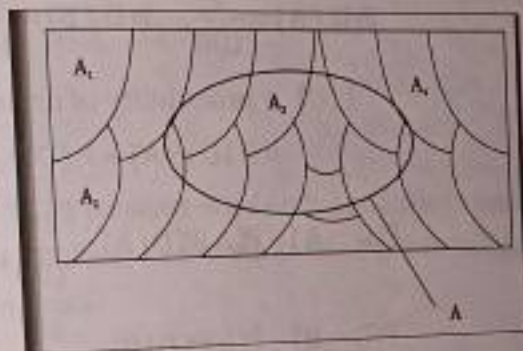
Proof: Consider the Venn diagram of the given set of events $A_1, A_2, \dots, A_n$ which are mutually exclusive and exhaustive

from the figure

$$A = (A \cap A_1) \cup (A \cap A_2) \cup \dots \cup (A \cap A_n)$$

$$P(A) = P(A \cap A_1) \cup P(A \cap A_2) \cup \dots \cup P(A \cap A_n)$$

$$P(A) = \sum_{i=1}^n P(A \cap A_i) \quad \dots(1)$$



... (2)

... (3)

$$\text{Let } P(A \cap A_i) = P(A) P(A_i|A)$$

$$\Rightarrow P(A \cap A_i) = P(A) P(A|A_i)$$

Then using (2) and (3) in (1), we get

$$P(A) = \sum_{i=1}^n P(A) P(A_i|A)$$

$$\text{Also } P(A) = \sum_{i=1}^n P(A_i) P(A|A_i)$$

$$\text{Again } P(A \cap A_i) = P(A) P(A_i|A) \quad \text{or} \quad P(A_i|A) = \frac{P(A \cap A_i)}{P(A)} \quad \dots(4)$$

$$\text{From (3) and (1)} \quad P(A_i|A) = \frac{P(A_i)P(A|A_i)}{\sum_{i=1}^n P(A \cap A_i)}$$

Finally, from (3) and (4), we have

$$P(A_i|A) = \frac{P(A_i)P(A|A_i)}{\sum_{i=1}^n P(A_i)P(A|A_i)}$$

Problems on Bayes' Theorem

Illustration 1

In a bolt factory 20%, 40% and 40% of the total is manufactured by machines A, B and C respectively, of which 6%, 5% and 3% are defective. What is the probability that a bolt drawn at random is (i) Defective bolt and (ii) it is manufactured by A

Solution

Given $P(A) = \frac{20}{100}$, $P(B) = \frac{40}{100}$ and $P(C) = \frac{40}{100}$

$$P(D/A) = \frac{6}{100}, \quad P(D/B) = \frac{5}{100} \quad \text{and} \quad P(D/C) = \frac{3}{100}$$

$P(D)$ = probability of drawing a defective bolt

$$= P(D \cap A) + P(D \cap B) + P(D \cap C)$$

$$P(D \cap A) = P(A)P(D|A) = \frac{20}{100} \cdot \frac{6}{100}$$

$$P(D \cap B) = P(B)P(D|B) = \frac{40}{100} \cdot \frac{5}{100}$$

$$P(D \cap C) = P(C)P(D|C) = \frac{40}{100} \cdot \frac{3}{100}$$

$$P(D) = \left(\frac{20}{100} \times \frac{6}{100} \right) + \left(\frac{40}{100} \times \frac{5}{100} \right) + \left(\frac{40}{100} \times \frac{3}{100} \right) + \frac{440}{10000} = \frac{44}{1000}$$

Again $P(D \cap A) = P(D) P(A|D)$

Probability of a bolt made by A and is defective

$$\begin{aligned} P(A|D) &= \frac{P(D \cap A)}{P(D)} = \frac{P(A)P(D|A)}{P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C)} \\ &= \frac{\frac{20}{100} \times \frac{6}{100}}{\left(\frac{20}{100} \times \frac{6}{100} \right) + \left(\frac{40}{100} \times \frac{5}{100} \right) + \left(\frac{40}{100} \times \frac{3}{100} \right)} = \frac{120}{440} \end{aligned}$$

Illustration 2

In a class 6% are boys 3% of girls are intelligent and 70% of the students are boys. Find the probability that a student selected at random is an intelligent girl.

Solution

$P(B)$ = Probability of selecting a boy = $\frac{70}{100}$

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Illustration

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Solution

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Illustration

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Solution

Here,

$$P(G) = \text{Probability of selecting a girl} = \frac{30}{100}$$

$$P(I|B) = \text{Probability of intelligent boy} = \frac{6}{100}; \quad P(I|G) = \frac{3}{100}$$

$P(I)$ = Probability of selecting an intelligent student

$$P(I) = P(I \cap B) + P(I \cap G) \\ = P(B) P(I|B) + P(G) P(I|G) = \frac{70}{100} \cdot \frac{6}{100} + \frac{30}{100} \cdot \frac{3}{100} = \frac{510}{10000}$$

Further
Where $P(I \cap B) = P(I) P(G|I)$

$P(G|I)$ = Probability of selecting a girl who is intelligent.

$$P(G|I) = \frac{P(I \cap G)}{P(I)} = \frac{P(G) \cdot P(I|G)}{P(I \cap G) + P(I \cap B)} = \frac{\frac{30}{100} \cdot \frac{3}{100}}{\frac{510}{10000}} = \frac{90}{510} = \frac{9}{51} = \frac{3}{17}$$

Illustration 3

In a certain college, there are 6% of men students and 3% of women students are taller than 1.8m. Furthermore, 40% of the students are women. If a student selected at random is taller than 1.8m, what is the probability that the student is a woman?

Solution

Let us denote the set of all men students taller than 1.8m by M and that of women students taller than 1.8m by W. Let $A = \{\text{Students taller than 1.8m}\}$

Then, $P(M) = \frac{6}{100} = 0.06$ $P(W) = \frac{3}{100} = 0.03$

Also $P(A|M) = \frac{60}{100} = 0.60$, $P(A|W) = \frac{40}{100} = 0.40$

$P(W|A)$, the probability that the student selected is taller than 1.8m is a woman. By Bayes' theorem, we get

$$P(W|A) = \frac{P(W)P(A|W)}{P(W)P(A|W) + P(M)P(A|M)} = \frac{(0.03)(0.40)}{(0.03)(0.40) + (0.06)(0.60)} = 0.25$$

Illustration 4

Three machines A, B and C produce respectively 65%, 25% and 20% of the total number of items of factory. The percentages of defective output of these machines are respective 3%, 5% and 5%. An item is selected at random and is found defective. Find the probability that the items were produced by machine C.

Solution

Here,

$$P(A) = \frac{65}{100} = 0.65,$$

$$P(B) = \frac{25}{100} = 0.25$$

$$P(C) = \frac{20}{100} = 0.20$$

Let $X = \{\text{defective item}\}$. Then

$$P(X|A) = \frac{3}{100} = 0.03, \quad P(X|B) = \frac{5}{100} = 0.05, \quad P(X|C) = \frac{5}{100} = 0.05$$

$P(C|X)$, is the probability that an item is produced by machine C given that the item is defective. Bayes' theorem.

$$P(C|X) = \frac{P(X|C)P(C)}{P(A)P(X|A) + P(B)P(X|B) + P(C)P(X|C)}$$

$$= \frac{(0.05)(0.20)}{(0.65)(0.03) + (0.25)(0.05) + (0.20)(0.05)} = \frac{0.01}{0.0195 + 0.0125 + 0.01} = \frac{0.01}{0.042} = 0.238$$

Illustration 5

Suppose 10 men out of 100 and 30 women out of 100 wear spectacles. A person wearing spectacles is chosen at random, what is the probability of his being a male? Assuming that males and females are in equal proportion.

Solution

Let E_1 be the event that men are wearing spectacles and E_2 be the event that women are wearing spectacles.

$$\text{Then } P(E_1) = 1/2 = P(E_2)$$

Let E be the event of choosing a person wearing spectacles.

$$\text{Then } P(E|E_1) = \frac{10}{100} \text{ and } P(E|E_2) = \frac{30}{100}$$

Probability of choosing men and wearing spectacles is

$$P(E \cap E_1) = P(E_1) \cdot P(E|E_1) = \frac{1}{2} \times \frac{10}{100} = \frac{10}{200} = 0.05$$

Similarly probability of choosing women, wearing spectacles is

$$P(E \cap E_2) = P(E_2) \cdot P(E|E_2) = \frac{1}{2} \times \frac{30}{100} = 0.15$$

The required probability is

$$P(E_1|E) = \frac{P(E \cap E_1)}{P(E \cap E_1) + P(E \cap E_2)} = \frac{0.05}{0.05 + 0.15} = \frac{0.05}{0.20} = 0.25$$

Illustration 6

Three machines A, B and C manufacture respectively 30%, 50% and 20% of the total production. The percentage of defective items produced by A, B and C are 3, 6 and 2 respectively. If an item is chosen at random and is found defective, find the probability of it being a product of machine A.

Solution

Let E_1, E_2 and E_3 be the events that the items are manufactured by the machines A, B and C respectively. Let E be the event that the item manufactured is defective,

Then $P(E_1) = 0.3$
Also $P(E_2) = 0.5$
 $P(E_3) = 0.2$
The probability of
 $P(E_1 \cap E)$
The required prob
 $P(E_3|E)$

Illustration 7

One factory F_1 produces 5000 articles, 400 of which are being defective. Another factory F_2 produces 6000 articles, 500 of which are being defective.

Solution

Let A_1, A_2 and A_3 be the events that an article is defective, respectively.

$$P(A_1) = \frac{400}{5000} = 0.08$$

Let B denote the event that an article is not defective.

$$P(B|A_1) = \frac{P(A_1 \cap B)}{P(A_1)} = \frac{4600}{5000} = 0.92$$

We have to find $P(A_1|B)$

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)}$$

$$P(E_1) = 0.3 \quad P(E_2) = 0.5 \quad P(E_3) = 0.2$$

$$P(E_1 \cap E) = P(E_1) P(E|E_1) = (0.3)(0.03) = 0.009$$

$$P(E_2 \cap E) = P(E_2) P(E|E_2) = (0.5)(0.06) = 0.03$$

$$P(E_3 \cap E) = P(E_3) P(E|E_3) = (0.2)(0.02) = 0.004$$

Probability of getting a defective item is

$$P(E_1 \cap E) + P(E_2 \cap E) + P(E_3 \cap E) = 0.009 + 0.03 + 0.004 = 0.043$$

Required probability is

$$P(E_3|E) = \frac{P(E_3 \cap E)}{P(E_1 \cap E) + P(E_2 \cap E) + P(E_3 \cap E)} = \frac{0.004}{0.043} = 0.0930$$

Illustration 7

Factory F_1 produces 2000 articles, 20 of them being defective, second factory F_2 produces 5000 articles, 40 of them being defective and third factory produces 3000 articles, 50 of them being defective. All these articles are put in one stock pile. One of them is chosen and is found to be defective. What is the probability that it is from the factory F_1 .

Solution

Let A_1 , A_2 and A_3 denote the event that the chosen article is produced by the factory F_1 , F_2 , and F_3 respectively.

$$P(A_1) = \frac{2000}{10000} = \frac{2}{10}; P(A_2) = \frac{5000}{10000} = \frac{1}{2}; P(A_3) = \frac{3000}{10000} = \frac{3}{10}$$

Let B denote the event that the article chosen is defective. Then

$$P(B|A_1) = \frac{20}{2000} = \frac{1}{100}; P(B|A_2) = \frac{40}{5000} = \frac{1}{125}; P(B|A_3) = \frac{50}{3000} = \frac{1}{60}$$

We have to find $P(A_1|B)$. By Baye's theorem

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{\sum_{i=1}^3 P(A_i)P(B|A_i)}$$

$$= \frac{\frac{2}{10} \times \frac{1}{100}}{\left(\frac{2}{10} \times \frac{1}{100}\right) + \left(\frac{1}{2} \times \frac{1}{125}\right) + \left(\frac{3}{10} \times \frac{1}{60}\right)}$$

$$= \frac{\frac{2}{10} \times \frac{1}{100}}{\frac{20 + 40 + 50}{10000}} = \frac{1}{500} \times \frac{10000}{110} = \frac{10}{11}$$

Illustration 8

A certain disease has an incidence rate of 2%. If the false negative rate is 10% and the false positive rate is 1%, compute the probability that a person who tests positive actually has the disease. [Assume 10,000 cases]

Solution

Imagine 10,000 people who are tested. Of these 10,000, 200 will have the disease; 10% of the 200, or 20, will test negative and the remaining 180 will test positive. Of the 9800 who do not have the disease, 1% of them, or 98, will test positive.

	Positive Test	Negative Test	Total
Have disease	180	20	200
Do not have disease	98	9702	9800
Total	278	9822	10,000

So of the 278 total people who test positive, 180 will have the disease. Thus

$$P(\text{disease} | \text{positive}) = \frac{180}{278} = 0.647$$

So about 65% of the people who test positive will have the disease.

Using Bayes theorem directly would give the same result:

$$P(\text{disease} | \text{positive}) = \frac{(0.02)(0.90)}{(0.02)(0.90) + (0.98)(0.01)} = \frac{0.018}{0.0278} = 0.647$$

Illustration 9

A company has found that 80% of its new management hires are meeting expectations, while 20% are not. Of the satisfactory hires, 75% had sales experience, while of the unsatisfactory hires, 55% had sales experience. What is the probability that a new hire with sales experience will meet expectations? [New hires = 100]

Solution

We can imagine 100 new hires. Of them, 80%, or 80, will meet expectations, and 20 will not. Of the 80 who meet expectations, 75%, or 60, had sales experience, and 20 did not. Of the 20 who did not meet expectations, 55%, or 11, had sales experience, and 9 did not.

Summarizing that in a table:

	Sales Experience	No Sales Experience	Total
Meeting expectations	60	20	80
Not Meeting expectations	11	9	20
Total	71	29	100

Now we can an

$P(\text{meet})$

So about 84.5%

Illustration

Three machin
the total daily
machines E_1 and
is picked up at

Solution

Let D be the eve

Let A_1, A_2 and A_3

$P(A_1)$

$P(A_2)$

Also, $P(D | A_1)$

$P(D | A_2)$

Putting these va

$P(D | A_3)$

Illustration

Refer to Examp

Solution

Now, we have to

$P(A_1 | D)$

and we can answer the question.

$$P(\text{meet expectations} | \text{sales experience}) = \frac{60}{71} = 0.845$$

about 84.5% of new hires with sales experience will meet expectations.

Illustration 10

Three machines E_1 , E_2 , E_3 in a certain factory produce 50%, 25% and 25%, respectively, of the total daily output of electric tubes. It is known that 4% of the tubes produced on each of machines E_1 and E_2 are defective, and that 5% of those produced on E_3 are defective. If one tube is picked up at random from a day's production, calculate the probability that it is defective.

Solution

Let D be the event that the picked up tube is defective

Let A_1 , A_2 and A_3 be the events that the tube is produced on machines E_1 , E_2 and E_3 , respectively.

$$P(D) = P(A_1)P(D|A_1) + P(A_2)P(D|A_2) + P(A_3)P(D|A_3) \quad \dots(1)$$

$$P(A_1) = \frac{50}{100} = \frac{1}{2}, \quad P(A_2) = \frac{1}{4}, \quad P(A_3) = \frac{1}{4}$$

$$\text{i.e. } P(D|A_1) = P(D|A_2) = \frac{4}{100} = \frac{1}{25}$$

$$P(D|A_3) = \frac{5}{100} = \frac{1}{20}$$

Putting these values in (1), we get

$$P(D) = \frac{1}{2} \times \frac{1}{25} + \frac{1}{4} \times \frac{1}{25} + \frac{1}{4} \times \frac{1}{20}$$

$$= \frac{1}{50} + \frac{1}{100} + \frac{1}{80} = \frac{8}{80} \times \frac{1}{50} + \frac{1}{100} \times \frac{4}{4} + \frac{1}{80} \times \frac{5}{5} = \frac{17}{400} = 0.0425$$

Illustration 11

Refer to Example 6. Calculate the probability that the defective tube was produced on machine E_1 .

Solution

Now, we have to find $P(A_1/D)$.

$$P(A_1/D) = \frac{P(A_1 \cap D)}{P(D)} = \frac{P(A_1)P(D/A_1)}{P(D)} = \frac{\frac{1}{2} \times \frac{1}{25}}{\frac{17}{400}} = \frac{8}{17}$$

Illustration 12

A car manufacturing factory has two plants, X and Y. Plant X manufactures 70% of cars and plant Y manufactures 30%. 80% of the cars at plant X and 90% of the cars at plant Y are of standard quality. A car is chosen at random and is found to be of standard quality. What is the probability that it has come from plant X?

Solution

Let E be the event that the car is of standard quality. Let B_1 and B_2 be the events that the car is manufactured in plants X and Y, respectively. Now

$$P(B_1) = \frac{70}{100} = \frac{7}{10},$$

$$P(B_2) = \frac{30}{100} = \frac{3}{10}$$

$P(E|B_1)$ = Probability that a standard quality car is manufactured in plant X = $\frac{80}{100} = \frac{4}{5}$

$$P(E|B_2) = \frac{90}{100} = \frac{9}{10}$$

$P(B_1|E)$ = Probability that a standard quality car has come from plant X

$$= \frac{P(B_1) \times P(E|B_1)}{P(B_1) \times P(E|B_1) + P(B_2) \times P(E|B_2)} = \frac{\frac{7}{10} \times \frac{4}{5}}{\frac{7}{10} \times \frac{4}{5} + \frac{3}{10} \times \frac{9}{10}} = \frac{56}{83}$$

Hence the required probability is $\frac{56}{83}$.

6.9 Review Questions**Short Answer Type Questions**

1. Define probability with an example.
2. What is a certain event? Give one example.
3. What is an impossible event? Give one example.
4. Define experiment in probability.
5. What is a random experiment? Give one example.
6. Define trial with an example.
7. Define outcome with an example.
8. What is sample space? Give an example of finite and infinite sample space.
9. Define an event in probability.
10. Give an example for a simple (elementary) event.
11. Give an example for a compound event.
12. What is a null event? Give one example.

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27. What is
28. State a
29. If $P(A)$

Long Answer

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2. Write
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13. Define sure event with example.
14. Define mutually exclusive events. Give one example.
15. Define exhaustive events with an example.
16. Define equally likely events with an example.
17. Define independent events with an example.
18. Define dependent events with an example.
19. Write the complement of event A when a die is rolled and $A = \{2, 4, 6\}$.
20. State the addition rule of probability (mutually exclusive case).
21. State the addition rule of probability (general case).
22. State the multiplication rule of probability (independent case).
23. State the multiplication rule of probability (dependent case).
24. Define conditional probability.
25. Write Bayes' Theorem formula.
26. What is relative frequency approach to probability?
27. What is the classical approach to probability?
28. State any two axioms of probability.
29. If $P(A) = 0.6$, find $P(A')$.

Long Answer Type Questions

1. Explain experiment, random experiment, trial, and outcome with examples.
2. Write a note on sample space with two examples each for finite and infinite sample space.
3. Explain the different types of events with suitable examples.
4. Distinguish between independent and dependent events with examples.
5. Explain the operations on events with examples (union, intersection, complement, difference).
6. Explain the classical approach to probability with examples.
7. Explain the empirical approach to probability with examples.
8. Explain the axiomatic approach to probability with examples.
9. State and explain the basic probability rules with examples.
10. Write short notes on:
 - a) Certain event and Impossible event
 - b) Equally likely and Exhaustive events
11. Explain conditional probability with two real-life examples.
12. Explain Bayes' Theorem and mention its applications.
13. A coin is tossed three times. Write its sample space and find the probability of getting (i) all heads (ii) exactly one head.
14. Explain different types of events with proper examples.

15. Explain the three approaches to probability with examples.
16. Explain the addition rule and multiplication rule of probability with examples.
17. Explain conditional probability with examples.
18. State and prove Bayes' Theorem.
19. A coin is tossed 3 times. Write the sample space. Find the probability of getting (i) at least one head (ii) no head (iii) exactly two heads.
20. Two dice are thrown together. Write the sample space and find the probability of getting (i) doublets (ii) total of 9 (iii) at least one 6.
21. A bag contains 3 red and 2 blue balls. Two balls are drawn at random. Find the probability of getting (i) both are red (ii) both are blue and one blue.
22. In a deck of 52 cards, one card is drawn at random. Let A = "card is red" and B = "card is a king." Show whether A and B are (i) mutually exclusive (ii) independent. Find $P(A \cup B)$.
23. A card is drawn from a pack of 52 cards. Find the probability of the following events: (i) A = king of hearts (ii) B = any king (iii) C = red card. Check whether A and B are mutually exclusive and whether A and C are independent.
24. An urn contains 5 white, 4 black, and 3 red balls. Two balls are drawn. Find the probability that both are (i) black (ii) not black (iii) of different colours.
25. A card is drawn from a pack of 52. Find the probability that it is (i) a king or a queen (ii) a king or a heart (iii) a face card or a spade.
26. In a group of students, 60% like cricket, 50% like football, and 20% like both. Find the probability that a student selected at random likes (i) cricket or football (ii) exactly one of the two games.
27. Out of 400 students in a college, 240 play cricket, 180 play football, and 100 play both. If a student is chosen at random, find the probability that he plays (i) at least one game (ii) only cricket (iii) neither cricket nor football.
28. A bag contains 3 white and 2 black balls. Two balls are drawn one after another without replacement. Find the probability that (i) both are white (ii) both are black (iii) first white and second black.
29. A bag contains 5 red and 4 green balls. Two balls are drawn one after another with replacement. Find the probability of getting (i) both red (ii) both green (iii) one red and one green.
30. The probability that a contractor will complete a road project in 9 months is 0.8. The probability that he will get an incentive bonus is 0.65. If the probability of both happening together is 0.5, find the probability that (i) he completes in 9 months but does not get the bonus (ii) he gets the bonus but does not complete in 9 months (iii) at least one of them happens.
31. A bag contains 4 red and 3 blue balls. Two balls are drawn one after another without replacement. Find the probability that the second ball is red if the first ball drawn was (i) red (ii) blue.
32. A die is thrown twice. Find the probability that the total is 9, given that the first throw resulted in 5.

38. A factory has two machines M1 and M2. M1 produces 40% of the items, M2 produces 60%. 4% of M1's output and 5% of M2's output are defective. If an item is chosen at random and is found defective, find the probability that it came from M1.
39. Three urns contain the following balls:
- Urn I: 3 white, 2 black
 - Urn II: 4 white, 1 black
 - Urn III: 2 white, 3 black
- One urn is chosen at random, and a ball is drawn. If it is white, find the probability that it came from Urn II.
40. A factory has 3 machines A, B, and C which produce 25%, 35%, and 40% of the total items. Their defective rates are 5%, 4%, and 2% respectively. If an item chosen at random is found defective, find the probability that it was produced by machine A.
41. A college admits students from three regions. Region I contributes 30% of students, Region II 45%, and Region III 25%. The percentages of scholarship holders are 10%, 8%, and 12% respectively. If a student is chosen at random and is a scholarship holder, find the probability that he belongs to Region III.
42. Two dice are thrown. Find the probability that the sum is (i) 8 (ii) divisible by 4 (iii) less than 5.
43. Three coins are tossed together. Find the probability of getting (i) all heads (ii) exactly two tails (iii) at least one head.
44. A die is rolled. Let $A = \text{"number is even,"}$ $B = \text{"number greater than 3."}$ Find (i) $P(A \cap B)$ (ii) $P(A \cup B)$ (iii) check whether A and B are independent.
45. A box contains 5 red, 4 green, and 6 blue balls. Two balls are drawn at random. Find the probability that they are of (i) same colour (ii) different colours.
46. Out of 600 families, 400 consume tea, 250 consume coffee, and 150 consume both. If one family is chosen at random, find the probability that it consumes (i) tea or coffee (ii) only tea (iii) neither tea nor coffee.
47. A purse contains 5 silver and 4 gold coins. Two coins are drawn at random. Find the probability of getting (i) both silver (ii) both gold (iii) one of each.
48. Find the Probability that a throw of an unbiased die results in (i) an odd number (ii) a multiple of three
49. One card is drawn from a well-shuffled pack of playing cards. Find the probability that one card drawn.
- (i) is a Heart
 - (ii) is a King
 - (iii) belongs to Red Suit
 - (iv) is a King or Heart
50. Find the Probability of getting a sum of 9 when two fair dice are thrown.
51. Find the Probability of drawing 3 white from a bag containing 6 white, 7 black & 3 red marbles.

47. Find the probability of drawing 5 marbles from a bag containing 7 white and 3 black marbles out of which
 (i) two are black (ii) at least two are black
48. A bag contains 5 black 2 red and 3 white marbles. Three marbles are drawn what is the probability that the marble drawn are
 (i) All are different colour (ii) Two are same colour
 (iii) All are same colour
49. Let A and B be events with $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$. Find (i) $P(A|B)$, (ii) $P(B|A)$, (iii) $P(\bar{A}|\bar{B})$ (iv) $P(\bar{B}|\bar{A})$.
50. The probability of occurrence of an event A is 0.7, the probability of non-occurrence of event B is 0.45 and the probability at least one of A and B not occurring is 0.6. Find the probability that at least one of A and B occurs.
 $P(A) = 0.7$, $P(\bar{B}) = 0.45$ and $P(\bar{A} \cup \bar{B}) = 0.6$
51. A fuse box contains 20 fuses of which 5 are defective. If 12 fuses are selected at random in succession without replacement, what is the probability that both the fuses are defective.
52. A certain problem is given to three students. The probabilities of their solving the problem are $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{5}$. Find the Probability that the problem is solved.
53. Ten pairs of shoes are in a closet. Four shoes are selected at random. Find the probability that there is atleast one pair among the four selected.
54. For two events A and B in a sample space S. Show that $P[(A \cap \bar{B}) \cup (B \cap \bar{A})] = P(A) + P(B) - P(A \cap B)$.
55. A six faced die is tossed. If the number shown is odd, what is the probability that it is a prime number?
56. A certain problem in mathematics is given to four students for solving. The probabilities of their solving the problem are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ respectively. Find the probability that the problem is solved.
57. A fair coin and a fair die are thrown. Find the probabilities of (i) head on the coin and number 6 on the die (ii) head on the coin and even number on the die.
58. In a city, 12% people read the news paper A, 25% people read the news paper B and 8% people read both news papers. Find the probability that randomly selected person reads at least one of these news papers.
59. The probability that a contractor will get plumbing contracts is $\frac{2}{3}$ probability that he will get an electrical contract is $\frac{5}{9}$. If the probability of getting at least one contract is $\frac{4}{5}$, what is the probability that he will get both?

80. The probabilities of a husband and a wife surviving for 20 more years are 0.8 and 0.9 respectively. Find the probability that after 20 years (i) both of them are alive (ii) at least one of them is alive.

81. X can hit a target 3 times in 5 shots; Y can hit 2 times in 5 shots; Z can hit 3 times in 4 shots. They fire a volley. What is the probability that 2 shots hit the target?

82. The chances that 4 students A, B, C, D solve a problem are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{4}$ respectively. If all of them

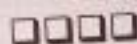
try to solve the problem, what is the probability that the problem is solved.

83. Let A and B be two events in a sample space such that $P(A) = 0.4$ and $P(B) = b$ and $P(A \cup B) = 0.7$. For what value of b (i) A and B are mutually exclusive (ii) A and B are independent.

84. Let A and B be two events in a sample space. If $P(A \cup B) = b/6$; $P(A \cap B) = 1/3$ and $P(\bar{B}) = 1/2$ find (i) $P(B)$ (ii) $P(A)$ (iii) Are $\bar{A}$ and $\bar{B}$ independent?

85. If A and B are two events and $P(B) \neq 1$ prove that

$$P(A|\bar{B}) = \frac{P(A) - P(A \cap B)}{1 - P(B)} \quad \text{Hence prove } P(A \cap B) \geq P(A) + P(B) - 1.$$



RANDOM VARIABLES

7.1 Random Variables

In probability and statistics, many events do not give a fixed result. The result is uncertain. A coin may show Head or Tail. A die may show any number from 1 to 6. Mathematics studies such uncertainty with the help of random variables. A **random variable** gives numbers to outcomes of a random experiment. This numeric form helps in calculation. It also helps in comparison and prediction.

Early study of probability focused only on events. Statements looked like "probability of getting 3 on a die" or "probability of getting a Head on a coin". Real life is richer and more complex. Daily rainfall changes from day to day. Marks in an exam differ from student to student. Share prices move up and down during market hours. A single event name is not enough for analysis. A random variable connects each outcome to a number. That number becomes the input for formulas and graphs and decisions.



Meaning : Random Variable

A random variable is a numerical quantity whose value depends on the outcome of a random experiment. It is a function or rule that assigns a number to every outcome in the sample space of the experiment.

- A random variable does not mean it is "variable at random." It is a well-defined rule.
- Its value changes depending on which outcome occurs in the experiment.
- It can take discrete values (countable like 0, 1, 2, 3, ...) or continuous values (any real number in a given interval).

Example

Random Variables

1. Coin Toss

- **Experiment:** Toss a coin once.
- **Sample space:** $S = \{H, T\}$.
- Define X as : $X=1$ if Head appears, $X=0$ if Tail appears.
- Here, X is a random variable.

2. Rolling a Die

- **Experiment:** Roll a die once.
- **Sample space:** $S = \{1, 2, 3, 4, 5, 6\}$.
- Define X as the number shown. Then X can take values 1, 2, 3, 4, 5, 6.

3. Two Tosses of a Coin

- **Experiment:** Toss two coins together.
- **Sample space:** $S = \{HH, HT, TH, TT\}$.
- Define X as the number of Heads. Then X can take values:
 - ★ $X=0$ for TT ,
 - ★ $X=1$ for HT or TH ,
 - ★ $X=2$ for HH .

Key Points:

- **Notation:** Random variables are typically denoted by capital letters, such as X , Y , or Z . For example, if we are conducting an experiment of rolling a fair six-sided die, we can define a random variable X to represent the number rolled on the die.
- **Range:** The range of a random variable is the set of all possible values that the variable can take on. For a discrete random variable, the range is a set of distinct values, while for a continuous random variable, the range is an interval or collection of intervals. It may be noted that different random variables may be associated with the same sample space S . The set of all real numbers of a random variable X is called the range of X .
- **Applications:** Random variables are used to model and analyze a wide range of random experiments including outcomes of games of chance, measurements in scientific experiments, and fluctuations in financial markets. They are fundamental to understanding and quantifying uncertainty in various fields.

Example Range of Random Variables

If a coin tossed twice, the sample space is

$$S = \{HH, HT, TH, TT\}.$$

X = Number heads that come up and
 Y = Number of tails that come up.

Sample Point	HH	HT	TH	TT
X	2	1	1	0

$\therefore$ Range of $X = \{2, 1, 0\}$

Sample Point	HH	HT	TH	TT
Y	0	1	1	2

$\therefore$ Range of $Y = \{0, 1, 2\}$

Random variables are often used to model real-world phenomena, such as the number of customers who enter a store in a given hour or the amount of rainfall in a particular region. By studying the properties of random variables, we can make predictions about the behavior of these phenomena and make informed decisions based on probability and statistics.

Some common properties of random variables include the mean, variance, and probability distribution. The mean of a random variable is the expected value of the variable, while the variance measures the spread of the variable's values around the mean. The probability distribution of a random variable describes the probabilities of each possible value of the variable.

Discrete and Continuous Random Variables

Random variables can be classified into two main types: **discrete random variables** and **continuous random variables**.

Discrete Random Variable: A discrete random variable is a random variable that can take only certain fixed values. These values are separate and distinct. They can be counted one by one. The values may be few (finite) or may continue without end (infinite), but they must still be countable. These values are often represented by integers or whole numbers. For example, the number of children in a family may be 0, 1, 2, 3, and so on. We can count each possibility. There will not be values like 1.5 children or 2.7 children. So, this is a discrete case. These values are often represented by integers or whole numbers.

Example 1 : Discrete Random Variables

1. Number on a Die :

When we roll a fair six-sided die once, the possible outcomes are the numbers on its faces.

- The random variable X can take values $\{1, 2, 3, 4, 5, 6\}$.
- These values are fixed and limited in number.
- We can easily count them one by one.

This is a discrete case because the set of possible values is finite and countable.

2. Number of Children in a Family

Suppose we take a random family and count the number of children in it.

- The random variable Y can take values $\{0, 1, 2, 3, \dots\}$.
- Here 0 means no children, 1 means one child, 2 means two children, and so on.
- There is no fractional value like 1.5 children or 2.3 children.
- Even if the maximum number is not known, the values are still countable one by one.

This is a discrete case because the possible values are whole numbers that can be listed in order.

3. Number of Calls Received in an Hour at a Call Center

Suppose we observe a call center for one hour and count how many calls it receives.

- The random variable Z can take values $\{0, 1, 2, 3, \dots\}$.
 - ★ If no call comes, $Z=0$.
 - ★ If 15 calls come, $Z=15$.
 - ★ If 200 calls come, $Z=200$.

- The values cannot be fractional, like 2.5 calls.

- Even though the maximum is not fixed, each value is still a whole number and can be counted.

This is also a discrete case because the values are integers that can be listed in sequence.

Example 2 : Understanding Discrete Random Variables**Step-by-Step Example: Tossing Two Coins****1. Experiment**

Toss two coins at the same time.

2. Sample Space

The Possible outcomes are :

$$S = \{HH, HT, TH, TT\}$$

where H = Head and T = Tail.

3. Define the Random Variable X

Let X represent the number of Heads obtained.

- If outcome is HH $\rightarrow$ number of Heads = 2 $\rightarrow$ so $X = 2$.
- If outcome is HT $\rightarrow$ number of Heads = 1 $\rightarrow$ so $X = 1$.
- If outcome is TH $\rightarrow$ number of Heads = 1 $\rightarrow$ so $X = 1$.
- If outcome is TT $\rightarrow$ number of Heads = 0 $\rightarrow$ so $X = 0$.

4. Possible Values of X

The random variable X can take values 0, 1, or 2.

These values are separate, distinct and countable.

2. Continuous Random Variable:

A continuous random variable is a random variable that can take any value inside a given range or interval. The values are not separate. They form a continuous line. The number of possible values is infinite and they cannot be listed one by one. These values are often represented by real numbers and can include decimal values.

For example, the height of a student can be 160.2 cm, 160.25 cm, 160.251 cm, and so on. Between 160 and 161, there are infinitely many possible heights. We cannot count them. This is why height is a continuous random variable.

Example 3: Continuous Random Variables**1. Height of Students in a Class**

Suppose we measure the height of students in a classroom.

- The random variable H represents height in centimeters.
- A student's height may be 160.5 cm, or 160.55 cm, or 160.555 cm or 160.5555551, and so on.
- Between 160 and 161 cm, there are infinitely many possible values.
- It is impossible to list all these values one by one.

This makes height a continuous random variable, because it can take any real value in a given interval, not just a fixed countable set.

2. Time Taken to Complete a Task

Suppose we record the time taken by a student to finish a mathematics problem.

- The random variable T represents time in seconds.
- The student may take 5.1 seconds, or 5.11 seconds, or 5.111 seconds, and so on.

- Between 5 and 6 seconds, there are infinitely many possible values.
- Time is usually measured with stopwatches or digital clocks, but in theory, it can be divided into smaller and smaller units without end.

Thus, time is a continuous random variable, since its values form an unbroken range and cannot be counted individually.

3. Weight of a Person

Suppose we measure the weight of a person.

- The random variable W represents weight in kilograms.
- A person's weight may be 55 kg, 55.2 kg, 55.25 kg, 55.257 kg, and so on.
- Between 55 and 56 kg, there are infinitely many possible values.
- Even though weighing machines round the values, in theory the weight can be any real number.

Therefore, weight is a continuous random variable, because it can take infinitely many values in a range.

Common Feature in All Three Cases

- The values are not separate and countable like in discrete cases.
- They form a continuous interval of real numbers.
- Probability is not assigned to a single exact value (for example, probability of height = 160.5 cm). Instead, probability is given for a range of values (for example, probability that height is between 160 cm and 165 cm). This is done using a Probability Density Function (PDF).

Example 4 : Understanding Continuous Random Variables

Step-by-Step Example: Continuous Random Variable

1. Experiment

Measure the time taken by a student to solve one mathematics problem.

2. Sample Space

The time can be any real number within a range. Suppose the Student usually takes between 2 minutes and 10 minutes.

So, the sample space is:

$$S = \{x : 2 \leq x \leq 10, x \in \mathbb{R}\}$$

This means all real numbers between 2 and 10 are possible outcomes.

3. Define the Random Variable X

Let X represent the time taken (in minutes).

- If the student finishes in 2.3 minutes $\rightarrow X = 2.3$
- If the student finishes in 5.75 minutes $\rightarrow X = 5.75$
- If the student finishes in 9.125 minutes $\rightarrow X = 9.125$

4. Possible Values of X

The random variable X can take any value in the interval $[2, 10]$

There are infinitely many such values - 2.01, 2.001, 2.0001 and so on - without limit. Here, X is a continuous random variable because its values are not separate and countable. Instead, they form a continuous range of real numbers.

7.3 Probability Distribution of a Random Variable

When a random variable is defined, we only know the set of values it can take. For example, in tossing a die, the random variable X may take values 1, 2, 3, 4, 5, or 6. But simply knowing these values is not enough in probability theory. We also need to know how likely each value is. This is known as probability. Probability is not just about listing possible outcomes; it is about measuring the chance of those outcomes.

To describe these chances in a systematic way, we use the concept of a probability distribution. A probability distribution provides a complete picture of how the total probability (which is always 1) is divided among the possible values of the random variable.

Thus, a probability distribution is like a "map of chances" that connects real-world experiments with mathematical analysis. It allows us to answer important questions such as:

- What is the most likely value?
- What is the chance of getting a particular value or a range of values?
- How are the outcomes spread out?

Because of this, probability distribution is considered one of the most important foundations of probability and statistics.

The concept of a probability distribution is similar to a frequency distribution.

- A frequency distribution tells us how the total frequency (number of observations) is divided among different values of a variable.
- A probability distribution, on the other hand, tells us how the total probability (which is always 1) is distributed among the possible values of a random variable.

Just like a frequency distribution can be shown in a tabular form, a probability distribution is also often represented as a table, where each possible value of the random variable is listed along with its probability.



Meaning : Probability Distribution

A probability distribution is a way of assigning probabilities to the values of a random variable so that the total probability is always equal to 1.

In simple words, a probability distribution is a set of rules that shows the possible values of a random variable and tells how much probability is given to each value.

- For discrete random variables, we use the Probability Mass Function (PMF).
- For continuous random variables, we use the Probability Density Function (PDF).

7.3.1

Discrete Probability Distribution

In many random experiments, the outcomes can be counted one by one. For example, the number of heads that appears on a die, the number of Heads in coin tosses, or the number of students present in a class. In all these cases, the random variable can take separate and countable values. Such random variables are called **discrete random variables**, and their probabilities are described using a discrete

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probability distribution. A discrete probability distribution shows how probabilities are assigned to each value of a discrete random variable.

- The random variable takes only specific values (finite or countably infinite).
- Each value has a probability linked with it.
- The sum of probabilities for all possible values is always equal to 1.

Since probabilities need to be assigned to each specific value of a discrete random variable, we use a special function called the Probability Mass Function (PMF). This function acts as the formal way to describe the distribution. It tells us, for every possible value of the random variable, what its probability is.

7.3.1.1 Probability Mass Function (PMF)

When we deal with a discrete random variable, we must assign probabilities to each of its possible values. To do this in a formal way, we use the **Probability Mass Function (PMF)**. The PMF is a mathematical function that tells us how the total probability of 1 is distributed across the values of a discrete random variable. It describes exactly how likely each value of the variable is. In simple words, the PMF is the rulebook of probabilities for a discrete random variable.

The probability distribution for a discrete random variable is often represented as a **probability mass function (p.m.f.)**, which assigns a probability to each possible outcome. The p.m.f. has the following properties:

1. The probability of each outcome is non-negative.
2. The sum of the probabilities of all possible outcomes is equal to 1.



Definition : Probability Mass Function (PMF)

If X is a discrete random variable, then its probability mass function is written as:

$$P(x_i) = P(X = x_i) \text{ for } i = 1, 2, \dots$$

This means that for every possible value x_i , the PMF gives the probability that the random variable X will take the value x_i .

For a valid probability mass function, the following conditions must always be satisfied:

1. $P(x_i) \geq 0$ for all values of i : The probability of each value must be zero or positive.

Example : This means that probabilities cannot be negative. For example, when rolling a die, the probability of getting a 3 cannot be -2.

2. $\sum P(x_i) = 1$: The sum of probabilities over all possible values must be 1.

Example : In a die, $P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 6 \times 1/6 = 1$

Example 1: Tossing a Coin

Step 1. Experiment

A fair coin is tossed once.

Step 2. Sample Space

The possible outcomes are:

$$S = \{H, T\}$$

Where H = Head and T = Tail.

Step 3. Define Random Variable X

Let X represent the outcome of the toss in numerical form:

- $X = 1$ if the coin shows Head.
- $X = 0$ if the coin shows Tail.

So,

$$X \in \{0, 1\}$$

Step 4. Assign Probabilities

Since the coin is fair, each outcome has probability $1/2$.

$$P(X = 0) = \frac{1}{2}, P(X = 1) = \frac{1}{2}$$

Step 5. Probability Distribution Table

X (Outcome)	0 (Tail)	1 (Head)
P(X)	0.5	0.5

Step 6. Verification

- Probabilities are non-negative.
- The sum is $0.5 + 0.5 = 1$.

Thus, it is a valid probability distribution.

This shows how the total probability of 1 is divided among the possible values of the random variable X. In a frequency distribution, the total number of observations is divided among categories.

Example 2: Rolling a Fair Die

Step 1. Experiment

A fair six-sided die is rolled once.

Step 2. Sample Space

The possible outcomes are the numbers on the die faces:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Step 3. Define Random Variable X

Let X represent the number that appears on the die.

So,

$$X \in \{1, 2, 3, 4, 5, 6\}$$

Step 4. Assign Probabilities

Since the die is fair, each outcome is equally likely. Therefore,

$$P(X = 1) = \frac{1}{6}, P(X = 2) = \frac{1}{6}, P(X = 3) = \frac{1}{6}, P(X = 4) = \frac{1}{6}, P(X = 5) = \frac{1}{6}, P(X = 6) = \frac{1}{6}$$

Step 5. Probability Distribution Table

X (Number on Die)	1	2	3	4	5	6
P(X)	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

Step 6. Verification

- All probabilities are non-negative.
- The total probability is:

$$\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$$

Thus, this is a valid probability distribution.

The example shows how a discrete probability distribution assigns equal probabilities to each outcome when the experiment is fair.

Example 3: Tossing Two Coins

Step 1. Experiment

Two fair coins are tossed at the same time.

Step 2. Sample Space

The possible outcomes are:

$$S = \{HH, HT, TH, TT\}$$

Step 3. Define Random Variable X

Let X represent the number of Heads obtained.

- For TT $\rightarrow X = 0$
- For HT $\rightarrow X = 1$
- For TH $\rightarrow X = 1$
- For HH $\rightarrow X = 2$

So,

$$X \in \{0, 1, 2\}$$

Step 4. Assign Probabilities

Each outcome in the sample space is equally likely, with probability $1/4$.

Now, collect outcomes for each value of X:

- $P(X = 0) = P(\{TT\}) = 1/4$
- $P(X = 1) = P(\{HT, TH\}) = 1/4 + 1/4 = 1/2$
- $P(X = 2) = P(\{HH\}) = 1/4$

Step 5. Probability Distribution Table

X (Number of Heads)	0	1	2
P(X)	1/4	1/2	1/4

Step 6. Verification

- Probabilities are non-negative.
- The sum is:

$$\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$$

Thus, this is a valid probability distribution. In this case, the distribution is not uniform because all values of X do not have the same probability. The most likely outcome is getting exactly 1 Head, with probability $1/2$.

Example 4: Number of Children in a Family**Step 1. Experiment**

Select a random family and note the number of children in that family.

Step 2. Sample Space

The possible outcomes are non-negative integers:

$$S = \{0, 1, 2, 3, 4, 5, \dots\}$$

Here, 0 means no children, 1 means one child, 2 means two children, and so on.

Step 3. Define Random Variable X

Let X represent the number of children in a family.

So,

$$X \in \{0, 1, 2, 3, \dots\}$$

Step 4. Assign Probabilities

The probabilities depend on real-life data. For example, suppose a survey in a city shows the following distribution:

- $P(X = 0) = 0.10$ (10% of families have no children)
- $P(X = 1) = 0.20$ (20% of families have one child)
- $P(X = 2) = 0.40$ (40% of families have two children)
- $P(X = 3) = 0.20$ (20% of families have three children)
- $P(X = 4) = 0.08$ (8% of families have four children)
- $P(X \geq 5) = 0.02$ (2% of families have five or more children)

Step 5. Probability Distribution Table

X (Number of Children)	0	1	2	3	4	5 or more
P(X)	0.10	0.20	0.40	0.20	0.08	0.02

Step 6. Verification

- All probabilities are non-negative.
- The total probability is:

$$0.10 + 0.20 + 0.40 + 0.20 + 0.08 + 0.02 = 1$$

Thus, it is a valid probability distribution.

- ✓ This example shows that even though the random variable X can, in theory, take infinitely many values ($0, 1, 2, 3, \dots$), the probabilities can still be assigned in a way that the total equals 1. This is a discrete probability distribution.

7.3.2**Discrete Probability Distribution**

In many random experiments, the outcomes cannot be counted one by one. Instead, the random variable can take any value within a given interval. For example, the height of students in a class, the time taken to complete a task, or the weight of a person. In all these cases, the random variables take uncountably infinite values. Such random variables are called **continuous random variables** and their probabilities are described using a **continuous probability distribution**.

continuous probability distribution shows how probabilities are spread over ranges of values of a continuous random variable.

- The random variable can take any value within an interval.

- The probability of a single exact value is always zero.

- The probability of the random variable lying within a certain interval is found using the **Probability Density Function (PDF)**.

- The total probability over the whole range of values is always equal to 1.

Since probabilities cannot be directly assigned to each individual value of a continuous random variable, we use a special function called the **Probability Density Function (PDF)**. This function gives the distribution of probabilities over intervals of values. It helps us calculate the probability that the random variable will lie between two given points.

7.1.1 Probability Density Function (PDF)

When we deal with a continuous random variable, the probability of any exact value is always zero. For example, the probability that a student's exact height is exactly 160.000 cm is zero. Instead, we talk about the probability that the height is between two values, such as between 160 cm and 165 cm.

To calculate such probabilities, we use a function called the **Probability Density Function (PDF)**. The PDF is a mathematical function that describes how the total probability of 1 is spread across the values of a continuous random variable. Probabilities are calculated as the area under the curve of the PDF, between the required limits. In simple words, the PDF is the curve of probabilities for a continuous random variable.

Properties of Probability Density Function (PDF)

The probability density function (PDF) has the following important properties:

1. Non-Negativity

$$f(x) \geq 0 \text{ for all values of } x$$

This means that the density (like probability) cannot be negative.

Example: When measuring time, the density of probability at any point must be zero or positive.

2. Total Probability = 1

The total area under the PDF curve, over the entire range of the random variable, must be equal to 1.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Example: If time taken to finish a job is always between 0 and 10 minutes then the total probability from 0 to 10 minutes must equal 1.

1. Probability over an Interval

The probability that the random variable lies between two values a and b is given by:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Example: If X is the time taken to solve a problem, the probability that it takes between 2 and 5 minutes is the area under the PDF curve from $x = 2$ to $x = 5$.



Definition : Probability Density Function (PDF)

If X is a continuous random variable, then its probability density function is written as $f(x)$ such that

1. $f(x) \geq 0$ for all x .
2. $\int_{-\infty}^{\infty} f(x) dx = 1$.
3. For any two real numbers a and b :

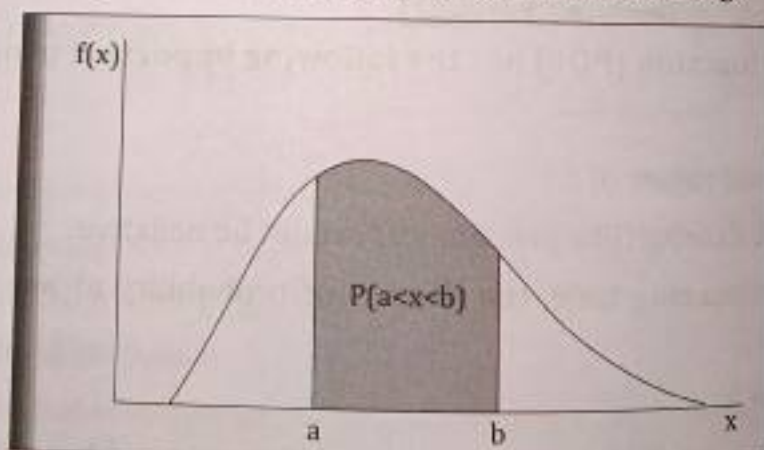
$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

This means that the PDF does not give the probability of a single value directly, but it helps us calculate probability over an interval.



Understanding Probability Density Function (PDF)

A random variable X is said to be a continuous random variable if it can take any value in any interval. The range of X may be finite or infinite. A probability density function is used to describe the probability distribution of a continuous random variable, which can take on any value within a certain range. The p.d.f. assigns a probability density to each possible value of the random variable, and the area under the curve of the p.d.f. over a certain range represents the probability of the random variable falling within that range.



The above diagram illustrates a continuous probability distribution function $f(x)$ plotted against x . The area under the curve between two points a and b is shaded, indicating the probability $P(a < x < b)$, which is the likelihood that the random variable x falls within that range. The graph illustrates a fundamental concept in probability and statistics, where the total area under the entire curve represents the probability of all possible outcomes and is equal to 1. The shaded region specifically represents the cumulative probability between two points a and b , and is used to find the likelihood of the variable falling within that specific interval.

Probability of

For a continuous random variable, the probability of falling within a certain interval. This is calculated by finding the area under the curve. Suppose we have a continuous random variable X and we want to find the probability that it falls within a certain interval (a, b) . We can use the probability density function $f(x)$ to calculate this probability.

Example 1:

Step 1. Experiment

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Step 2. Probability

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Step 3. Verification

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Probability of an Interval (A)

For a continuous random variable X with a probability density function $f(x)$, the probability of X falling within a specific interval $A = (a, b)$ can be calculated as the area under the p.d.f. curve over that interval. This is represented as $P(A)$ or $P(a < X < b)$.

Suppose we have a continuous random variable X representing the time it takes for a certain process to be completed. If we want to find the probability that the process takes between 5 and 10 minutes, we can use the p.d.f. $f(x)$ to calculate $P(5 < X < 10)$ by finding the area under the curve of $f(x)$ over the interval $(5, 10)$.

Example 1: a. Uniform Distribution of Time Taken

Step 1. Experiment

Suppose a student takes between 0 and 10 minutes to complete a problem.

The time can be any real value in this interval.

So, the random variable X = time taken (in minutes).

Step 2. Probability Density Function (PDF)

Since all times between 0 and 10 minutes are equally likely, this is a uniform distribution.

The PDF is defined as:

$$f(x) = \begin{cases} \frac{1}{10}, & 0 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

Here, the value $\frac{1}{10}$ is obtained by using the formula of a uniform distribution:

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

Where $a = 0$ and $b = 10$. Since the interval length is $10 - 0 = 10$, the constant height of the PDF is $1/10$.

This means the "height" of the probability density curve is constant between 0 and 10.

Step 3. Verification of PDF Properties

1. Non - Negativity:

$$f(x) = 1/10 \geq 0. \text{ Condition satisfied.}$$

2. Total Probability = 1:

$$\int_0^{10} \frac{1}{10} dx = \frac{1}{10} \times (10 - 0) = 1$$

Condition satisfied.

Step 4. Find Required Probability

Suppose we want the probability that the student takes between 2 and 5 minutes.

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{10} dx = \frac{1}{10} \times (5 - 2) = \frac{3}{10} = 0.3$$

So, the probability is 0.3 (or 30%)

Step 5. Interpretation

This result means that there is a 30% probability that the student will take between 2 and 3 minutes to complete the problem.

The probability for any exact value (like exactly 3 minutes) is 0, but for an interval it is possible.

This example clearly shows how a Probability Density Function (PDF) works:

- The total area under the curve = 1.
- The probability of an interval = area under the curve for that interval.
- A single exact value has probability 0.

Example 2: b. Uniform Distribution of Bus travel Time**Step 1. Experiment**

Suppose a bus takes between 15 and 25 minutes to travel from one stop to the next, the travel time can be any real value in this interval.

So, the random variable X = travel time (in minutes).

Step 2. Probability Density Function (PDF)

Since all travel times between 15 and 25 minutes are equally likely, this is a uniform distribution. The PDF is defined as:

$$f(x) = \begin{cases} \frac{1}{25-15} = \frac{1}{10}, & 15 \leq x \leq 25 \\ 0, & \text{otherwise} \end{cases}$$

Here, the value $\frac{1}{10}$ is obtained using the formula of a uniform distribution:

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

where $a = 15$ and $b = 25$, since the interval length is $25 - 15 = 10$, the constant height of the PDF is $\frac{1}{10}$.

This means the probability density curve is flat between 15 and 25 minutes.

Step 3. Verification of PDF Properties

1. Non - Negativity:

$$f(x) = 1/10 \geq 0. \text{ Condition satisfied}$$

2. Total Probability = 1.

$$\int_{15}^{25} \frac{1}{10} dx = \frac{1}{10} \times (25 - 15) = 1$$

Condition satisfied.

Step 4. Find Required Probability

Suppose we want the probability that the bus takes between 18 and 22 minutes.

$$\begin{aligned} P(18 \leq X \leq 22) &= \int_{18}^{22} \frac{1}{10} dx \\ &= \frac{1}{10} \times (22 - 18) = \frac{4}{10} = 0.4 \end{aligned}$$

So, the probability is 0.4 (or 40%)

Example 5 Interpretation

This result means there is a 40% probability that the bus will take between 18 and 22 minutes to complete the journey.
The probability for any exact value (like exactly 20 minutes) is 0, but for an interval it is positive.

1.4 Expectation (or) Mean of a Random Variable

The expectation or expected value of a random variable is the long-run average value that we expect to obtain if an experiment is repeated many times. It is also called the **mean** of the random variable.

- For a discrete random variable, the expectation is a weighted average of all possible values, where the weights are their probabilities.
- For a continuous random variable, the expectation is found by integrating the product of the variable and its probability density function.

In simple words, expectation tells us the "average outcome" of a random experiment if it were repeated indefinitely.

Formula to Find the Expectation or Mean of a Random Variable

Formulas

1. For a Discrete Random Variable

If X is a discrete random variable with possible values $x_1, x_2, \dots, x_n$ and probabilities $p(x_1), p(x_2), \dots, p(x_n)$, then the expectation is:

$$E[X] = \sum_{i=1}^n x_i \cdot p(x_i)$$

This means we multiply each value of X by its probability and then add them all.

2. For a Continuous Random Variable

If X is a continuous random variable with probability density function $f(x)$, then the expectation is:

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

This means we multiply x by the density at x , and then integrate across all possible values.

Example 1: a. Discrete Random Variable (Rolling a Die)

Step 1. Experiment

A fair die is rolled once.

Step 2. Random Variable

Let X = number that appears.

So, $X \in \{1, 2, 3, 4, 5, 6\}$.

Step 3. Probabilities

Since the die is fair:

$$P(X = k) = \frac{1}{6}, \quad k = 1, 2, 3, 4, 5, 6$$

Step 4. Mean (Expectation) Formula

$$\mu = E[X] = \sum_{k=1}^6 k \cdot P(X=k)$$

Step 5. Calculation

$$\mu = \left(1 \cdot \frac{1}{6}\right) + \left(2 \cdot \frac{1}{6}\right) + \left(3 \cdot \frac{1}{6}\right) + \left(4 \cdot \frac{1}{6}\right) + \left(5 \cdot \frac{1}{6}\right) + \left(6 \cdot \frac{1}{6}\right)$$

$$\mu = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

Step 6. Interpretation

The Mean (μ) = 3.5,

This means if we roll the die many times, the long-run average outcome will approach 3.5.

Example 2: b. Continuous Random Variable (Uniform Distribution)**Step 1. Experiment**

A student takes between 0 and 10 minutes to complete a problem, uniformly distributed.

Step 2. PDF

$$f(x) = \frac{1}{10}, \quad 0 \leq x \leq 10$$

Step 3. Mean (Expectation) Formula

$$\mu = E[X] = \int_0^{10} x \cdot f(x) dx$$

Step 4. Calculation

$$\mu = \int_0^{10} x \cdot \frac{1}{10} dx = \frac{1}{10} \int_0^{10} x dx$$

$$\frac{1}{10} \left[\frac{x^2}{2} \right]_0^{10} = \frac{1}{10} \times \frac{100}{2} = \frac{50}{10} = 5$$

Step 5. Interpretation

The mean (μ) = 5 minutes.

This means, on average, the student takes 5 minutes to complete the problem.

7.5 Variance of a Random Variable

The variance measures the spread or variability of a distribution. It quantifies how much the values of the random variable X deviate from the mean (μ) or Expectation (E). The variance of a random variable is a measure of how much the values of the random variable deviate from the mean (expected value).

- It shows the spread or dispersion of the values around the mean.
- If the variance is small, the values of the random variable are close to the mean.
- If the variance is large, the values are spread out widely.
- The square root of variance is called the standard deviation.

In simple words, variance tells us how far the outcomes are from the average value, on the whole.

1. For a Discrete Random Variable

If X is a discrete random variable with possible values $x_1, x_2, \dots, x_n$, probabilities $P(x_1), P(x_2), \dots, P(x_n)$, and mean $\mu = E[X]$, then the variance is:

$$V(X) = \sum (x_i - \mu)^2 \cdot P(x_i)$$

This definition says: subtract the mean from each value, square it, multiply by probability, and then add them up.

Alternative Formula

The Variance can also be expressed as:

$$V(X) = \sum x_i^2 P(x_i) - \left[\sum x_i P(x_i) \right]^2$$

or

$$V(X) = E[X^2] - (E[X])^2$$

This form is often easier to use in calculations.

2. For a Continuous Random Variable

If X is a continuous random variable with probability density function $f(x)$ and mean $\mu = E[X]$, then the variance is:

$$V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx$$

Alternative Formula

$$V(X) = \int_{-\infty}^{\infty} x^2 f(x) dx - \left(\int_{-\infty}^{\infty} x f(x) dx \right)^2$$

or

$$V(X) = E[X^2] - (E[X])^2$$

Example 1: a. (Discrete): Variance of a fair die - first approach only, no table

Ans. Find

$$V(X) = \sum (x_i - \mu)^2 \cdot P(x_i)$$

for a fair die, without using the shortcut and without a table.

Step 1. Random variable and probabilities

A fair die is rolled once, Let X be the number shown.

So $P(X = k) = \frac{1}{6}$ for $k = 1, 2, 3, 4, 5, 6$.

Step 2. Mean (μ)

$$\mu = E[X] = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

Step 3. Variance by the definition

$$V(X) = \sum_{k=1}^6 (k - \mu)^2 \cdot \frac{1}{6} = \frac{1}{6} [(1-3.5)^2 + (2-3.5)^2 + (3-3.5)^2 + (4-3.5)^2 + (5-3.5)^2 + (6-3.5)^2]$$

Compute the squared deviations directly:

$$(1-3.5)^2 = (-2.5)^2 = 6.25, \quad (2-3.5)^2 = (-1.5)^2 = 2.25,$$

$$(3-3.5)^2 = (-0.5)^2 = 0.25, \quad (4-3.5)^2 = (0.5)^2 = 0.25,$$

$$(5-3.5)^2 = (1.5)^2 = 2.25, \quad (6-3.5)^2 = (2.5)^2 = 6.25.$$

Add Them

$$6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25 = 17.5.$$

Therefore,

$$V(X) = \frac{1}{6} \times 17.5 = \frac{17.5}{6} = \frac{35}{12} \approx 2.9167.$$

Result

$$V(X) = \frac{35}{12} \approx 2.9167$$

$$\text{Standard deviation} = \sqrt{35/12} = 1.71.$$

Example 2: b. (Discrete): Variance of Number of Heads in Two Tosses**Step 1. Experiment**

Two fair coins are tossed.

Step 2. Random Variable and Probabilities

Let X = number of Heads obtained.

Possible values: $X \in \{0, 1, 2\}$.

$$\bullet P(X=0) = \frac{1}{4} \text{ (TT)}$$

$$\bullet P(X=1) = \frac{2}{4} = \frac{1}{2} \text{ (HT or TH)}$$

$$\bullet P(X=2) = \frac{1}{4} \text{ (HH)}$$

Step 3. Mean ($\mu = E[X]$)

$$\mu = \sum x_i P(x_i) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 0 + 0.5 + 0.5 = 1$$

So, $\mu = 1$

Step 4. Variance (first approach)

$$V(X) = \sum (x_i - \mu)^2 P(x_i)$$

$$= (0-1)^2 \cdot \frac{1}{4} + (1-1)^2 \cdot \frac{1}{2} + (2-1)^2 \cdot \frac{1}{4}$$

$$= (1) \cdot \frac{1}{4} + (0) \cdot \frac{1}{2} + (1) \cdot \frac{1}{4}$$

$$= \frac{1}{4} + 0 + \frac{1}{4} = \frac{1}{2}$$

Answer

$$V(X) = 0.5$$

Standard Deviation = $\sqrt{0.5} = 0.707$.**Example 3: c. Continuous Random Variable (Uniform Distribution)****Step 1. Experiment**

A student takes between 0 and 10 minutes to complete a problem, uniformly distributed.

Step 2. PDF

$$f(x) = \frac{1}{10}, \quad 0 \leq x \leq 10$$

Step 3. Mean (Expectation) Formula

$$\mu = E[X] = \int_0^{10} x \cdot f(x) dx$$

Step 4. Calculation

$$\mu = \int_0^{10} x \cdot \frac{1}{10} dx = \frac{1}{10} \int_0^{10} x dx$$

$$\frac{1}{10} \left[\frac{x^2}{2} \right]_0^{10} = \frac{1}{10} \times \frac{100}{2} = \frac{50}{10} = 5$$

Step 5. InterpretationThe mean (μ) = 5 minutes.

This means, on average, the student takes 5 minutes to complete the problem.

Step 6. Variance (Definition)

Formula

$$V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Substitute, Here $f(x) = \frac{1}{10}$ for $0 \leq x \leq 10$ and $\mu = 5$. Hence

$$V(X) = \int_0^{10} (x - 5)^2 \cdot \frac{1}{10} dx.$$

Step 7. Calculation

Expand and integrate term by term.

$$V(X) = \frac{1}{10} \int_0^{10} (x^2 - 10x + 25) dx = \frac{1}{10} \left[\frac{x^3}{3} - 5x^2 + 25x \right]_0^{10}$$

Evaluate the bracket at the limits.

$$\left(\frac{1000}{3} - 500 + 250\right) - 0 = \frac{250}{3}$$

Multiply by $\frac{1}{10}$.

$$V(X) = \frac{1}{10} \cdot \frac{250}{3} = \frac{25}{3} = 8.33$$

Step 8. Interpretation

The Variance is $\frac{25}{3}$, Hence the spread of times around the mean 5 minutes is fairly wide interval $[0, 10]$. The standard deviation is $\sqrt{25/3} \approx 2.89$ minutes.

7.6 Problems and Solutions

Illustration 1

A coin is tossed twice. A random variable X represent the number of heads turnings up. Find the discrete probability distribution for X . Also find its mean and variance.

Solution:

$$S = \{HH, HT, TH, TT\}$$

Random variable

$$X = \{2, 1, 1, 0\}$$

$$P\{HH\} = \frac{1}{4}, P\{HT\} = \frac{1}{4}, P\{TH\} = \frac{1}{4}, P\{TT\} = \frac{1}{4},$$

$$P(X = 0, \text{no head}) = P(TT) = \frac{1}{4}$$

$$P(X = 1, 1 \text{ head}) = P(HT \cup TH) = P(HT) + P(TH) = \frac{1}{2}$$

$$P(X = 2, 2 \text{ heads}) = P(HH) = \frac{1}{4}$$

The discrete probability distribution for X is as follows

$X = x_i$	0	1	2
$P(x_i)$	$1/4$	$1/2$	$1/4$

From table we observe that $P(x_i) > 0$ and $\sum P(x_i) = 1$

$$\text{Mean} = \mu = \sum x_i P(x_i) = \left[0 \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{2} + 2 \times \frac{1}{4}\right] = 1$$

$$\begin{aligned}\text{Variance } = V &= \sum (x_i - \mu)^2 P(x_i) \\ &= (0-1)^2 \frac{1}{4} + (1-1)^2 \frac{1}{2} + (2-1)^2 \frac{1}{4} = \frac{1}{2}\end{aligned}$$

$$\text{Mean } (\mu) = 1, \text{ Variance } (V) = \frac{1}{2}.$$

Illustration 2

A pair of dice be thrown and the random variable X is the sum of the numbers that appears on the two dice. Find mean and standard deviation.

Solution:

$$S = \{(x, y) \text{ where } x = 1, 2, 3, \dots, 6, y = 1, 2, 3, \dots, 6\}$$

$$S = \{(1, 1) (1, 2) (1, 3) \dots (6, 4), (6, 5) (6, 6)\}$$

Number of elements in $S = 36$

The random variable X is the sum of the numbers on the two dice are $\{2, 3, 4, \dots, 10, 11, 12\}$

$$P(X=2) = P\{(1, 1)\} = \frac{1}{36}$$

$$P(X=3) = P\{(1, 2) (2, 1)\} = \frac{2}{36}$$

$$P(X=4) = P\{(1, 3) (2, 2) (3, 1)\} = \frac{3}{36}$$

$$P(X=5) = P\{(1, 4) (2, 3) (3, 2) (4, 1)\} = \frac{4}{36}$$

$$P(X=6) = P\{(1, 5) (2, 4) (3, 3) (4, 2) (5, 1)\} = \frac{5}{36}$$

$$P(X=7) = P\{(1, 6) (2, 5) (3, 4) (4, 3) (5, 2) (6, 1)\} = \frac{6}{36}$$

$$P(X=8) = P\{(2, 6) (3, 5) (4, 4) (5, 3) (6, 2)\} = \frac{5}{36}$$

$$P(X=9) = P\{(3, 6) (4, 5) (5, 4) (6, 3)\} = \frac{4}{36}$$

$$P(X=10) = P\{(4, 6) (5, 5) (6, 4)\} = \frac{3}{36}$$

$$P(X=11) = P\{(5, 6) (6, 5)\} = \frac{2}{36}$$

$$P(X=12) = P\{(6, 6)\} = \frac{1}{36}$$

The probability distribution X is

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$P(X)$ or P_i	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$P(x_i) > 0 \text{ and } \sum P(x_i) = 1$$

$$\text{Mean} = \mu = \sum x_i P(x_i) = 2 \times \frac{1}{36} + 3 \times \frac{2}{36} + 4 \times \frac{3}{36} + 5 \times \frac{4}{36} + 6 \times \frac{5}{36} + 7 \times \frac{6}{36} + 8 \times \frac{5}{36} + 9 \times \frac{4}{36} + 10 \times \frac{3}{36} + 11 \times \frac{2}{36} + 12 \times \frac{1}{36} = 7$$

$$\text{Variance} = V = \sum (x_i - \mu)^2 P(x_i)$$

$$V = (2-7)^2 \times \frac{1}{36} + (3-7)^2 \times \frac{2}{36} + (4-7)^2 \times \frac{3}{36} + (5-7)^2 \times \frac{4}{36} + (6-7)^2 \times \frac{5}{36} + (7-7)^2 \times \frac{6}{36} + (8-7)^2 \times \frac{5}{36} + (9-7)^2 \times \frac{4}{36} + (10-7)^2 \times \frac{3}{36} + (11-7)^2 \times \frac{2}{36} + (12-7)^2 \times \frac{1}{36}$$

$$\therefore \text{Mean} = 7, \text{S.D.} = \sqrt{V} = \sqrt{\frac{110}{36}}$$

Illustration 3

Find the variance and mean of the number obtained on a throw of an unbiased dice.

Solution:

The sample space of the experiment $S = \{1, 2, 3, 4, 5, 6\}$

Let X denote the number obtained on the throw. Then X is a random variable which take values 1, 2, 3, 4, 5 or 6

$$\therefore P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

The probability distribution,

X_i	1	2	3	4	5	6
$P(x_i)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$P(x_i) > 0 \text{ and } \sum P(x_i) = 1$$

$$\text{Mean} = \mu = \sum x_i P(x_i) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5 \times \frac{1}{6} + 6 \times \frac{1}{6} = \frac{21}{6} = \frac{7}{2}$$

$$\text{Variance} = V = \sum (x_i - \mu)^2 P(x_i)$$

$$= \left(1 - \frac{7}{2}\right)^2 \times \frac{1}{6} + \left(2 - \frac{7}{2}\right)^2 \times \frac{1}{6} + \left(3 - \frac{7}{2}\right)^2 \times \frac{1}{6} + \left(4 - \frac{7}{2}\right)^2 \times \frac{1}{6} + \left(5 - \frac{7}{2}\right)^2 \times \frac{1}{6} + \left(6 - \frac{7}{2}\right)^2 \times \frac{1}{6}$$

$$= \left(\frac{-5}{2}\right)^2 \times \frac{1}{6} + \left(\frac{-3}{2}\right)^2 \times \frac{1}{6} + \left(\frac{-1}{2}\right)^2 \times \frac{1}{6} + \left(\frac{1}{2}\right)^2 \times \frac{1}{6} + \left(\frac{3}{2}\right)^2 \times \frac{1}{6} + \left(\frac{5}{2}\right)^2 \times \frac{1}{6}$$

Mean (μ)

Illustration - The Probability

X
P(x)

1. Find P(X)
2. What will

Solution:

1. If X is a r

i.e., K +

⇒ 49

∴ P(X)

P(X)

P(X)

2. P(X ≤ 2)

∴ M

Illustration

A random va

x
P(x)

1. Find th
2. Evalua
3. P(0 < x)

$$\frac{25}{4} \times \frac{1}{6} + \frac{9}{4} \times \frac{1}{6} + \frac{1}{4} \times \frac{1}{6} + \frac{1}{4} \times \frac{1}{6} + \frac{9}{4} \times \frac{1}{6} + \frac{25}{4} \times \frac{1}{6} = \frac{35}{12}$$

$$\text{Mean } (\mu) = \frac{21}{6} \quad \text{Variance } (V) = \frac{35}{12}$$

Illustration 4

Probability density function of a variate X is

X	0	1	2	3	4	5	6
P(x)	K	3K	5K	7K	9K	11K	13K

1. Find $P(X < 4)$, $P(X \geq 5)$, $P(3 < X \leq 6)$

2. What will be the minimum value of K so that $P(\leq 2) > 3$.

Solution:

1. If X is a random variable then $\sum_{i=0}^6 P(x_i) = 1$

$$\text{i.e. } K + 3K + 5K + 7K + 9K + 11K + 13K = 1$$

$$\Rightarrow 49K = 1 \quad \Rightarrow \quad K = \frac{1}{49}$$

$$P(X < 4) = K + 3K + 5K + 7K = 16K = 16 \times \frac{1}{49} = \frac{16}{49}$$

$$P(X \geq 5) = 11K + 13K = 24K = \frac{24}{49}$$

$$P(3 < X \leq 6) = 9K + 11K + 13K = 33K = \frac{33}{49}$$

$$2. P(X \leq 2) = K + 3K + 5K = 9K > 3$$

$$K > \frac{1}{3}$$

$$\therefore \text{Minimum value of } K = \frac{1}{3}$$

Illustration 5

Random variable X has the following Probability function:

x	0	1	2	3	4	5	6	7
P(x)	0	K	2K	3K	K	K ²	2K ²	7K ² +2K

1. Find the value of K

2. Evaluate $P(X < 6)$, $P(X \geq 6)$

3. $P(0 < X < 5)$.

Solution:

1. If
- X
- is a random variable, then

$$\sum_{i=0}^7 P(x_i) = 1$$

$$0 + K + 2K + 3K + K + K^2 + 2K^2 + 7K^2 + 2K = 1$$

$$10K^2 + 9K - 1 = 0$$

$$(10 - K)(K + 1) = 0$$

$$K = \frac{1}{10}$$

$$\begin{aligned} 2. \quad P(X < 6) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) \\ &= 0 + K + 2K + 3K + K^2 + K^2 \\ &= 2K^2 + 6K \end{aligned}$$

$$= 2\left(\frac{1}{10}\right)^2 + 6\left(\frac{1}{10}\right) = \frac{2}{100} + \frac{6}{10} = \frac{62}{100} = \frac{31}{50}$$

$$P(X \geq 6) = P(X = 6) + P(X = 7)$$

$$= 2K^2 + 7K^2 + 2K = 9K^2 + 2K$$

$$= 9\left(\frac{1}{10}\right)^2 + 2 \times \frac{1}{10} = \frac{9}{100} + \frac{2}{10} = \frac{29}{100}$$

$$3. \quad P(0 < x < 5) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$= K + 2K + 3K + K^2 = K^2 + 6K$$

$$= \left(\frac{1}{10}\right)^2 + \frac{6}{10} = \frac{1}{100} + \frac{60}{100} = \frac{61}{100}$$

Illustration 6

A die is tossed thrice. A success is 'getting 1 or 6' on a toss. Find the mean and variance of number of successes.

Solution:

$$\text{Probability of a success} = \frac{2}{6} = \frac{1}{3}$$

$$\text{Probability of failures} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\therefore \text{Probability of no. success} = \text{Probability of all 3 failures} = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$$

Probability of one successes and 2 failures = $3C_1 \times \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$

Probability of two successes and one failure = $3C_2 \times \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$

Probability of three successes = $\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$

∴ Probability distribution of X is

x_i	0	1	2	3
$P(x_i)$	$\frac{8}{27}$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{1}{27}$

Mean = $\mu = \sum x_i P(x_i) = 0 \times \frac{8}{27} + 1 \times \frac{4}{9} + 2 \times \frac{2}{9} + 3 \times \frac{1}{27}$

$\mu = 1$

Variance = $V = \sum (x_i - \mu)^2 P(x_i) = (0-1)^2 \times \frac{8}{27} + (1-1)^2 \times \frac{4}{9} + (2-1)^2 \times \frac{2}{9} + (3-1)^2 \times \frac{1}{27}$
 $= \frac{8}{27} + \frac{4}{9} + \frac{2}{9} + \frac{4}{27} = \frac{8+12+6+4}{27} = \frac{30}{27} = \frac{10}{9} = \frac{2}{3}$

Illustration 7

Find the probability that in 10 throws of a fair die a score which is a multiple of 3 will be obtained in at least 8 of the throws.

Solution:

Here success is a score which is a multiple of 3 i.e., 3 or 6.

∴ $p(3 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$

The probability of r successes in 10 throws is given by

$P(r) = {}^{10}C_r \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^{10-r}$

∴ $P(\text{at least 8 successes}) = P(8) + P(9) + P(10)$

$= {}^{10}C_8 \left(\frac{1}{3}\right)^8 \left(\frac{2}{3}\right)^2 + {}^{10}C_9 \left(\frac{1}{3}\right)^9 \left(\frac{2}{3}\right)^1 + {}^{10}C_{10} \left(\frac{1}{3}\right)^{10}$
 $= \frac{1}{3^{10}} [45 \times 4 + 10 \times 2 + 1] = \frac{201}{3^{10}}$

Illustration 8

Four balls are to be drawn without replacement from a box containing 8 red and 4 white balls. If X denotes the number of red balls drawn, find the probability distribution of X .

Solution:

Since 4 balls have to be drawn, therefore, X can take the values 0, 1, 2, 3, 4.

$${}^{12}C_4 = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495$$

$$P(X=0) = P(\text{no red ball}) = P(4 \text{ white balls}) = \frac{{}^4C_4}{{}^{12}C_4} = \frac{1}{\frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1}} = \frac{1}{495}$$

$$P(X=1) = P(1 \text{ red ball and 3 white balls}) = \frac{{}^8C_1 \times {}^4C_3}{{}^{12}C_4} = \frac{8 \times \frac{4 \times 3 \times 2}{3 \times 2 \times 1}}{495} = \frac{32}{495}$$

$$P(X=2) = P(2 \text{ red balls and 2 white balls}) = \frac{{}^8C_2 \times {}^4C_2}{{}^{12}C_4} = \frac{\frac{8 \times 7}{2 \times 1} \times \frac{4 \times 3}{2 \times 1}}{495} = \frac{168}{495}$$

$$P(X=3) = P(3 \text{ red balls and 1 white ball}) = \frac{{}^8C_3 \times {}^4C_1}{{}^{12}C_4} = \frac{\frac{8 \times 7 \times 6}{3 \times 2 \times 1} \times 4}{495} = \frac{224}{495}$$

$$P(X=4) = P(4 \text{ red balls}) = \frac{{}^8C_4}{{}^{12}C_4} = \frac{\frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1}}{495} = \frac{70}{495}$$

Thus the following is the required probability distribution of X

X	0	1	2	3	4
$P(X)$	$\frac{1}{495}$	$\frac{32}{495}$	$\frac{168}{495}$	$\frac{224}{495}$	$\frac{70}{495}$

Illustration 9

Determine variance and standard deviation of the number of heads in three tosses of a coin.

Solution:

Let X denote the number of heads tossed. So, X can take the values 0, 1, 2, 3. When a coin is tossed three times, we get

Sample space $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$$P(X=0) = P(\text{no head}) = P(TTT) = \frac{1}{8}$$

$$P(X=1) = P(\text{one head}) = P(HTT, THT, TTH) = \frac{3}{8}$$

$$P(X=2) = P(\text{two heads}) = P(HHT, HTH, THH) = \frac{3}{8}$$

$$P(X=3) = P(\text{three heads}) = P(HHH) = \frac{1}{8}$$

the probability distribution of X is:

X	0	1	2	3
P(X)	—	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$X = \sum x_i^2 p_i - \mu^2, \quad \dots(1)$$

where μ is the mean of X given by

$$\mu = \sum x_i p_i = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = \frac{3}{2} \quad \dots(2)$$

$$\sum x_i^2 p_i = 0^2 \times \frac{1}{8} + 1^2 \times \frac{3}{8} + 2^2 \times \frac{3}{8} + 3^2 \times \frac{1}{8} = 3 \quad \dots(3)$$

From (1), (2) and (3), we get

$$\sigma^2 = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$$

$$\text{Standard Deviation} = \sqrt{\sigma^2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

Illustration 10

A random variable has the following probability distribution

Values of X	0	1	2	3	4	5	6	7	8
P(X)	a	3a	5a	7a	9a	11a	13a	15a	17a

1. Determine the value of a

2. Find:

(i) $P(X < 3)$

(ii) $P(X \leq 3)$

(iii) $P(X > 7)$

(iv) $P(2 \leq X \leq 5)$

(v) $P(2 < X < 5)$

3. Find the cumulative distribution function of X.

Solution:

1. Since p_i is the probability mass function of discrete random variable X ,
We have:

$$\therefore a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$$

$$81a = 1$$

$$a = 1/81$$

2. (i) $P(x < 3) = P(x=0) + P(x=1) + P(x=2)$

$$= a + 3a + 5a = 9a$$

$$= 9\left(\frac{1}{81}\right) = \frac{1}{9}$$

- (ii) $P(x \leq 3) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$

$$= a + 3a + 5a + 7a = 16a = \frac{16}{81}$$

- (iii) $P(x > 7) = P(x=8) = 17a = \frac{17}{81}$

- (iv) $P(2 \leq x \leq 5) = P(x=2) + P(x=3) + P(x=4) + P(x=5)$

$$= 5a + 7a + 9a + 11a$$

$$= 32a = \frac{32}{81}$$

- (v) $P(2 < x < 5) = P(x=3) + P(x=4)$

$$= 7a + 9a$$

$$= 16a = \frac{16}{81}$$

3. The distribution function is as follows:

$X=x$	0	1	2	3	4	5	6	7
$F(x) = P(X \leq x)$	a	$4a$	$9a$	$16a$	$25a$	$36a$	$49a$	$64a$
(or) $F(x)$	$\frac{1}{81}$	$\frac{4}{81}$	$\frac{9}{81}$	$\frac{16}{81}$	$\frac{25}{81}$	$\frac{36}{81}$	$\frac{49}{81}$	$\frac{64}{81}$

Illustration 11

An urn contains 6 red and 4 white balls. Three balls are drawn at random. Obtain the probability distribution of the number of white balls drawn.

Solution:

The total number of balls in the urn is 10. Let X denote the number of white balls drawn. If three balls are drawn, the random variable takes the value $X = 0, 1, 2, 3$

probability of getting white balls from the urn containing 10 balls (red and white) with the following

probability are

$$P(\text{no white, 3 red balls}) = \frac{{}^6C_3 \cdot {}^4C_0}{{}^{10}C_3} = \frac{1 \times 120}{720} = \frac{5}{30}$$

$$P(\text{1 white, 2 red}) = \frac{{}^4C_1 \cdot {}^6C_2}{{}^{10}C_3} = \frac{15}{30}$$

$$P(\text{2 white, 1 red}) = \frac{{}^4C_2 \cdot {}^6C_1}{{}^{10}C_3} = \frac{9}{30}$$

$$P(\text{3 white, no red}) = \frac{{}^4C_3 \cdot {}^6C_0}{{}^{10}C_3} = \frac{1}{30}$$

the probability of X is given by

C	0	1	2	3
$P(X=x)$	$\frac{5}{30}$	$\frac{15}{30}$	$\frac{9}{30}$	$\frac{1}{30}$

Illustration 12

Find the expected value of x, where x represents the outcome when a die is thrown.

Solution:

Each of the outcome (i.e., number) 1, 2, 3, 4, 5 and 6 occurs with probability $1/6$. Thus the probability distribution of X will be

x	1	2	3	4	5	6
$P(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

the expected value of X is

$$E(X) = \sum x_i p_i = x_1 p_1 + x_2 p_2 + x_3 p_3 + x_4 p_4 + x_5 p_5 + x_6 p_6$$

$$E(X) = \left[1 \times \frac{1}{6}\right] + \left[2 \times \frac{1}{6}\right] + \left[3 \times \frac{1}{6}\right] + \left[4 \times \frac{1}{6}\right] + \left[5 \times \frac{1}{6}\right] + \left[6 \times \frac{1}{6}\right] = \frac{7}{2}$$

$$E(X) = 3.5$$

Illustration 13

An urn containing 7 white and 3 red balls. Two balls are drawn together at random from the urn. Find the expected number of white balls drawn.

Solution:

In the urn containing 7 white and 3 red balls, two balls can be drawn in ${}^{10}C_2$ ways. Let X denote the number of white balls in the draw. X can take the values 0, 1 and 2.

8
81a
$\frac{81}{81}$

probability

The probability distribution of X is obtained as follows:

$P(0)$ = Probability that neither of two balls is white.

Probability that both balls drawn are red.

$$\frac{{}^3C_2}{{}^{10}C_2} = \frac{3 \times 2}{10 \times 9} = \frac{1}{15}$$

$P(1)$ = Probability of getting 1 white and 1 red ball.

$$= \frac{{}^7C_1 \times {}^3C_1}{{}^{10}C_2} = \frac{7 \times 3 \times 2}{10 \times 9} = \frac{7}{15}$$

$P(2)$ = Probability of getting two white balls

$$= \frac{{}^7C_2}{{}^{10}C_2} = \frac{7 \times 6}{10 \times 9} = \frac{7}{15}$$

Hence expected number of white balls drawn is

$$E(x) = \sum x_i P(x_i) = \left[0 \times \frac{1}{15}\right] + \left[1 \times \frac{7}{15}\right] + \left[2 \times \frac{7}{15}\right]$$

$$= \frac{7}{5} = 1.4$$

Illustration 14

A dealer in television sets estimates from his past experience the probabilities of his selling television sets in a day is given below. Find the expected number of sales in a day.

Number of TV sold in a day	0	1	2	3	4	5	6
Probability	0.02	0.10	0.21	0.32	0.20	0.09	0.06

Solution:

We observe that the number of television sets sold in a day is a random variable which can assume the values 0, 1, 2, 3, 4, 5, 6 with the respective probabilities given in the table.

Now the expectation of $x = E(X) = \sum x_i p_i$

$$= x_1 p_1 + x_2 p_2 + x_3 p_3 + x_4 p_4 + x_5 p_5 + x_6 p_6$$

$$= (0)(0.02) + (1)(0.10) + (2)(0.21) + (3)(0.32) + (4)(0.20) + (5)(0.09) + (6)(0.06)$$

$$E(X) = 3.09$$

The expected number of sales per day is 3.

Illustration 15

Let x be a discrete random variable with the following probability distribution

X	-3	6	9
$P(X=x)$	$1/6$	$1/2$	$1/3$

Find the mean and variance

$$E(X) = \sum x_i P_i$$

$$= (-3)\left(\frac{1}{6}\right) + (6)\left(\frac{1}{2}\right) + 9\left(\frac{1}{3}\right) = \left(\frac{11}{2}\right)$$

$$E(X^2) = \sum x_i^2 P_i$$

$$= (-3)^2\left(\frac{1}{6}\right) + (6)^2\left(\frac{1}{2}\right) + (9)^2\left(\frac{1}{3}\right) = \left(\frac{93}{2}\right)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \left[\frac{93}{2}\right] - \left[\frac{11}{2}\right]^2 = \left[\frac{93}{2}\right] - \left[\frac{121}{4}\right]$$

$$= \frac{186 - 121}{4} = \frac{65}{4}$$

Review Questions

Short Answer Type Questions

1. Define a random variable with one example.
2. Differentiate between discrete and continuous random variables with examples.
3. Write the probability distribution of tossing a coin once.
4. State the properties of a probability mass function (PMF).
5. State the properties of a probability density function (PDF).
6. Define expectation (mean) of a random variable.
7. Define variance of a random variable.
8. Write the formula for variance in terms of expectation.
9. Give one real-life example of a discrete random variable and one of a continuous random variable.
10. Write the range of random variable X = number of heads in tossing 2 coins.
11. If a die is rolled once, what is the probability distribution of X = number shown?
12. Write the conditions for a valid probability distribution.
13. If $P(X=0)=0.4$, $P(X=1)=0.3$, $P(X=2)=0.3$, verify whether it is a probability distribution.
14. What is the expectation of rolling a fair die?
15. Write the probability distribution of the number of heads in a single coin toss.
16. Define Probability Mass Function (PMF).
17. State two properties of a PMF with example.

es of his selling
day.

5	6
0.09	0.06

variable which can
be table.

(0.06)

on

18. Define Probability Density Function (PDF).
19. State two properties of a PDF.
20. Write one difference between PMF and PDF.
21. If X is the outcome of rolling a die, write its PMF.
22. If X = number of heads in tossing one coin, write its PMF.
23. What is the probability of getting exactly 3 minutes in a PDF of uniform distribution between 0 and 10?
24. If a random variable is continuous, why is $P(X = a) = 0$ in PDF?
25. Write the formula for probability of interval $(a < X < b)$ in terms of PDF.

Long Answer Type Questions

1. Define a random variable. Explain with examples: (a) Toss of a coin (b) Roll of a die (c) Toss of two coins.
2. Explain the difference between discrete and continuous random variables with three examples for each.
3. Define range of a random variable. Illustrate with the case of tossing two coins.
4. A coin is tossed twice. Let X = number of heads. Write the probability distribution of X . Find its mean and variance.
5. A die is rolled once. Write the probability distribution of X = number shown. Find mean and variance.
6. A die is rolled twice. Let X = sum of the numbers obtained. Write the probability distribution of X . Find mean and standard deviation.
7. A random variable X has the following distribution:

X	0	1	2	3	4
$P(X)$	K	$2K$	$3K$	$4K$	$5K$

Find (i) the value of K (ii) $P(X < 3)$ (iii) Mean of X .

8. Define PDF. State its properties with one example.
9. The time taken by a student to solve a problem is uniformly distributed between 0 and 10 minutes. Find the probability that the student takes (i) less than 4 minutes (ii) between 3 and 7 minutes. Also find the mean time.
10. A bus travel time between two stops is uniformly distributed between 15 and 25 minutes. Find (i) the probability that the bus takes more than 20 minutes (ii) the probability that the bus takes between 18 and 22 minutes.
11. Define expectation of a random variable. Derive the formula for a discrete case.
12. A fair die is rolled. Find the expectation of the outcome.

13. An urn contains 7 white and 3 red balls. Two balls are drawn together. Let X = number of white balls. Find the expected number of white balls.
14. A shopkeeper sells 0, 1, 2, or 3 pens with probabilities 0.2, 0.3, 0.4, 0.1 respectively. Find the expected number of pens sold.
15. Define variance of a random variable. Derive the alternative formula $\text{Var}(X) = E(X^2) - [E(X)]^2$.
16. A die is rolled once. Find the variance of the outcome.
17. Two coins are tossed. Let X = number of heads. Find the mean and variance.
18. Three coins are tossed. Find the variance and standard deviation of number of heads.
19. A fair die is tossed thrice. A success is getting 1 or 6. Find the mean and variance of number of successes.
20. A random variable X has the following distribution:

X	0	1	2	3	4	5
$P(X)$	K	$2K$	$3K$	$4K$	$5K$	$7K$

- (i) Find K . (ii) Find $P(X < 3)$ (iii) Find mean and variance.

21. The daily sales of a TV dealer follow the distribution:

No. of TVs Sold	0	1	2	3	4	5	6
Probability	0.02	0.08	0.20	0.30	0.25	0.10	0.05

Find the expected number of TVs sold per day.

22. An urn has 6 red and 4 white balls. Three balls are drawn without replacement. Write the probability distribution of the number of white balls. Find mean.
23. A random variable X has the distribution:

X	0	1	2	3	4	5	6
$P(X)$	K	$2K$	$3K$	$4K$	$5K$	$6K$	$7K$

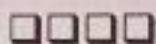
Find K , mean, and variance.

24. Define PMF. Explain its properties with examples.
25. A coin is tossed twice. Let X = number of heads. Write the PMF of X . Verify the conditions of PMF.
26. A fair die is rolled once. Let X = number shown. Write its PMF. Find mean and variance.
27. An urn contains 3 red and 2 white balls. Two balls are drawn at random. Let X = number of red balls. Write the PMF of X .
28. A die is rolled twice. Let X = sum of the numbers obtained. Write the PMF of X . Find mean.
29. A random variable X has the following distribution:

X	0	1	2	3
$P(X)$	K	$2K$	$3K$	$4K$

Find the value of K . Write the PMF and verify its conditions.

30. Define PDF. Explain its properties with examples.
31. The time taken by a student to complete a problem is uniformly distributed between 0 and 5 minutes. Write the PDF. Find $P(2 < X < 5)$.
32. The travel time of a bus is uniformly distributed between 15 and 25 minutes. Write the PDF. Find (i) $P(X < 20)$ (ii) $P(18 \leq X \leq 22)$.
33. Suppose the height of students in a class is normally distributed with mean 160 cm and standard deviation 10 cm. Write the PDF of $X = \text{height}$. Find the probability that height is between 150 cm and 170 cm.
34. If X is uniformly distributed over $(0, 1)$, write its PDF. Find $P(0.2 < X < 0.6)$.



STANDARD PROBABILITY DISTRIBUTIONS

8.1 Binomial Distribution

The binomial distribution is also known as Bernoulli distribution is associated with the name of a Swiss mathematician James Bernoulli (1654 – 1755). The binomial distribution describes the likelihood of achieving a specific number of successes in a set of trials, where each trial has only two possible outcomes: success or failure. For instance, when flipping a coin 10 times, the binomial distribution can provide the probability of obtaining a particular number of heads.

This distribution is characterized by two key parameters: the probability of success in a single trial (denoted as p) and the number of trials (denoted as n). For example, in the case of flipping a fair coin, the probability of success (getting heads) is 0.5, and the number of trials (flips) is 10.



Definition : Binomial Distribution

The binomial distribution is a probability distribution that is used to model scenarios where there are a fixed number of independent trials, each with a binary outcome (success or failure), and where the probability of success is constant across all trials.

If p is probability of success and q is the Probability of failure, the Probability of x success out of n trials is given by,

$$P(x) = {}^nC_x \cdot p^x \cdot q^{n-x}$$

(or)

$$P(x) = {}^nC_x \cdot p^x \cdot (1-p)^{n-x}$$

Given that the binomial distribution deals with situations where each trial has only two possible outcomes (success or failure), the probability of failure is complementary to the probability of success. Therefore, if the probability of success in a single trial is denoted by " p ," then the probability of failure in a single trial is represented by " $1 - p$."

For example, if the probability of success (e.g., getting heads in a coin toss) is 0.7, then the probability of failure (getting tails in the same coin toss) would be $1 - 0.7 = 0.3$.

We form the following Probability distribution of $[x, P(x)]$ where $x = 0, 1, 2, \dots, n$.

x	0	1	2	...	n
$P(x)$	q^n	${}^nC_1 \cdot q^{n-1} p$	${}^nC_2 \cdot q^{n-2} p^2$	...	p^n

Example: Areas Where Binomial Distribution is Applied

The Binomial Distribution can be used to model the following scenarios:

The binomial distribution can be used to model scenarios involving a fixed number of trials, each with a binary outcome (success or failure), and where the trials are independent of each other. Here are some examples of scenarios that can be modeled using the binomial distribution:

1. Number of successful attempts in a series of independent trials, such as the number of heads in a series of coin flips.
2. Number of defective items in a sample of products from a production line.
3. Number of customers who make a purchase in a series of sales interactions.
4. Number of correct answers in a multiple-choice test with a fixed number of questions.
5. Number of patients who respond positively to a new treatment in a clinical trial.
6. Number of voters who support a particular candidate in a series of independent opinion polls.

In each of these examples, the binomial distribution can be used to model the probability of obtaining a specific number of successes in a fixed number of independent trials, given the probability of success on each trial. This makes the binomial distribution a valuable tool for analyzing and making predictions about binary outcomes (success or failure) in a wide range of practical scenarios.

The Mean, Variance and Standard Deviation of the Binomial Distribution

The mean and variance of the binomial distribution are important measures of central tendency and spread, respectively. The mean of the binomial distribution is given by np , and the variance is given by $np(1-p)$.

$$\text{Mean } (\mu) = np$$

$$\text{Variance } (V) = npq \text{ (or) } np(1-p)$$

$$\text{S.D. } (\sigma) = \sqrt{V} = \sqrt{npq} \text{ (or) } \sqrt{np(1-p)}$$

The binomial distribution finds applications in various fields such as finance, engineering, and biology. For instance, it can be utilized to model the number of defective products in a manufacturing process, the number of successful trades in a financial portfolio, or the number of offspring with a specific genetic trait in a population.



Applications of Binomial Distribution

Applications of the binomial distribution are widespread, and it is commonly used in various fields such as finance, engineering, biology, and quality control. Here are some elaborations on its applications:

1. **Quality Control:** In manufacturing and production processes, the binomial distribution is used to model the number of defective items in a sample of products. By understanding the probability of a certain number of defects occurring in a fixed number of items, companies can make informed decisions about quality control measures and process improvements.
2. **Biology and Genetics:** In genetics, the binomial distribution is used to model the inheritance of genetic traits. For example, it can be applied to predict the likelihood of specific genetic combinations occurring in offspring based on the probability of inheriting certain traits from parents.

3. Finance and portfolio management: The binomial distribution can be used to model the probability of a stock price increasing or decreasing over a period of time, which is a key component of financial risk management.

4. Survey results: The binomial distribution can be used to model the probability of a certain number of responses in a survey, which is useful for analyzing survey data.

5. Clinical trials: The binomial distribution can be used to model the probability of a certain number of patients responding positively to a new treatment, which is useful for analyzing clinical trial results.

The mean and variance of the binomial distribution are important measures of central tendency and spread of the distribution, which are useful for analyzing and making predictions about binary outcomes in various scenarios.

Problems

Illustration

Suppose a coin is tossed 4 times.

Solution

We can use the binomial distribution to find the probability of getting a certain number of heads or tails.

$$P(x) = \binom{n}{x} p^x q^{n-x}$$

Given $n = 4$, $p = 0.5$, and $q = 0.5$, we can find the probability of getting a certain number of heads or tails.

Using the formula, we can find the probability of getting 0, 1, 2, 3, or 4 heads out of 4 coin tosses.

Therefore, the probability of getting 0 heads is $\frac{1}{16}$, the probability of getting 1 head is $\frac{4}{16}$, the probability of getting 2 heads is $\frac{6}{16}$, the probability of getting 3 heads is $\frac{4}{16}$, and the probability of getting 4 heads is $\frac{1}{16}$. To find the probability of getting a certain number of tails, we can use the same formula with $p = 0.5$ and $q = 0.5$.

3. **Finance and Risk Management:** In finance, the binomial distribution is utilized in risk assessment and portfolio management. It helps in analyzing the probability of various investment outcomes and is a key component of option pricing models, such as the binomial options pricing model.

4. **Survey Sampling:** When conducting surveys or opinion polls, the binomial distribution can be used to model the probability of obtaining a certain number of positive responses (successes) out of a fixed number of survey participants. This is essential for understanding the reliability and accuracy of survey results.

5. **Clinical Trials:** In medical research and clinical trials, the binomial distribution is employed to analyze the outcomes of treatment interventions. It helps in assessing the probability of a certain number of patients responding positively to a treatment under investigation.

The mean and variance of the binomial distribution provide valuable insights into the central tendency and spread of the distribution, allowing for the quantification of uncertainty and risk in various real-world scenarios.

Problems on Binomial Distribution

Illustration 1

Suppose a coin is flipped 4 times. What is the probability of getting exactly 2 heads?

Solution

We can use the binomial distribution formula to calculate the probability of x success out of n trials:

$$P(x) = {}^nC_x \cdot p^x \cdot q^{(n-x)}$$

Given:

$$n = 4 \text{ (number of coin flips)}$$

$$p = 0.5 \text{ (probability of getting heads)}$$

$$q = 1 - p = 0.5 \text{ (probability of getting tails)}$$

$$x = 2 \text{ (number of heads)}$$

Using the formula for the binomial distribution, we can find the probability of getting exactly 2 heads out of 4 coin flips:

$$P(x = 2) = {}^nC_2 \cdot p^2 \cdot q^{(4-2)}$$

$${}^nC_2 = {}^4C_2 = 4! / (2! \cdot 2!) = 6$$

$$P(x = 2) = 6 \cdot 0.5^2 \cdot 0.5^2 = 0.375$$

Therefore, the probability of getting exactly 2 heads out of 4 coin flips is 0.375. To find the mean, variance, and standard deviation of the binomial distribution, we can use the formulas:

$$\text{Mean: } \mu = np$$

$$\text{Variance: } \sigma^2 = npq$$

$$\text{Standard deviation: } \sigma = \sqrt{npq}$$

For this example: $n = 4$, $p = 0.5$, $q = 0.5$

Mean: $\mu = 4 \times 0.5 = 2$

Variance: $\sigma^2 = 4 \times 0.5 \times 0.5 = 1$

Standard deviation: $\sigma = \sqrt{4 \times 0.5 \times 0.5} = 1$

Therefore, for the scenario of flipping a coin 4 times, the mean number of heads, variance, and standard deviation of the binomial distribution are 2, 1, and 1, respectively.

Illustration 2

Eight coins are tossed simultaneously. Find the probability of getting at least six heads.

Solution

Here number of trials, $n = 8$, p denotes the probability of getting a head.

$$p = \frac{1}{2} \text{ and } q = \frac{1}{2}$$

If the random variable X denotes the number of heads, then the probability of a success in n trials is given by

$$P(X = x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$${}^8C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{8-x} = {}^8C_x \left(\frac{1}{2}\right)^8 = \frac{1}{2^8} {}^8C_x$$

Probability of getting at least six heads is given by

$$P(X \geq 6) = P(X = 6) + P(X = 7) + P(X = 8)$$

$$= \frac{1}{2^8} {}^8C_6 + \frac{1}{2^8} {}^8C_7 + \frac{1}{2^8} {}^8C_8$$

$$= \frac{1}{2^8} [28 + 8 + 1] = \frac{37}{256}$$

Illustration 3

Ten coins are tossed simultaneously. Find the probability of getting (i) at least seven heads (ii) exactly seven heads (iii) at most seven heads.

Solution

p = Probability of getting a head = $1/2$

q = Probability of not getting a head = $1/2$

The probability of getting x heads throwing 10 coins simultaneously is given by

$$P(X = x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$= {}^{10}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x} = \frac{1}{2^{10}} {}^{10}C_x$$

(i) Probability of getting at least seven heads

$$P(X \geq 7) = P(x=7) + P(x=8) + P(x=9) + P(x=10)$$

$$= \frac{1}{2^{10}} [{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}]$$

$$= \frac{1}{1024} [120 + 45 + 10 + 1] = \frac{176}{1024}$$

(ii) Probability of getting exactly 7 heads 11

$$P(X=7) = \frac{10}{2^{10}} {}^{10}C_7 = \frac{1}{2^{10}} (120) = \frac{120}{1024}$$

(iii) Probability of getting atmost 7 heads

$$P(X \leq 7) = 1 - P(X > 7)$$

$$= 1 - \{P(X=8) + P(X=9) + P(X=10)\}$$

$$= 1 - \frac{1}{2^{10}} ({}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10})$$

$$= 1 - \frac{1}{2^{10}} [45 + 10 + 1] = 1 - \frac{56}{1024} = \frac{968}{1024}$$

Illustration 4

20 wrist watches in a box of 100 are defective. If 10 watches are selected at random, find the probability that (i) 10 are defective (ii) 10 are good (iii) at least one watch is defective (iv) at most 3 are defective.

Solution

20 out of 100 wrist watches are defective

Probability of defective wrist watch

$$p = \frac{20}{100} = \frac{1}{5} \quad q = 1 - p = \frac{4}{5}$$

Since 10 watches are selected at random, $n = 10$

$$P(X=x) = {}^nC_x p^x q^{n-x}, x=0,1,2,\dots,10$$

$$= {}^{10}C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{10-x}$$

(i) Probability of selecting 10 defective watches

$$P(X=10) = {}^{10}C_{10} \left(\frac{1}{5}\right)^{10} \left(\frac{4}{5}\right)^0 = 1 \cdot \left(\frac{1}{5}\right)^{10} \cdot 1 = \frac{1}{5^{10}}$$

(ii) Probability of selecting 10 good watches (i.e. no defective)

$$P(X=0) = {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} = 1 \cdot 1 \cdot \left(\frac{4}{5}\right)^{10} = \left(\frac{4}{5}\right)^{10}$$

(iii) Probability of selecting at least one defective watch

$$\begin{aligned} P(x \leq 1) &= 1 - P(x < 1) \\ &= 1 - {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} \\ &= 1 - \left(\frac{4}{5}\right)^{10} \end{aligned}$$

(iv) Probability of selecting at most 3 defective watches

$$\begin{aligned} P(x \leq 3) &= P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) \\ &= {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} + {}^{10}C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^9 + {}^{10}C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^8 + {}^{10}C_3 \left(\frac{1}{5}\right)^3 \left(\frac{4}{5}\right)^7 \\ &= 1 \cdot 1 \cdot \left(\frac{4}{5}\right)^{10} + 10 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^9 + \frac{10 \cdot 9}{1 \cdot 2} \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^8 + \frac{10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3} \left(\frac{1}{5}\right)^3 \left(\frac{4}{5}\right)^7 \\ &= 1(0.107) + 10(0.026) + 45(0.0062) + 120(0.0016) = 0.859(\text{approx.}) \end{aligned}$$

Illustration 5

Suppose a basketball player has a free throw percentage of 80%. If the player takes 10 free throws, what is the probability that they make exactly 7 of them?

Solution

We can use the binomial distribution formula to calculate the probability of 'x' success out of 'n' trials

$$P(x) = {}^nC_x \cdot p^x \cdot q^{(n-x)}$$

Given:

$n = 10$ (number of free throws)

$p = 0.8$ (probability of making a free throw)

$q = 1 - p = 0.2$ (probability of missing a free throw)

$x = 7$ (number of successful free throws)

Using the formula for the binomial distribution, we can find the probability of making exactly 7 out of 10 free throws:

$$P(x = 7) = {}^nC_7 \cdot p^7 \cdot q^{(10-7)}$$

$${}^nC_7 = {}^{10}C_7 = 10! / (7! \cdot 3!) = 120$$

$$P(x = 7) = 120 \cdot 0.8^7 \cdot 0.2^3 \approx 0.2013$$

Therefore, the probability of making exactly 7 out of 10 free throws with an 80% success rate is approximately 0.2013.

To find the mean, variance, and standard deviation of the binomial distribution, we can use the formulas:

$$\text{Mean: } \mu = np$$

$$\text{Variance: } \sigma^2 = npq$$

$$\text{Standard deviation: } \sigma = \sqrt{npq}$$

For this example: $n = 10$, $p = 0.8$, $q = 0.2$

$$\text{Mean: } \mu = 10 \times 0.8 = 8$$

$$\text{Variance: } \sigma^2 = 10 \times 0.8 \times 0.2 = 1.6$$

$$\text{Standard deviation: } \sigma = \sqrt{10 \times 0.8 \times 0.2} \approx 1.26$$

Therefore, for the scenario of a basketball player with an 80% free throw percentage taking 10 free throws, the mean number of successful free throws, variance, and standard deviation of the binomial distribution are approximately 8, 1.6, and 1.26, respectively.

Illustration 6

10 eggs are drawn successively with replacement from a lot containing 10% defective eggs. Find the Probability that there is atleast one defective egg.

Solution

Let X denote the number of defective eggs in the 10 eggs drawn. Since the drawing is done with replacement the trials are Bernoulli trials. X has the binomial distribution with $n = 10$ and

$$p = \frac{10}{100} = \frac{1}{10}$$

$$\therefore q = 1 - p = \frac{9}{10}$$

$$P(\text{atleast one defective egg}) = P(X \geq 1)$$

$$= 1 - P(X = 0)$$

$$= 1 - {}^{10}C_0 \left(\frac{9}{10}\right)^{10} = 1 - \frac{9^{10}}{10^{10}}$$

Illustration 7

Fit the binomial Probability distribution which has mean 2 and variance $4/3$.

Solution

$$\text{Given mean} = np = 2, \text{ Variance} = npq = \frac{4}{3}$$

$$\therefore 2q = \frac{4}{3} \text{ or } q = \frac{2}{3}$$

$$\therefore p = 1 - q = \frac{1}{3}$$

$$\text{Since } np = 2, \text{ we have } n\left(\frac{1}{3}\right) = 2$$

$$\therefore n = 6$$

The binomial Probability function $P(x) = {}^nC_x p^x q^{n-x}$

$$P(x) = {}^6C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x}$$

The Probability distribution is

x	0	1	2	3	4	5
P(x)	$\left(\frac{2}{3}\right)^5$	${}^5C_1 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^4$	${}^5C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^3$	${}^5C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^2$	${}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)$	${}^5C_5 \left(\frac{1}{3}\right)^5$

Illustration 8

When a coin is tossed 4 times, find the Probability of getting

(1) Exactly one heads

(2) atmost 3 heads

(3) atleast 2 heads.

Solution

$$p = \frac{1}{2} = 0.5, q = 1 - p = 1 - 0.5 = 0.5, n = 4$$

The binomial Probability distribution is

$$P(x) = {}^nC_x p^x q^{n-x}$$

$$(1) P(1 \text{ head}) = P(x=1) = {}^4C_1 (0.5)^1 (0.5)^3 = 0.25$$

$$(2) P(\text{atmost 3 heads}) = P(x=0) + P(x=1) + P(x=2) + P(x=3)$$

$$= {}^4C_0 (0.5)^0 (0.5)^4 + {}^4C_1 (0.5)^1 (0.5)^3 + {}^4C_2 (0.5)^2 (0.5)^2 + {}^4C_3 (0.5)^3 (0.5)$$

$$= (0.5)^4 (1 + 4 + 6 + 4) = 0.9375.$$

$$(3) P(\text{atleast 2 heads}) = 1 - [P(x=0) + P(x=1)]$$

$$= 1 - [(0.5)^4 + {}^4C_1 (0.5) (0.5)^3] = 0.6875.$$

Illustration 9

In 256 sets of 12 tosses of a coin, in how many cases one can expect 8 heads and 4 tails.

Solution

$$P(\text{head}) = \frac{1}{2} \text{ and } P(\text{tail}) = \frac{1}{2}$$

By Binomial distribution, Probability of 8 heads and 4 tails in 12 trials $P(X=8)$

$$= {}^{12}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^4 = \frac{12!}{8!4!} \times \frac{1}{2^{12}} = \frac{495}{4096}$$

$\therefore$ The expected number of such cases in 256 sets is

$$= 256 \times P(X=8) = 256 \times \frac{495}{4096} = 30.9 \approx 31.$$

Illustration 10

The number of telephone lines busy at an instant of time is a binomial variate with Probability 0.1 that a line is busy. If 10 lines are chosen at random, what is the Probability that (1) no line is busy (2) all lines are busy (3) at most 2 lines are busy (4) at least one line is busy.

Solution

By data $p = 0.1, q = 1 - p = 0.9, n = 10$.

$$P(x) = {}^{10}C_x p^x q^{10-x} = {}^{10}C_x (0.1)^x (0.9)^{10-x}$$

1. Probability that no line is busy $P(x=0) = (0.9)^{10} = 0.3487$

2. Probability that all lines are busy $P(10) = (0.1)^{10}$

3. Probability that at most 2 lines are busy $= P(X=0) + P(x=1) + P(x=2)$
 $= (0.9)^{10} + {}^{10}C_1 (0.1)^1 (0.9)^9 + {}^{10}C_2 (0.1)^2 (0.9)^8$
 $= 0.9298$

4. Probability that at least one line is busy $= 1 - P(X=0) = 1 - 0.3487$
 $= 0.6513.$

Illustration 11

Air line knows that 5% of the people making reservations on a certain flight will not turn up. Consequently their policy is to sell 52 tickers for a flight that can only hold 50 people. What is the Probability that there will be a seat for every passenger who turns up?

Solution

The Probability (P) that a passenger will not turn up is

$$p = 0.05 \text{ and } q = 1 - p = 1 - 0.05 = 0.95$$

By Binomial distribution

$$P(x) = {}^nC_x p^x q^{n-x} \text{ where } n = 52$$

$$P(x) = {}^{52}C_x (0.05)^x (0.95)^{52-x}$$

A seat is assured for every passenger who turns up if the number of passengers who to turn up is more than or equal to 2.

$$\therefore P(X \geq 2) = 1 - P(x < 2)$$

$$= 1 - [P(x=0) + P(x=1)]$$

$$= 1 - [(0.95)^{52} + 52 (0.05) (0.95)^{51}] = 0.7405$$

Probability that a seat is available for every Passenger is 0.7405.

Illustration 12

Team A has Probabilities of $\frac{2}{3}$ of winning when ever it plays. If it plays 4 games Find the Probability that it wins (1) 2 games (2) at least one game.

Solution

$$p = \frac{2}{3}, n = 4, q = 1 - p = 1 - \frac{2}{3} = \frac{1}{3}$$

$$P(x) = {}^nC_x p^x q^{n-x}$$

$$= {}^4C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{4-x}$$

$$(1) P(\text{team A wins 2 games}) = P(x = 2) = {}^4C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2 = 0.2963$$

$$\begin{aligned} (2) P(\text{A wins at least one game}) &= 1 - P(\text{A wins none}) \\ &= 1 - P(x = 0) \\ &= 1 - {}^4C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^4 = 1 - 0.0123 = 0.9871 \end{aligned}$$

Illustration 13

The mean of a Binomial distribution is 6. The standard deviation is $\sqrt{2}$. Find n and p .

Solution

$$\text{Mean} = np = 6, \text{ Variance} = npq = (\sqrt{2})^2 = 2$$

$$\frac{\text{Variance}}{\text{Mean}} = \frac{npq}{np} = \frac{2}{6} = \frac{1}{3} \Rightarrow \therefore q = \frac{1}{3}$$

$$p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{Mean} = np = n \times \frac{2}{3} = 6$$

$$n = 9 \Rightarrow \therefore n = 9, p = \frac{2}{3}$$

Illustration 14

In a large consignment of electric lamps 5% are defective. A random sample of 8 lamps is taken for an inspection. What is the Probability that it has one or more defectives?

Solution

$$n = 8, p = \frac{5}{100} = 0.05, q = 1 - p = 1 - 0.05 = 0.95$$

$$P(x) = {}^8C_x (0.05)^x (0.95)^{8-x}$$

$$\begin{aligned} P(\text{Sample has one or more defective}) &= 1 - P(\text{no defective}) \\ &= 1 - P(x = 0) \\ &= 1 - {}^8C_0 (0.05)^0 (0.95)^8 \\ &= 0.3366. \end{aligned}$$

Illustration 15

In a Binomial distribution mean = 6 and Variance = 1.5.

Find (1) $P(x = 2)$ (2) $P(x \leq 2)$

$$\text{Mean} = np = 6$$

$$\text{Variance} = npq = 1.5$$

$$\frac{\text{Variance}}{\text{Mean}} = \frac{npq}{nP} = \frac{1.5}{6} = \frac{1}{4}$$

$$q = \frac{1}{4}, p = \frac{3}{4}$$

$$\text{Mean} = n \times \frac{3}{4} = 6$$

$$\therefore n = 8$$

$$P(x) = {}^nC_x \left(\frac{3}{4}\right)^x \left(\frac{1}{4}\right)^{n-x}$$

$$(1) P(x=2) = {}^8C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^6 = 0.003845$$

$$(2) P(x \leq 2) = P(0) + P(1) + P(2)$$

$$= {}^8C_0 \left(\frac{3}{4}\right)^0 \left(\frac{1}{4}\right)^8 + {}^8C_1 \left(\frac{3}{4}\right)^1 \left(\frac{1}{4}\right)^7 + {}^8C_2 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^6 = 0.004227.$$

Illustration 16

Die were thrown 96 times and the number of times of an odd number actually turned out in the experiment is given. Fit a binomial distribution to this data and calculate the expected frequencies.

No. of dice showing 1 or 3 or 5	0	1	2	3	4	5
Observed frequency	1	10	24	35	18	8

Solution

$$p = \text{Probability of getting 1 or 3 or 5} = \frac{3}{6} = \frac{1}{2}$$

$$\therefore q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$$

Let x denote the number of times an odd number turning out.

$$P(x) = {}^nC_x p^x q^{n-x}, n = 5$$

$$P(x) = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x} = \frac{1}{2^5} \cdot {}^5C_x$$

Since 5 dice were thrown 96 times, expected frequencies are obtained from $f(x) = 96 P(x)$ where $x = 0, 1, 2, 3, 4, 5$.

$$f(x) = 96 \cdot \frac{1}{2^5} \cdot {}^5C_x = 3 \cdot {}^5C_x$$

$$f(0) = 3 \cdot {}^5C_0 = 3,$$

$$f(1) = 3 \cdot {}^5C_1 = 3 \times 5 = 15,$$

$$f(2) = 3 \cdot {}^5C_2 = 3 \times 10 = 30,$$

$$f(3) = 3 \cdot {}^5C_3 = 3 \times 10 = 30,$$

$$f(4) = 3 \cdot {}^5C_4 = 3 \times 5 = 15,$$

$$f(5) = 3 \cdot {}^5C_5 = 3 \times 1 = 3$$

The expected frequencies are 3, 15, 30, 30, 15, 3.

Illustration 17

In 800 families with 5 children each how many families would be expected to have
(1) 3 boys (2) 5 girls (3) either 2 or 3 (4) atmost 2 girls by assuming probabilities for boys and girls to be equal.

Solution

$$p = \text{Probability of having a boy} = \frac{1}{2}$$

$$q = \text{Probability of having a girl} = \frac{1}{2}$$

By Binomial distribution, $P(x) = {}^nC_x p^x q^{n-x}$ where $n = 5$

$$P(x) = {}^5C_x \left(\frac{1}{2}\right)^x \cdot \left(\frac{1}{2}\right)^{5-x} = \frac{1}{2^5} \cdot {}^5C_x = \frac{{}^5C_x}{32}$$

Since we have to find the expected number in respect of 800 families we have,

$$800 P(x) = 800 \cdot \frac{{}^5C_x}{32} = 25 \cdot {}^5C_x = f(x)$$

$$1. f(3) = 25 \cdot {}^5C_3 = 25 \times 10 = 250$$

Expected number of families with 3 boys is 250.

$$2. f(0) = 25 \cdot {}^5C_0 = 25 \times 1 = 25$$

Expected number of families with 5 girls is 25

$$\begin{aligned} 3. f(2) + f(3) &= 25 \cdot {}^5C_2 + 25 \cdot {}^5C_3 = 25 \times \frac{5 \times 4}{2 \times 1} + 25 \times \frac{5 \times 4 \times 3}{3 \times 2 \times 1} \\ &= 25 \times 10 + 25 \times 10 \\ &= 500 \end{aligned}$$

Expected number of families with 2 or 3 boys is 500.

4. Atmost 2 girls means that, families can have 5 boys and 0 girls or 4 boys and 1 girl or 3 boys and 2 girls.

$$\begin{aligned} f(5) + f(4) + f(3) &= 25 \cdot {}^5C_5 + 25 \cdot {}^5C_4 + 25 \cdot {}^5C_3 \\ &= 25 \times 1 + 25 \times 5 + 25 \times 10 \\ &= 25 + 125 + 250 \\ &= 400. \end{aligned}$$

Expected number of families with atmost 2 girls is 400.

Illustration
Find p for binomial

Solution:
The probability
 $P(X)$

Here, $P(X)$
 $P(X)$
 $P(X)$

Given that,
 $9 \cdot P(X)$
 $9 \cdot P(X)$
 $9 \cdot 15$

Taking positive

8.2 Poisson

The Poisson distribution is used to find the probability of time or space where events occur.

When to use

- The event occurs at a constant rate.
- The average number of events is known.
- Two events cannot occur at the same time.
- The probability of an event occurring is small.

Example Solve the following

- Number of events occurring in a given interval.
- Number of events occurring in a given interval.
- Number of events occurring in a given interval.
- Number of events occurring in a given interval.
- Number of events occurring in a given interval.

Example 18:

Find p for binomial random variable X if $n=6$ and $9.P(X=4)=P(X=2)$

Solution:

Probability mass function of binomial random variable X is given by

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

$$n = 6$$

$$P(X=x) = {}^6C_x p^x q^{6-x}$$

$$P(x=4) = {}^6C_4 p^4 q^2$$

$$P(x=2) = {}^6C_2 p^2 q^4$$

Given that,

$$9.P(x=4) = P(x=2)$$

$$9.{}^6C_4 p^4 q^2 = {}^6C_2 p^2 q^4$$

$$9.15p^2 = 15.q^2$$

$$9.p^2 = q^2$$

Taking positive square root on both sides we get,

$$3p = q$$

$$3p = 1 - p$$

$$4p = 1$$

$$p = \frac{1}{4} = 0.25$$

11. Poisson Distribution

The Poisson distribution gives the chance of a certain number of events happening in a fixed interval of time or space. It is named after the French mathematician Simeon Denis Poisson. It suits situations where events are rare and independent and occur at a steady average rate.

How to use it

- The events are independent of each other and one event does not affect another.
- The average rate of occurrence is constant over the interval.
- Two events cannot occur at exactly the same instant in the model.
- The interval is fixed such as one hour or one kilometre or one day.

Sample Scenarios;

- Number of phone calls received by a call centre in one hour.
- Number of customers who arrive at a bank counter in ten minutes.
- Number of typing errors on one printed page.
- Number of road accidents at a junction in one day.
- Number of emails received by an office in one hour.

The Poisson model uses one parameter called **lambda**. Lambda is the average number of events in the chosen interval. The expected value is lambda and the variance is also lambda. So the mean and the spread are the same in this model. If the interval is doubled then the average count roughly doubles and the model remains Poisson with a larger lambda.



Definition : Poisson Distribution

A discrete random variable X is said to follow a Poisson distribution if it assumes only non-negative integer values and its probability density function is given by

$$p(x) = P(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & \text{if } x = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

where:

- $P(X = x)$ is the probability of observing x events,
- e is the base of the natural logarithm (approximately 2.71828),
- λ is the average rate of occurrence of the event,
- x is the number of events observed,
- $x!$ denotes the factorial of x .

The parameter λ represents the average rate of occurrence of the event within the specified interval. For example, in the context of a call center, λ could represent the average number of calls received per hour.

Remarks

1. λ is the parameter of Poisson distribution. Poisson distribution has only one parameter.
2. A Poisson distribution with parameter λ is denoted by $P(x : \lambda)$ or $P(\lambda)$.
3. Poisson distribution may be considered as an approximation of the binomial distribution under the following conditions. (Binomial distribution tends to Poisson distribution under the following conditions.)
 - (i) p is very small ($p \rightarrow 0$)
 - (ii) n is very large ($n \rightarrow \infty$) and
 - (iii) $np = \lambda$ is fixed
4. Poisson distribution with parameter λ can be displayed as follows:

$x :$	0	1	2	3
$P(x) :$	$\frac{e^{-\lambda} \lambda^0}{0!}$	$\frac{e^{-\lambda} \lambda^1}{1!}$	$\frac{e^{-\lambda} \lambda^2}{2!}$	$\frac{e^{-\lambda} \lambda^3}{3!}$

5. The sum of the probabilities is

$$\sum_{x=0}^{\infty} P(x) = \frac{e^{-\lambda} \lambda^0}{0!} + \frac{e^{-\lambda} \lambda^1}{1!} + \frac{e^{-\lambda} \lambda^2}{2!} + \dots$$

6. Various theorems by N (Total frequency) The frequency

$$T_x = N$$

The Mean, Variance

The mean, variance parameter λ . The probability of observing a narrower distribution. The mean, variance

1. Mean : λ
2. Variance : λ
3. Standard Deviation : $\sqrt{\lambda}$

The mean and variance of occurrence and rare events in various



Application

The Poisson distribution

1. Telecommunication within a specific timeframe
2. Insurance claims within a specific timeframe
3. Environmental events like earthquakes
4. Biostatistics for certain populations
5. Quality control in manufacturing



Calculation

Take $e = 2.718$ and

How to calculate

- For

$$= e^{-\lambda} \left[\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \dots \right] = e^{-\lambda} \cdot e^{\lambda} = e^0 = 1$$

Various theoretical Poisson frequencies are obtained by multiplying the Poisson probabilities by N (Total frequency).
The frequency function is given by

$$T_x = N \cdot p(x) = N \cdot \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

Mean, Variance and Standard Deviation of the Poisson Distribution

Mean, variance, and standard deviation of the Poisson distribution are all dependent on the parameter λ . The larger the value of λ , the greater the spread of the distribution and the higher the probability of observing larger values of the random variable. Conversely, smaller values of λ result in a narrower distribution with a lower probability of observing larger values of the random variable. Mean, variance, and standard deviation of the Poisson distribution are given by:

1. Mean : λ

2. Variance : λ

3. Standard Deviation : $\sigma = \sqrt{\lambda}$

Mean and variance of the Poisson distribution provide valuable insights into the average rate of occurrence and the spread of the distribution, making it a useful tool for analyzing and predicting events in various real-world scenarios.

Applications of Poisson Distribution

Poisson distribution has a wide range of applications across various fields:

- Telecommunications** : It can be used to model the number of phone calls arriving at a call center within a specific time period.
- Insurance** : It is employed to estimate the number of insurance claims filed within a given timeframe.
- Environmental Studies** : It can be utilized to analyze the occurrence of natural disasters, such as earthquakes or floods, within a region.
- Biostatistics** : In epidemiology, it is used to study the incidence of diseases in a population over a certain period.
- Quality Control** : It finds application in monitoring the number of defects in a production process.

Calculating 'e' value

Let $e = 2.718$ as fixed

How to get powers of e :

- For positive powers, keep multiplying by e .

$$e^3 = e \times e \times e. \text{ Hence } e^3 = e^2 \times e$$

- For negative powers take the reciprocal of the positive power.

$$e^{-1} = 1/e. \text{ Hence } e^{-2} = 1/e^2. \text{ Hence } e^{-3} = 1/e^3$$

Quick results using $e = 2.718$:

- $e^1 = 2.718$
- $e^2 \approx 2.718 \times 2.718 = 7.389$
- $e^3 \approx 7.389 \times 2.718 = 20.086$
- $e^{-1} \approx 1/2.718 = 0.368$
- $e^{-2} \approx 1/7.389 = 0.135$
- $e^{-3} \approx 1/20.086 = 0.0498$

Problems on Poisson Distribution

Illustration 1

A call center receives an average of 10 calls per hour. What is the probability of receiving exactly 15 calls in the next hour?

Solution

$$\lambda = 10 \text{ and } x = 15.$$

Using the Poisson distribution formula, we get: ($e = 2.718$)

$$P(X = 15) = (e^{-10} \cdot 10^{15}) / 15!$$

$$P(X = 15) \approx 0.034$$

Therefore, the probability of receiving exactly 15 calls in the next hour is approximately 0.034.

Illustration 2

A manufacturing company produces an average of 5 defective products per day. What is the probability of producing no defective products in a day?

Solution

$\lambda = 5$ and $x = 0$. Using the Poisson distribution formula, we get:

$$P(X = 0) = (e^{-5} \cdot 5^0) / 0!$$

$$P(X = 0) \approx 0.007$$

Therefore, the probability of producing no defective products in a day is approximately 0.007.

Illustration 3

A hospital receives an average of 3 emergency cases per hour. What is the probability of receiving more than 5 emergency cases in the next 2 hours?

Solution

$\lambda = 6$ (since 3 cases per hour for 2 hours) and $x = 6$.

Using the Poisson distribution formula, we get:

$$P(X > 5) =$$

$$P(X \leq 5) =$$

$$P(X \leq 5) =$$

$$P(X > 5) =$$

$$P(X > 5) =$$

Therefore, the
approximately

Illustration

A company rec
receiving exact

Solution

$\lambda = 14$ (since 2 c
Using the Poisse

$$P(X$$

$$P(X$$

Therefor

Illustration

Find the prob
experience sho

Solution

p = probab

$$n = 200$$

$$\lambda = np = 2$$

Let X denote the

Then the distrib

So, probability

$$P(X \leq 5) = P$$

$$= e$$

$$= e$$

$$= 0$$

$$P(X > 5) = 1 - P(X \leq 5)$$

$$P(X \leq 5) = \sum_{x=0}^5 P(X=x)$$

$$P(X \leq 5) \approx 0.4454$$

$$P(X > 5) \approx 1 - 0.4454$$

$$P(X > 5) \approx 0.5546$$

therefore, the probability of receiving more than 5 emergency cases in the next 2 hours is approximately 0.5546

Illustration 4

A company receives an average of 2 customer complaints per day. What is the probability of receiving exactly 10 complaints in a week?

Solution

$\lambda = 14$ (since 2 complaints per day for 7 days) and $x = 10$.

Using the Poisson distribution formula, we get:

$$P(X = 10) = (e^{-14} \cdot 14^{10}) / 10!$$

$$P(X = 10) \approx 0.066$$

Therefore, the probability of receiving exactly 10 complaints in a week is approximately 0.066.

Illustration 5:

Find the probability that at most 5 defective fuses will be found in a box of 200 fuses if experience shows that 2 percent of such fuses are defective. ($e^{-4} \approx 0.0183$)

Solution

p = probability that a fuse is defective = $2/100$

$n = 200$

$$\lambda = np = 2/100 \times 200 = 4$$

Let X denote the number of defective fuses found in a box

Then the distribution is given by $P(X=x) = p(x) = \frac{e^{-4} 4^x}{x!}$

So, probability that at most 5 defective fuses will be found in a box of 200 fuses.

$$P(X \leq 5) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= e^{-4} + \frac{e^{-4}}{1!} + \frac{e^{-4} 4^2}{2!} + \frac{e^{-4} 4^3}{3!} + \frac{e^{-4} 4^4}{4!} + \frac{e^{-4} 4^5}{5!}$$

$$= e^{-4} \left[1 + \frac{4}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!} + \frac{4^5}{5!} \right]$$

$$= 0.0183 \times \frac{643}{15} = 0.785$$

Illustration 6:

The number of accidents in a year attributed to taxi drivers in a city follows poisson distribution with mean 3. Out of 1000 taxi drivers, find the approximate number of drivers with

- no accident in a year
- more than 3 accidents in a year

Solution

Here

$$\lambda = np = 3$$

$$n = 1000$$

Then the distribution is

$$P(X = x) = \frac{e^{-3} \cdot 3^x}{x!} \text{ where } X \text{ denotes the number accidents}$$

$$(i) P(\text{no accidents in a year}) = P(X = 0) = e^{-3} = 0.05$$

$$\therefore \text{Number of drivers with no accident} = 1000 \times 0.05 = 50$$

$$(ii) P(\text{that more than 3 accident in a year}) = P(X > 3)$$

$$= 1 - P(X \leq 3)$$

$$= 1 - \left[e^{-3} + \frac{e^{-3} 3^1}{1!} + \frac{e^{-3} 3^2}{2!} + \frac{e^{-3} 3^3}{3!} \right]$$

$$= 1 - e^{-3} [1 + 3 + 4.5 + 4.5]$$

$$= 1 - e^{-3}(13) = 1 - 0.0498 \times 13 = 1 - 0.65 = 0.35$$

$$\rightarrow \text{Number of drivers with more than 3 accidents} = 1000 \times 0.35 = 350$$

Illustration 7:

A local electrical appliances shop has found from experience that the demand for tube lights is distributed as Poisson with a mean of 4 tube lights per week. If the shop keeps 6 tubes during a particular week, what is the probability that the demand will exceed the supply during that week? ($e^{-4} = 0.0183$)

Solution:

Hint: Let the demand for the tube be taken as success.

Probability that the demand will exceed the supply:

$$= 1 - e^{-4} \left[\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!} + \frac{4^5}{5!} + \frac{4^6}{6!} \right] = 0.1114$$

Illustration 8:

In a certain factory blades are manufactured and packed in a package of 10. There is 0.2% probability for any blade to be defective. Using Poisson distribution calculate approximate number of packets containing two defective blades in a consignment of 20,000 packets (given $e^{-0.02} = 0.9802$).

Solution:
Average number
Required number

8.3 Normal

The Normal distribution is a bell-shaped curve. The mean and standard deviation are the parameters of the distribution. The normal distribution is used to model the spread of phenomena and errors.



De

A random variable
this distribution

$f(x)$

Where:

- $f(x)$
- e is
- π is
- μ is
- σ^2 is
- x is
- $-\infty$

The Normal distribution is the center of the mean.

Example

The Normal distribution is diverse

1. He
- The
- inc

number of defective blades in 1 packet of 10 blades (λ) = $np = 10 \times 0.002 = 0.02$

$$\begin{aligned} \text{number of packets} &= 20,000 \times \frac{e^{-0.02} \times (0.02)^2}{2} \\ &= 20,000 \times \frac{0.9802 \times 0.0004}{2} = 3.9208 = 4(\text{approx}) \end{aligned}$$

Normal Distribution

Normal Distribution is also known as the **Gaussian distribution** is a continuous probability distribution that is widely used in statistics, science, and engineering. It is characterized by a bell-shaped curve that is symmetric around the mean, with the majority of the data clustered around the mean and fewer data points further away from the mean.

The normal distribution is defined by two parameters: the mean (μ) and the standard deviation (σ). The mean represents the central tendency of the distribution, while the standard deviation represents the spread or variability of the data. The normal distribution is often used to model real-world phenomena that are naturally distributed, such as heights, weights, test scores, and measurement errors.

Definition : Normal Distribution

A random variable X is said to have a Normal distribution with Parameters μ and σ^2 , if its density function for X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where

- $f(x)$ is the probability density function
- e is the base of the natural logarithm (approximately 2.71828)
- π is the mathematical constant pi (approximately 3.14159)
- μ is the mean of the distribution
- σ^2 is the variance of the distribution
- x is the random variable.

$$-\infty < x < \infty, -\infty < \mu < \infty \text{ and } \sigma > 0.$$

The normal distribution is characterized by its symmetric, bell-shaped curve, with the mean μ representing the center of the distribution and the variance σ^2 determining the spread or dispersion of the data around the mean.

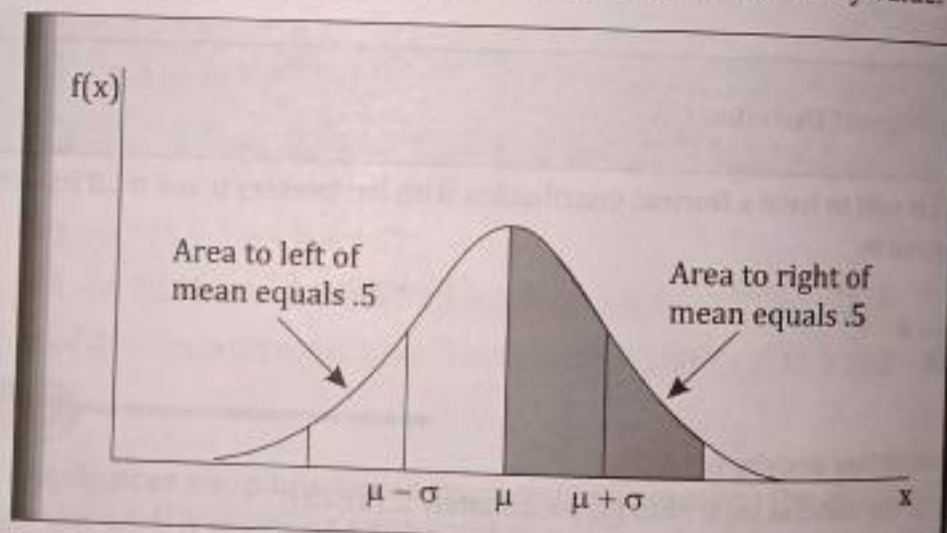
The Normal distribution can be used to model the following scenarios:

1. **Heights of Adult Males:** The heights of adult males in a population often follow a Normal distribution. This distribution can be used to model the variability in heights and analyze the probability of individuals falling within certain height ranges.

2. **IQ Scores:** Intelligence quotient (IQ) scores in a population are often assumed to follow a normal distribution. This allows for the analysis of intelligence levels and the determination of probabilities associated with specific IQ ranges.
3. **Stock Prices:** Daily or monthly stock returns are often assumed to be Normally distributed. This assumption allows for the analysis of stock price movements and the calculation of probabilities associated with specific return levels.
4. **Test Scores:** Scores on standardized tests, such as SAT or GRE, are often assumed to follow a normal distribution. This distribution is used to analyze the performance of test-takers and determine probabilities associated with specific score thresholds.
5. **Blood Pressure:** Blood pressure measurements in a population can be modeled using a normal distribution. This allows for the analysis of blood pressure variability and the determination of probabilities associated with specific blood pressure ranges.

Normal Distribution Curve

The below diagram depicts a normal distribution curve, which is a graphical representation of a normal probability distribution. The curve is symmetrical and bell-shaped, centered around the mean (μ), which is the highest point on the curve indicating the most likely value.



The key features are as follows:

1. **Mean (μ):** It's the average value and the center of the distribution. For a normal distribution, it's also the median and mode.
2. **Standard Deviation (σ):** It's a measure of the spread of the distribution. The curve changes width based on the standard deviation. A smaller σ leads to a steeper curve, and a larger σ leads to a flatter curve.
3. **Area Under the Curve:** The total area under the curve represents the entirety of the data points within the distribution. For a normal distribution, the area under the curve equals 1.
4. **Area to the Left of the Mean:** The area to the left of the mean ($\mu - \sigma$) is 0.5, which means that 50% of the data falls below the mean. This is because, in a normal distribution, half of the observations are less than the mean.
5. **Area to the Right of the Mean:** Similarly, the area to the right of the mean ($\mu + \sigma$) is also 0.5, indicating that 50% of the data is above the mean.

Inflection Points: The points at $\mu - \sigma$ and $\mu + \sigma$ are known as inflection points where the curve changes direction. These points mark one standard deviation from the mean, and about 68% of the data falls within one standard deviation of the mean in a normal distribution.

Mean and Standard Deviation of the Normal Distribution

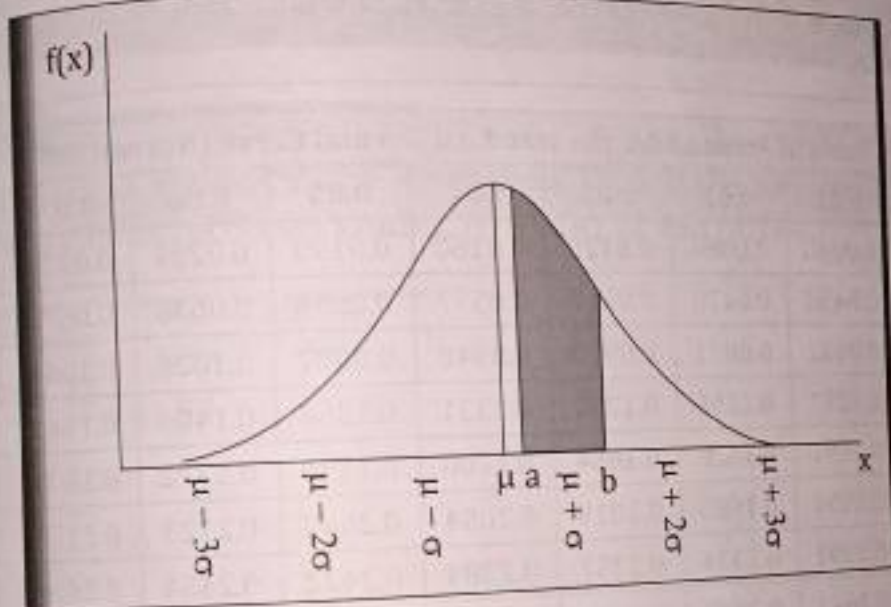
$$\text{Mean } (\mu) = \int_{-\infty}^{\infty} xf(x)dx$$

$$\text{Variance } (\sigma)^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$$

Variance and standard deviation of the normal distribution is equal to the variance and S.D of the distribution.

Properties of Normal Distribution

A normal distribution with parameters μ and σ has the following properties.



1. The curve is Bell-shaped.

(a) It is a symmetrical (Non-skew). That is, $\beta_1 = 0$

(b) The mean, median and mode are equal.

2. The curve is asymptotic to the X-axis. That is, the curve touches the X-axis only at $-\infty$ and $+\infty$

3. The curve has points of inflexion at $\mu - \sigma$ and $\mu + \sigma$.

4. For the distribution

(a) Standard Deviation = σ

(b) Quartile Deviation = $\frac{2}{3}\sigma$ (approx)

(c) Mean Deviation = $\frac{4}{5}\sigma$ (approx)

5. For the distribution

(a) The odd order moments are equal to zero

(b) The even order moments are given by

$$\mu_{2r} = 1 \times 3 \times 5 \dots \times (2r - 1) \sigma^{2r}$$

Thus, $\mu_2 = \sigma^2$ and $\mu_4 = 3\sigma^4$

6. The distribution is mesokurtic. That is $\beta_2 = 3$.



Applications of Normal Distribution

The Normal distribution has widespread applications in various fields, including:

- 1. Statistics and Data Analysis:** It is commonly used in statistical inference, hypothesis testing, and confidence interval estimation.
- 2. Finance and Economics:** It is employed to model stock prices, asset returns, and economic indicators.
- 3. Natural and Social Sciences:** It is used to describe measurements such as height, weight, IQ scores, and test scores in educational assessments.
- 4. Quality Control and Process Monitoring:** It is utilized in manufacturing and industrial processes to monitor product quality and process variability.
- 5. Risk Management:** It is applied in risk assessment and insurance to model and analyze potential losses and risks.

A Table of Areas under the standard Normal Curve (Normal Table)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3828
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817

2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

Problems on Normal Distribution

Illustration 1

On a final examination in statistics, the mean, was 72, and the S.D. was 15. Determine the standard scores of students receiving grades, (i) 95 (ii) 72 and (iii) 64.

Solution

$$(i) z = \frac{x - \bar{x}}{\sigma} = \frac{95 - 72}{15} = +1.53$$

$$(ii) z = \frac{x - \bar{x}}{\sigma} = \frac{72 - 72}{15} = 0$$

$$(iii) z = \frac{x - \bar{x}}{\sigma} = \frac{64 - 72}{15} = -0.53$$

Illustration 2

Find the area under the normal curves in each of the cases

$$(i) z = 0 \text{ and } z = 1.4$$

$$(iii) z = -0.46 \text{ and } z = 2.21$$

$$(v) \text{ To the left of } z = -0.5$$

$$(vii) \text{ Right of } z = 0.46$$

$$(ii) z = 0.78 \text{ and } z = 0$$

$$(iv) z = 0.81 \text{ and } z = 1.84$$

$$(vi) \text{ Right of } z = 1.48$$

Solution

$$(i) \text{ Area between } z = 0 \text{ and } z = 1.4 \\ = 0.4192$$



$$\begin{aligned} \text{(ii) Area between } z = -0.78 \text{ and } z = 0 \\ &= \text{Area } (0 \text{ and } 0.78) \\ &= 0.2823 \end{aligned}$$

$$\begin{aligned} \text{(iii) Area } -0.46 \text{ and } 2.21 \\ &= \text{Area between } [-0.46 \text{ and } 0] + \text{Area between } (0 \text{ and } 2.21) \\ &= \text{Area between } (0 \text{ and } 0.46) + \text{Area between } (0 \text{ and } 2.21) \\ &= 0.1772 + 0.4865 = 0.6637. \end{aligned}$$

$$\begin{aligned} \text{(iv) Area } (0.81 \text{ and } 1.83) \\ &= \text{Area } (0 \text{ and } 1.84) - \text{Area } (0 \text{ and } 0.81) \\ &= 0.4671 - 0.2910 = 0.1761 \end{aligned}$$

Note : Area $(-0.81 \text{ and } -1.84) = 0.1761$

$$\begin{aligned} \text{(v) Area of } (-\infty \text{ and } -0.5) \\ &= \text{Area } (-\infty \text{ and } 0) - \text{Area } (0 \text{ and } 0.5) \\ &= 0.5 - 0.1915 = 0.3085 \end{aligned}$$

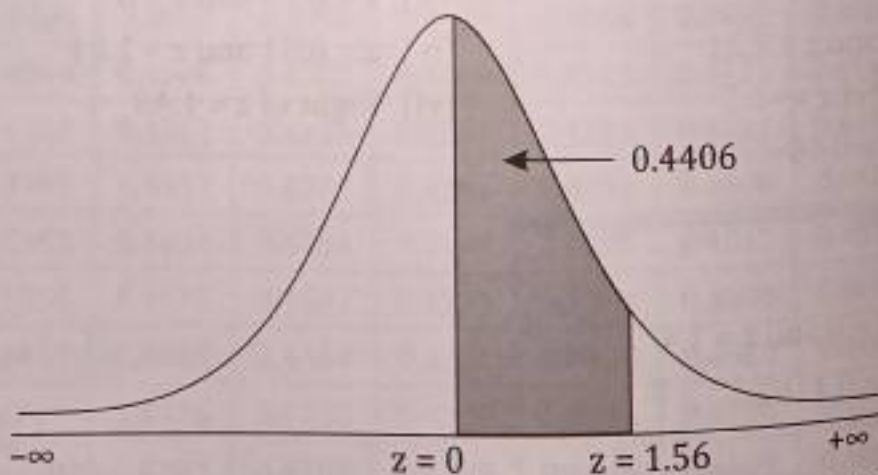
$$\begin{aligned} \text{(vi) Area } (1.48 \text{ and } \infty) \\ &= \text{Area } (0 \text{ and } \infty) - \text{Area } (0 \text{ and } 1.48) \\ &= 0.5 - 0.4306 = 0.0694 \end{aligned}$$

$$\begin{aligned} \text{(vii) Area of } (-0.46 \text{ and } \infty) \\ &= \text{Area } (-0.46 \text{ and } 0) + \text{Area } (0 \text{ and } \infty) \\ &= \text{Area } (0 \text{ and } 0.46) + 0.5 \\ &= 0.1772 + 0.5 = 0.6772 \end{aligned}$$

Illustration 3:

Find the probability that the standard normal variate lies between 0 and 1.56

Solution:



$$P(0 < z < 1.56) = \text{Area between } z = 0 \text{ and } z = 1.56 = 0.4406$$

Illustration
Find the area of
Solution:

Area b
 $P(-1.9$

Illustration
Find the area
Solution:

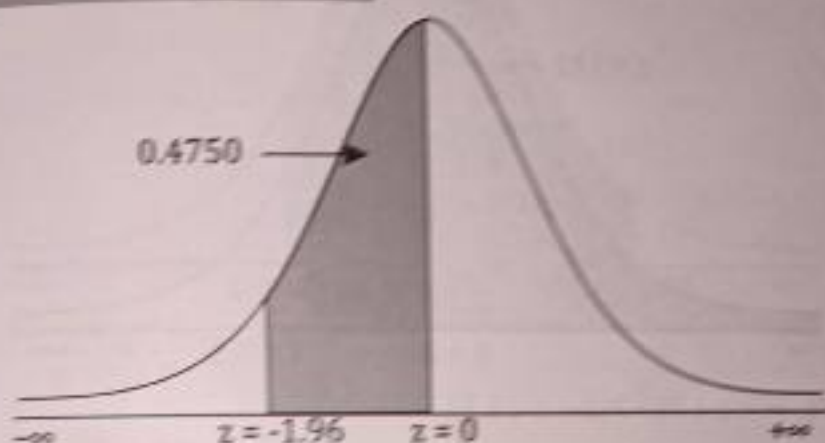
Illustration
Find the

[From table]

Illustration 4:

Find the area of the standard normal variate from -1.96 to 0 .

Solution:



Area between $z = 0$ & $z = 1.96$ is same as the area $z = -1.96$ to $z = 0$

$$P(-1.96 < z < 0) = P(0 < z < 1.96) \\ = 0.4750$$

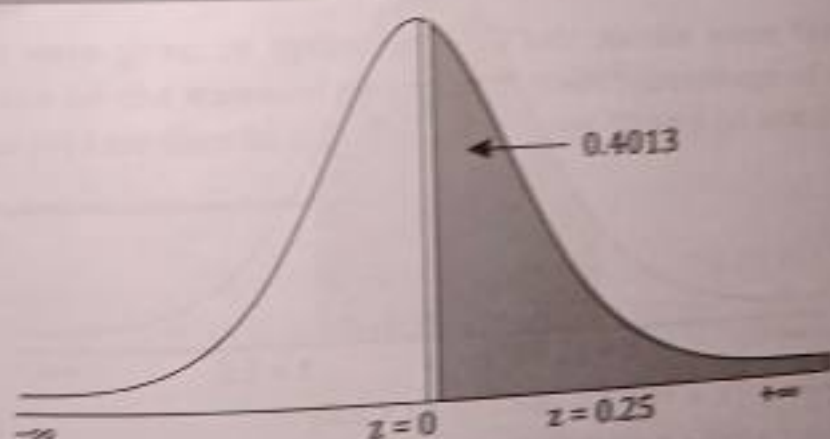
[By symmetry]

[From the table]

Illustration 5:

Find the area to the right of $z = 0.25$

Solution:

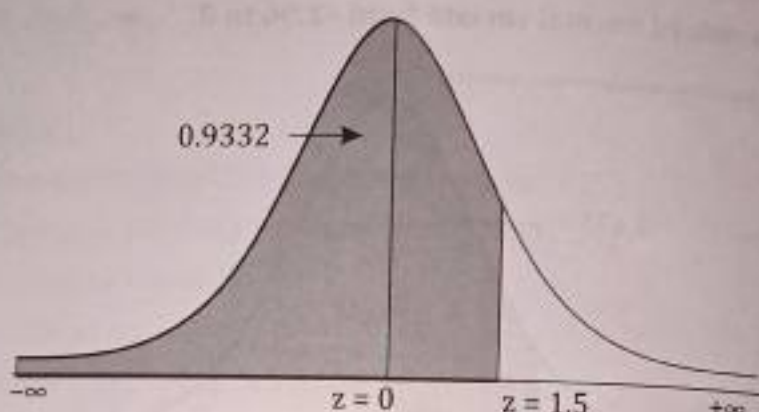


$$P(z > 0.25) = P(0 < z < \infty) - P(0 < z < 0.25) \\ = 0.5000 - 0.0987 = 0.4013$$

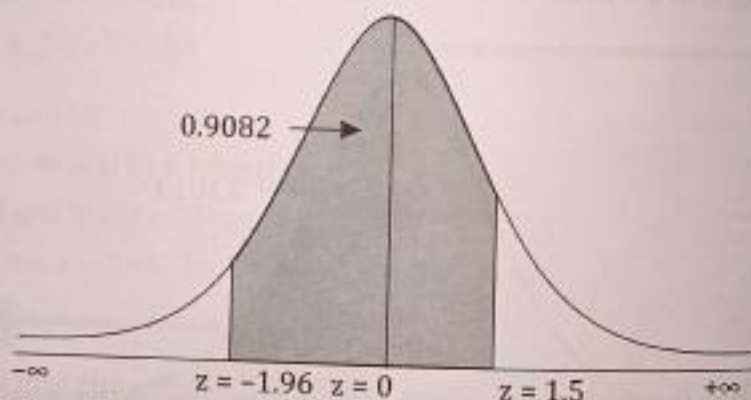
[From the table]

Illustration 6:

Find the area to the left of $z = 1.5$

Solution:

$$\begin{aligned} P(z > 1.5) &= P(-\infty < z < 0) + P(0 < z < 1.5) \\ &= 0.5 + 0.4332 = 0.9332 \end{aligned}$$

Illustration 7:Find the area of the standard normal variate between -1.96 and 1.5 **Solution:**

$$\begin{aligned} P(-1.96 < z < 1.5) &= P(-1.96 < z < 0) + P(0 < z < 1.5) \\ &= P(0 < z < 1.96) + P(0 < z < 1.5) \\ &= 0.4750 + 0.4332 = 0.9082 \end{aligned}$$

Illustration 8:Given a normal distribution with $\mu = 50$ $\sigma = 8$, find the probability that x assumes a value between 42 and 64.**Solution:**Given that $\mu = 50$ and $\sigma = 8$

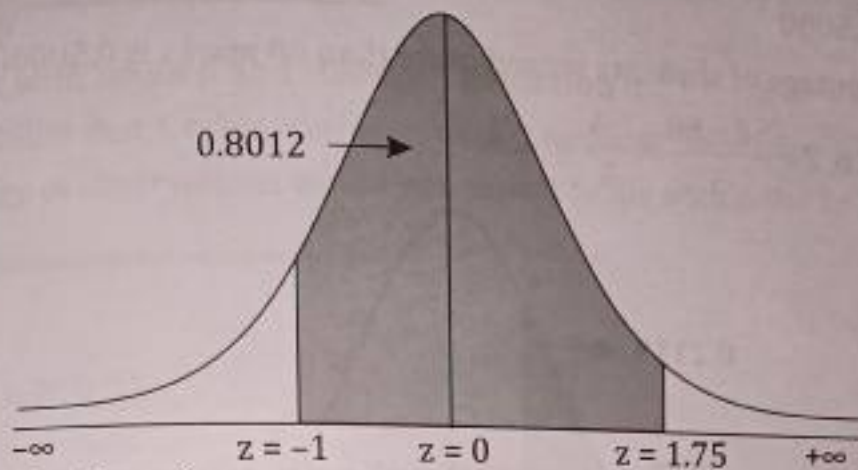
The standard normal variate

$$z = \frac{x - \mu}{\sigma}$$

If $x = 42$,If $x = 64$, $\therefore P(42 < x < 64)$ **Illustration 9**

Students of a class are distributed with mean 60 marks. More than 60 marks.

Solution:Given that mean (μ) = 60
i. The standard deviation (σ) = 8If $x = 60$ $\therefore P(x > 60)$



$$\text{If } X = 42, Z_1 = \frac{42 - 50}{8} = \frac{-8}{8} = -1$$

$$\text{If } X = 64, Z_2 = \frac{64 - 50}{8} = \frac{14}{8} = 1.75$$

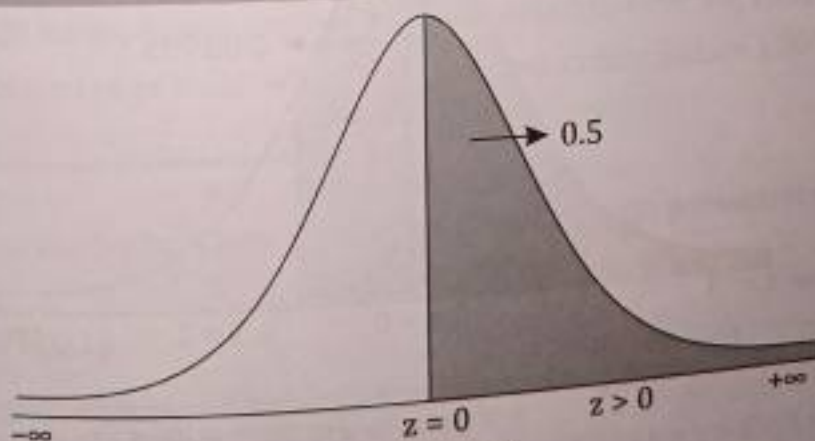
$$\begin{aligned} \therefore P(42 < X < 64) &= P(-1 < Z < 1.75) \\ &= P(-1 < Z < 0) + P(0 < Z < 1.75) \\ &= P(0 < Z < 1) + P(0 < Z < 1.75) \\ &= 0.3413 + 0.4599 = 0.8012 \end{aligned}$$

[By symmetry]

[From the table]

Illustration 9:

Students of a class were given an aptitude test. Their marks were found to be normally distributed with mean 60 and standard deviation 5. What percentage of students scored? (i) More than 60 marks (ii) Less than 56 marks (iii) Between 45 and 65 marks.

Solution:

Given that mean (μ) = 60 and standard deviation = σ = 5

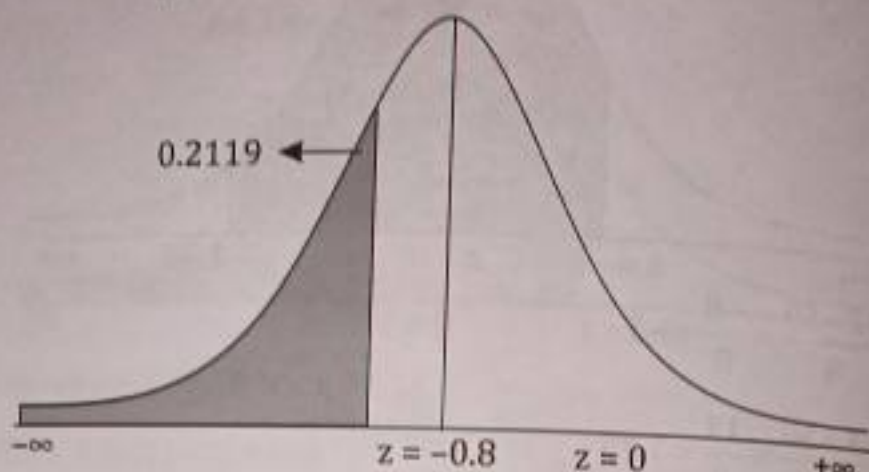
1. The standard normal variate $Z = \frac{X - \mu}{\sigma}$

$$\text{If } X = 60, Z = \frac{X - \mu}{\sigma} = \frac{60 - 60}{5} = 0$$

$$\begin{aligned} \therefore P(X > 60) &= P(Z > 0) \\ &= P(0 < Z < \infty) \end{aligned}$$

Hence the percentage of students scored more than 60 marks is $0.5000(100) = 50\%$.

ii. If $X = 56, Z = \frac{56 - 60}{5} = \frac{-4}{5} = -0.8$



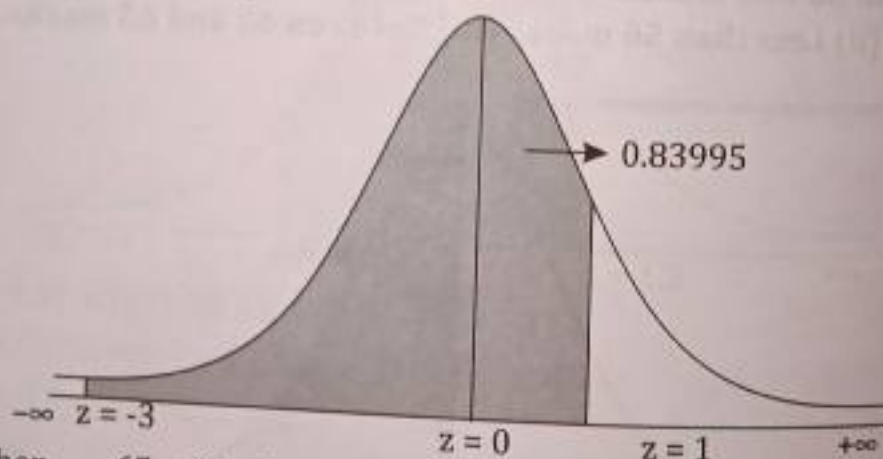
$$\begin{aligned} P(X < 56) &= P(Z < -0.8) \\ &= P(-\infty < Z < 0) - P(-0.8 < Z < 0) \\ &= P(0 < Z < \infty) - P(0 < Z < 0.8) \\ &= 0.5 - 0.2881 \\ &= 0.2119 \end{aligned}$$

[By Symmetry]

[From the table]

Hence the percentage of students score less than 56 marks is $0.2119(100) = 21.19\%$.

iii. If $X = 45$, then $Z = \frac{45 - 60}{5} = \frac{-15}{5} = -3$



$X = 65$ then $Z = \frac{65 - 60}{5} = \frac{5}{5} = 1$

$$\begin{aligned} P(45 < X < 65) &= P(-3 < Z < 1) \\ &= P(-3 < Z < 0) + P(0 < Z < 1) \\ &= P(0 < Z < 3) + P(0 < Z < 1) \\ &= 0.4986 + 0.3413 \\ &= 0.8399 \end{aligned}$$

[By symmetry]
[From the table]

Hence the percentage of students scored between 45 and 65 marks is $0.8399(100) = 83.99\%$

Illustration 10

X is a normal variate with mean μ and standard deviation σ

- Find the probability that X takes a value in the 2σ neighborhood of μ .
- What percentage of observations would you expect to lie within the 2σ limits?

Solution

Let $Z = \frac{X - \mu}{\sigma}$. Then, Z is $N(0, 1)$

- Probability that X takes a value in the 2σ neighborhood of μ is

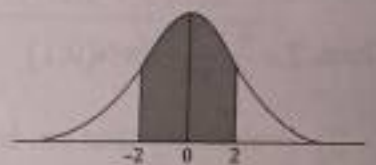
$$P[\mu - 2\sigma < X < \mu + 2\sigma]$$

$$= P\left[\frac{\mu - 2\sigma - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{\mu + 2\sigma - \mu}{\sigma}\right]$$

$$= P[-2 < Z < 2]$$

$$= [\text{area from } -2 \text{ to } 0] + [\text{area from } 0 \text{ to } 2]$$

$$= 0.4772 + 0.4772 = 0.9544$$



- Percentage of observation lying within 2σ limits is

$$100 \times P[\mu - 2\sigma < X < \mu + 2\sigma] = 100 \times 0.9544 = 95.44$$

Illustration 11

Mean life of electric bulbs manufactured by a firm is 1200 hrs. The standard deviation is 200 hrs.

- In a lot of 10,000 bulbs, how many bulbs are expected to have life 1050 hrs. or more?
- What is the percentage of bulbs which are expected to fuse before 1500 hrs. of service?

Solution

Let X denote the life of the bulbs. Then X is a normal variate with parameters $\mu = 1200$ hrs. and $\sigma = 200$ hrs.

$$Z = \frac{X - 1200}{200} \text{ is } N(0, 1)$$

- Probability that life of a bulb is 1050 hrs. or more is

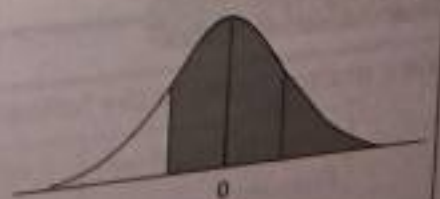
$$P[X \geq 1050] = P\left[\frac{X - 1200}{200} \geq \frac{1050 - 1200}{200}\right]$$

$$= P[Z \geq -0.75]$$

$$= 0.2734 + 0.5 = 0.7734$$

In a lot of $N = 10,000$ bulbs, expected number of bulbs with life 1050 hrs. or more is

$$N \times P[X \geq 1050] = 100 \times 0.7734 = 7734$$



(ii) Probability that life of a bulb is less than 1500 hrs. is

$$\begin{aligned} P[X < 1500] &= P\left[\frac{X - 1200}{200} \geq \frac{1500 - 1200}{200}\right] \\ &= P[Z < 1.5] \\ &= 0.5 + 0.4332 = 0.9332 \end{aligned}$$

The percentage of bulbs with life less than 1500 hrs is

$$100 \times P[X < 1500] = 100 \times 0.9332 = 93.32.$$

Illustration 12

Height of students is normally distributed with mean 165 cms. and standard deviation 5 cms. Find the probability that height of a student is (i) more than 177 cms (ii) less than 162 cms.

Solution

Let X denote height. Then, X is a normal variate with parameters $\mu = 165$ cms. and $\sigma = 5$ cms.

Then, $Z = \frac{X - 165}{5}$ is $N(0,1)$

(i) Probability that student is more than 177 cms tall is

$$\begin{aligned} P[X > 177] &= P\left[\frac{X - 165}{5} > \frac{177 - 165}{5}\right] \\ &= P[Z > 2.4] \\ &= \text{area from } 2.4 \text{ to } \infty \\ &= [\text{area from } 0 \text{ to } \infty] - [\text{area from } 0 \text{ to } 2.4] \\ &= 0.5 - 0.4918 = 0.0082 \end{aligned}$$

(ii) Probability that the student is less than 162 cms. tall is

$$\begin{aligned} P[X < 162] &= P\left[\frac{X - 165}{5} > \frac{162 - 165}{5}\right] \\ &= P[Z < -0.6] \\ &= \text{area from } (-\infty) \text{ to } (-0.6) \\ &= \text{area from } 0.6 \text{ to } \infty \\ &= 0.5 - 0.2258 = 0.2742 \end{aligned}$$

Illustration 13

Fit a normal curve to the following data

Length of line (in Cm)	8.60	8.59	8.58	8.57	8.56	8.55	8.54	8.53	8.52
Frequency	2	3	4	9	10	8	4	1	1

x	f	$d = x - a$	fd	fd^2
8.60	2	0.04	0.08	0.0032
8.59	3	0.03	0.09	0.0027
8.58	4	0.02	0.08	0.0016
8.57	9	0.01	0.09	0.0009
8.56	10	0	0	0
8.55	8	-0.01	-0.08	0.0008
8.54	4	-0.02	-0.08	0.0016
8.53	1	-0.03	-0.03	0.0009
8.52	1	-0.04	-0.04	0.0016
	$\Sigma f = 42$		$\Sigma fd = 0.11$	$\Sigma fd^2 = 0.0133$

$$\text{Mean}(\mu) = a + \frac{\Sigma fd}{\Sigma f} = 8.56 + \frac{0.11}{42} = 8.56262$$

$$SD(\sigma) = \sqrt{\frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f}\right)^2}$$

$$\sigma = \sqrt{\frac{0.0133}{42} - \left(\frac{0.11}{42}\right)^2} = 0.0176$$

Hence the equation of the normal curve fitted to the given data is

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad -\infty \leq x \leq \infty$$

Where $\mu = 8.56262$

$\sigma = 0.0176$

Illustration 14

Let x be a normally distributed random variable with a mean of 10 and a standard deviation of 2. Find the probability that x lies between 11 and 13.6.

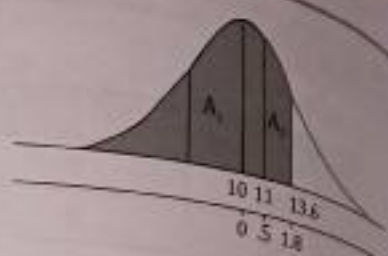
Solution

The interval from $x = 11$ to $x = 13.6$ must be standardized using the formula for z . When $x = 11$,

$$z = \frac{x - \mu}{\sigma} = \frac{11 - 10}{2} = .5 \text{ and when } x = 13.6$$

$$z = \frac{x - \mu}{\sigma} = \frac{13.6 - 10}{2} = 1.8$$

The desired probability is therefore $P(.5 \leq z \leq 1.8)$, the area lying between $z = .5$ and $z = 1.8$, as shown in below Figure. From Appendix A, you find that the area to the left of $z = .5$ is .6915, and the area to the left of $z = 1.8$ is .9641. The desired probability is the difference between these two probabilities, or $P(.5 \leq z \leq 1.8) = .9641 - .6915 = .2726$



Comparison of Binomial, Poisson and Normal Distribution

Comparison	Binomial Distribution	Poisson Distribution	Normal Distribution
Definition	The binomial distribution is a probability distribution that is used to model scenarios where there are a fixed number of independent trials, each with a binary outcome (success or failure), and where the probability of success is constant across all trials	The Poisson distribution is a probability distribution that is used to model scenarios where the number of occurrences of an event in a fixed interval of time or space follows a Poisson process.	The normal distribution is a statistical concept used to describe the way data is distributed. It is characterized by a symmetrical, bell-shaped curve, with the mean at the center and the majority of the data clustered around the mean. The standard deviation measures the spread of the data points around the mean.
Type of Variable	Discrete	Discrete	Continuous
Parameters	Number of trials (n) and probability of success (p) in each trial.	Mean rate of occurrence (λ)	Mean (μ) and standard deviation (σ)
Key Properties	Fixed number of trials. Two possible outcomes (success/failure). Trials are independent. Probability of success is constant for all trials.	Number of events in an interval is independent of the number in another non-overlapping interval Events occur singly. Mean rate doesn't change over time.	Symmetrical and bell-shaped Mean, median, and mode are equal Asymptotic tails approaching the x-axis
Mean	$n * p$	λ	μ
Variance	$n * p * (1 - p)$	λ	σ^2
Usage Examples	1. Number of heads when flipping a coin multiple times 2. Number of defective items in a batch	1. Number of calls received by a call center in an hour 2. Number of decay events from a radioactive source per second	1. Heights of individuals 2. IQ scores 3. Central Limit Theorem applications

Applications

8.4 Review

Short Answer

1. Define E
2. State tw
3. State th
4. Mention
5. Define E
6. State th
7. Write tw
8. Define M
9. Write an
10. Write th
11. State th
12. Mention
13. Differen

Long Answer

1. Explain
2. Discuss
3. A coin is (i) exa
4. A basket use the mean an
5. In 256 s expected

Applications	The binomial distribution is commonly used in quality control to analyze the number of defective items in a sample, in finance to model the probability of success or failure in investment decisions, and in biology to study genetic variations and the probability of specific traits in offspring.	The Poisson distribution finds applications in queuing theory to model the number of arrivals at a service point within a given time interval, in insurance to estimate the number of claims filed within a specific period, and in telecommunications to analyze the number of phone calls or data packets received by a system.	The normal distribution is widely used in finance to model stock prices and investment returns, in psychology to analyze test scores and intelligence quotient (IQ) measurements, and in physics to describe measurement errors and natural phenomena such as heights and weights of individuals.
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8.4 Review Questions

Short Answer Type Questions

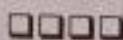
1. Define Binomial distribution.
2. State two conditions for the Binomial distribution.
3. State the mean and variance of a Binomial distribution.
4. Mention two applications of the Binomial distribution.
5. Define Poisson distribution.
6. State the parameter of a Poisson distribution and its mean and variance.
7. Write two conditions under which Binomial tends to Poisson distribution.
8. Define Normal distribution.
9. Write any two properties of the Normal curve.
10. Write the formula of the probability density function (PDF) of the Normal distribution.
11. State the relation between mean, median, and mode in a Normal distribution.
12. Mention two applications of the Normal distribution.
13. Differentiate between Binomial, Poisson, and Normal distributions (any two points).

Long Answer Type Questions

1. Explain the assumptions of Binomial distribution. Derive its mean and variance.
2. Discuss the practical applications of Binomial distribution.
3. A coin is tossed 4 times. Using Binomial distribution, find the probability of getting:
 - (i) exactly 1 head, (ii) at most 3 heads, (iii) at least 2 heads.
4. A basketball player has a success probability of 0.8 for free throws. If he attempts 10 throws, use the Binomial distribution to find the probability of exactly 7 successful throws. Also find mean and variance.
5. In 256 sets of 12 tosses of a coin, use the Binomial distribution to estimate how many cases are expected to show exactly 8 heads.

6. A company finds that 5% of electric lamps are defective. If a sample of 8 lamps is taken, use the Binomial distribution to find the probability that the sample has one or more defective lamps.
7. State and explain the properties of Poisson distribution.
8. Explain how Poisson distribution is derived as a limiting case of Binomial distribution.
9. A call center receives on average 10 calls per hour. Using Poisson distribution, find the probability of receiving exactly 15 calls in the next hour.
10. A factory produces on average 5 defective items per day. Using Poisson distribution, find the probability that on a particular day no defectives are produced.
11. A hospital receives on average 3 emergency cases per hour. Using Poisson distribution, find the probability of receiving more than 5 cases in 2 hours.
12. A company receives on average 2 complaints per day. Using Poisson distribution, find the probability that exactly 10 complaints are received in one week.
13. Explain the properties of Normal distribution with the help of a diagram.
14. Discuss the applications of Normal distribution in statistics, finance, and quality control.
15. Derive the mean and variance of Normal distribution.
16. On a statistics exam, the mean score was 72 and the standard deviation was 15. Using Normal distribution, find the Z-scores for students scoring (i) 95, (ii) 72, and (iii) 64.
17. The mean life of electric bulbs is 1200 hours with standard deviation 200 hours. In a lot of 10,000 bulbs, use Normal distribution to find:
 - (i) how many bulbs are expected to last at least 1050 hours,
 - (ii) percentage of bulbs expected to fuse before 1500 hours.
18. Heights of students are normally distributed with mean 165 cm and standard deviation 5 cm. Using Normal distribution, find the probability that a student's height is:
 - (i) more than 177 cm,
 - (ii) less than 162 cm.
19. Let X be normally distributed with mean 10 and standard deviation 2. Using Normal distribution, find the probability that X lies between 11 and 13.6.
20. Fit a Normal distribution to the following data and calculate the parameters:

Length of line (cm)	8.60	8.59	8.58	8.57	8.56	8.55	8.54	8.53	8.52
Frequency	2	3	4	9	10	8	4	1	1



Sampling is gold by example. Cashews are better by plucking based on a sample. A doctor studies a supervisor in most practical samples.

In this chapter, population, samples behave reliable conclusions.

In statistics, individuals are analyzed. Since it contains analysis.

A population describe. It includes:
 • Population
 • It may
 • Parameter